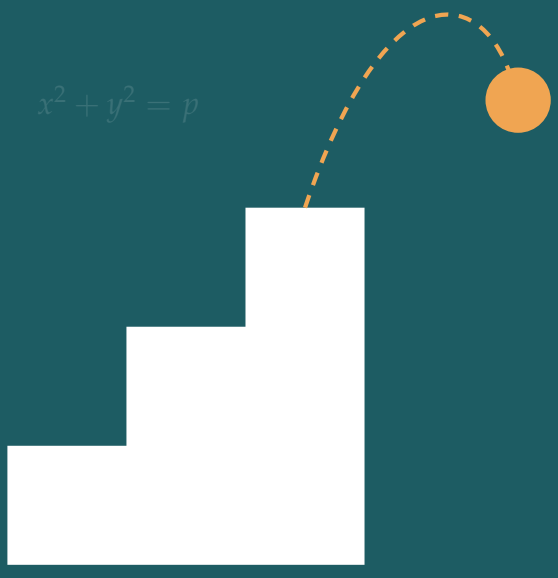
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2010 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2010 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2010 Worked Solutions

Worked solutions to all 39 questions of the 2010 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

First published online November 2026.

Everything in this PDF that is the author's own work, including the text, solutions, figures, exercises and answers, is dedicated to the public domain under the Creative Commons CC0 1.0 Universal licence: you may copy, adapt and share it for any purpose, without asking permission. See `creativecommons.org/publicdomain/zero/1.0`.

The STEP questions discussed here are the copyright of Cambridge University Press & Assessment. They are not reproduced: each one links to the question at an official source, the STEP Support Programme's question database or OCR's past papers.

Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

Contents

STEP I

Q1 Completing the Square Pure . . . 3
Q2 Curve Sketching Pure . . . 6
Q3 Sum-to-Product Formulae Pure . . . 10
Q4 Integration by Substitution Pure . . . 13
Q5 Binomial Coefficient Identities Pure . . . 16
Q6 Reduction of Order Pure . . . 19
Q7 Vectors and Intersecting Lines Pure . . . 22
Q8 Infinite Descent Pure . . . 26
Q9 Friction and Equilibrium Mechanics . . . 29
Q10 Vector Kinematics Mechanics . . . 33
Q11 Collisions on a Circle Mechanics . . . 37
Q12 Expectation via Tail Sums Probability and Statistics . . . 40
Q13 Poisson Waiting Times Probability and Statistics . . . 44

STEP II

Q1 Curvature and Osculating Circles Pure . . . 48
Q2 Trigonometric Integration Pure . . . 51
Q3 Fibonacci Numbers and Recurrences Pure . . . 54
Q4 Integration by Substitution Pure . . . 57
Q5 Vector Angles and Cones Pure . . . 60
Q6 Regular Tetrahedron Geometry Pure . . . 64
Q7 Cubic Equations Pure . . . 68
Q8 Areas Between Curves Pure . . . 72
Q9 Colliding Projectiles Mechanics . . . 75
Q10 Elastic Collisions Mechanics . . . 78
Q11 Three-Force Equilibrium Mechanics . . . 82
Q12 Mean and Median Probability and Statistics . . . 85
Q13 Independent Events Probability and Statistics . . . 89

STEP III

Q1 Mean and Variance Pure . . . 93

Q2 Definite Integrals Pure . . . 97

Q3 Roots of Unity Pure . . . 101

Q4 Common Roots of Polynomials Pure . . . 105

Q5 Coordinate Geometry Construction Pure . . . 109

Q6 Rotations in Three Dimensions Pure . . . 113

Q7 Maclaurin Series Pure . . . 116

Q8 Quotient Rule Integration Pure . . . 120

Q9 Motion on a Circle Mechanics . . . 124

Q10 Small Oscillations Mechanics . . . 127

Q11 Friction and Relative Motion Mechanics . . . 130

Q12 Geometric Distribution Expectation Probability and Statistics . . . 134

Q13 Correlation of Linear Combinations Probability and Statistics . . . 138

WORKED SOLUTION

STEP I 2010, Question 1

Pure • Completing the Square

STEP I 2010, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q1

The key idea

The identity rewrites an awkward quadratic as a sum of two squares plus a constant. Once you notice that the second equation can be written using the *same* two brackets, $x-y+2$ and $2x-y$, the problem becomes two simple equations in two new unknowns. These are linear in the squares of the new unknowns, so we can solve them directly.

Solution

The identity

Expanding the right-hand side,

$$\begin{aligned} &a(x-y+2)^2+b(cx+y)^2+d \\ &\quad = a(x^2+y^2-2xy+4x-4y+4)+b(c^2x^2+2cxy+y^2)+d \\ &\quad = (a+bc^2)x^2+(a+b)y^2+(2bc-2a)xy+4ax-4ay+(4a+d). \end{aligned}$$

Since this is an identity, we can compare coefficients with $5x^2+2y^2-6xy+4x-4y$:

$$\begin{aligned} x: &\quad 4a=4, && \text{so } a=1; \\ y: &\quad -4a=-4, && \text{which agrees;} \\ y^2: &\quad a+b=2, && \text{so } b=1; \\ xy: &\quad 2bc-2a=-6, && \text{so } 2c=-4,\ c=-2; \\ x^2: &\quad a+bc^2=5, && \text{check: } 1+4=5; \\ \text{constant}: &\quad 4a+d=0, && \text{so } d=-4. \end{aligned}$$

Note that the x^2 coefficient on its own only gives $c^2=4$; it is the xy coefficient that fixes the sign. So

$$a=1, \qquad b=1, \qquad c=-2, \qquad d=-4,$$

that is,

$$5x^2+2y^2-6xy+4x-4y \equiv (x-y+2)^2+(2x-y)^2-4,$$

where we have written $(-2x+y)^2$ as $(2x-y)^2$.

The simultaneous equations

By the identity, the first equation becomes

$$(x-y+2)^2+(2x-y)^2=13.$$

Ask yourself The second equation looks different. Can it be written using the same two brackets?

Its terms in x and y are $8x-8y$, twice those of the first equation. This suggests trying $2(x-y+2)^2$, which has the right linear terms:

$$2(x-y+2)^2=2x^2+2y^2-4xy+8x-8y+8.$$

The rest of the left-hand side is

$$6x^2+3y^2-8xy+8x-8y-2(x-y+2)^2=4x^2+y^2-4xy-8=(2x-y)^2-8.$$

So the second equation is $2(x-y+2)^2+(2x-y)^2-8=14$, or

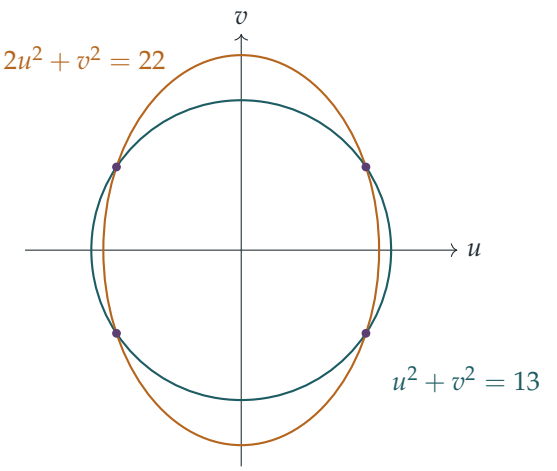
$$2(x-y+2)^2+(2x-y)^2=22.$$

(You could also find this the way we found the identity, by comparing coefficients in $\alpha(x-y+2)^2+\beta(\gamma x+y)^2+\delta$.)

Now let $u=x-y+2$ and $v=2x-y$. The system becomes

$$u^2+v^2=13, \qquad 2u^2+v^2=22.$$

Subtracting gives $u^2=9$, and then $v^2=4$. So $u=\pm3$ and $v=\pm2$, with the signs chosen independently, which gives four pairs (u,v).



To get back to x and y, subtract: $v-(u-2)=(2x-y)-(x-y)=x$, so

$$x=v-u+2, \qquad y=2x-v=v-2u+4.$$

Each (u,v) gives exactly one (x,y), because this change of variables is linear and invertible. The four solutions are:

u	v	x	y
3	2	1	0
3	-2	-3	-4
-3	2	7	12
-3	-2	3	8

So

$$(x,y)=(1,0), \quad (-3,-4), \quad (7,12), \quad (3,8).$$

Checks

Substitute $(1,0)$: the first equation gives $5+4=9$ and the second $6+8=14$. For $(7,12)$: $245+288-504+28-48=9$ and $294+432-672+56-96=14$. A computer confirmed the two identities at a thousand random rational points and confirmed that all four pairs satisfy both equations. A brute-force search over all integer pairs with $|x|,|y|\leq 50$ found exactly these four solutions. Two conics meet in at most four points, so we cannot have missed any.

Common slips

- Taking $c=2$ from $c^2=4$ without checking the xy coefficient.
- Keeping only the positive square roots, and so finding one solution instead of four.
- Eliminating a variable directly from the original equations. This gives a quartic, which is very hard work by hand.

Beyond STEP

Writing a quadratic form as a combination of squares is the start of the theory of quadratic forms. By diagonalising a symmetric matrix you can always find a linear change of variables that turns $Ax^2+Bxy+Cy^2$ into $\lambda_1u^2+\lambda_2v^2$, and this is how conics are classified. Here both equations happened to share the same pair of squares, which is why they decoupled so neatly. Strang's *Introduction to Linear Algebra* covers symmetric matrices and quadratic forms clearly.

WORKED SOLUTION

STEP I 2010, Question 2

Pure • Curve Sketching

STEP I 2010, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q2

The key idea

Differentiate with the quotient rule and pull out the factor e^x, which is never zero. The stationary points are then the roots of a quadratic, so "two stationary points" is a statement about its discriminant. For the sketches, the sign of that same quadratic tells us where the curve rises and falls, and the behaviour near the asymptote and as $x \to \pm\infty$ fills in the rest.

Solution

Write $y = \dfrac{x-a}{x-b}e^x$, defined for $x \neq b$. By the product rule, with $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{x-a}{x-b}\right) = \dfrac{(x-b)-(x-a)}{(x-b)^2} = \dfrac{a-b}{(x-b)^2}$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x-a}{x-b}e^x + \frac{a-b}{(x-b)^2}e^x = \frac{e^x\left[(x-a)(x-b)+(a-b)\right]}{(x-b)^2}.$$

Since $e^x > 0$ and $(x-b)^2 > 0$ for $x \neq b$, the sign of $\mathrm{d}y/\mathrm{d}x$ is the sign of

$$q(x) = (x-a)(x-b)+(a-b) = x^2 - (a+b)x + ab + a - b,$$

and the stationary points are exactly the roots of q other than $x = b$.

? **Ask yourself** Could a root of q sit at $x = b$, where the curve isn't defined?

No harm can come from this: $q(b) = a - b$, so $x = b$ is a root only when $a = b$. But then the curve is just $y = e^x$ (with a hole at $x = b$), which has no stationary points at all. So in every case the stationary points are precisely the real roots of q, and there are two of them exactly when q has two distinct real roots, that is, when its discriminant is positive:

$$(a+b)^2 - 4(ab+a-b) = (a-b)^2 - 4(a-b) = (a-b)(a-b-4) > 0.$$

A product of two factors is positive when both are positive or both negative. Both negative means $a-b<0$; both positive means $a-b>4$ (which already forces $a-b>0$). Hence two stationary points require

$$a-b<0 \quad \text{or} \quad a-b>4,$$

as required. (Conversely, either condition gives two stationary points.)

Part (i)

With $a=0$, $b=\frac{1}{2}$ we have $a-b=-\frac{1}{2}<0$, and

$$y=\frac{x\,\mathrm{e}^x}{x-\frac{1}{2}}, \qquad q(x)=x^2-\tfrac{1}{2}x-\tfrac{1}{2}=(x-1)\big(x+\tfrac{1}{2}\big).$$

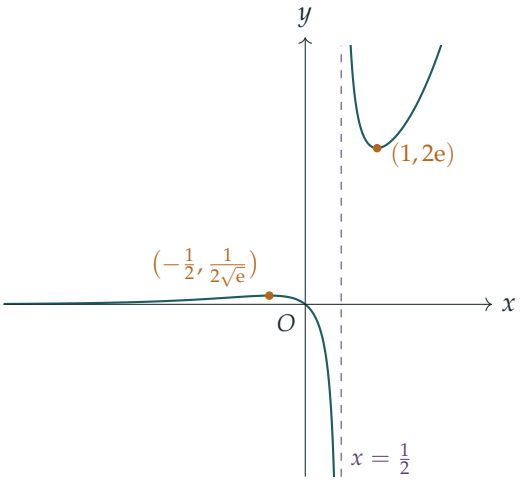
The stationary points are at $x=-\frac{1}{2}$ and $x=1$, one on each side of the vertical asymptote $x=\frac{1}{2}$, as required. Their y-values are

$$y\big(-\tfrac{1}{2}\big)=\frac{-\frac{1}{2}\,\mathrm{e}^{-1/2}}{-1}=\frac{1}{2\sqrt{\mathrm{e}}}\approx 0.30, \qquad y(1)=\frac{\mathrm{e}}{\frac{1}{2}}=2\mathrm{e}\approx 5.44.$$

Since $q>0$ for $x<-\frac{1}{2}$, $q<0$ for $-\frac{1}{2}<x<1$ (with $x\neq\frac{1}{2}$) and $q>0$ for $x>1$, the point $\big(-\frac{1}{2},\frac{1}{2\sqrt{\mathrm{e}}}\big)$ is a local maximum and $(1,2\mathrm{e})$ is a local minimum.

Other features:

- As $x\to-\infty$, $\frac{x}{x-1/2}\to 1$ and $\mathrm{e}^x\to 0$, so $y\to 0$. For $x<0$ both x and $x-\frac{1}{2}$ are negative, so $y>0$: the curve approaches the negative x-axis from above.
- The curve meets the axes only at the origin, and $y<0$ for $0<x<\frac{1}{2}$.
- As $x\to\frac{1}{2}^-$, $y\to-\infty$; as $x\to\frac{1}{2}^+$, $y\to+\infty$ (the numerator tends to $\frac{1}{2}\mathrm{e}^{1/2}>0$).
- As $x\to\infty$, $y\to\infty$, growing like e^x.



Part (ii)

With $a = \frac{9}{2}$, $b = 0$ we have $a - b = \frac{9}{2} > 4$, so again there are two stationary points. Now

$$y = \frac{x - \frac{9}{2}}{x}\,\mathrm{e}^x, \qquad q(x) = x^2 - \tfrac{9}{2}x + \tfrac{9}{2} = \tfrac{1}{2}(2x - 3)(x - 3),$$

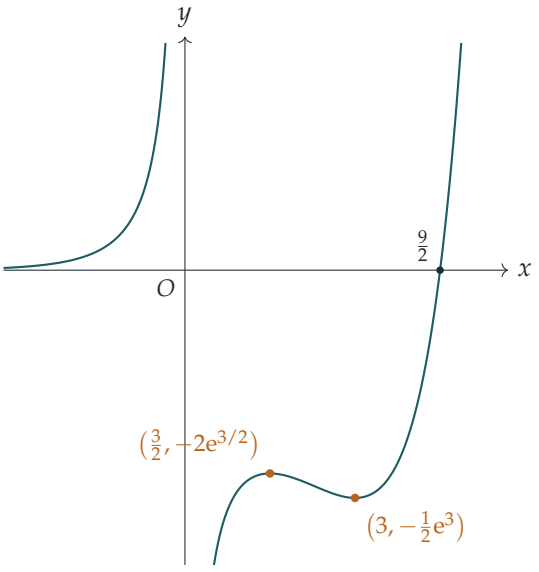
so the stationary points are at $x = \frac{3}{2}$ and $x = 3$, both to the right of the asymptote $x = 0$. Their y-values are

$$y\left(\tfrac{3}{2}\right) = \frac{-3}{\frac{3}{2}}\,\mathrm{e}^{3/2} = -2\mathrm{e}^{3/2} \approx -8.96, \qquad y(3) = \frac{-\frac{3}{2}}{3}\,\mathrm{e}^3 = -\tfrac{1}{2}\mathrm{e}^3 \approx -10.04.$$

The sign of q is positive for $x < \frac{3}{2}$ ($x \neq 0$), negative on $\left(\frac{3}{2}, 3\right)$ and positive for $x > 3$, so $\left(\frac{3}{2}, -2\mathrm{e}^{3/2}\right)$ is a local maximum and $\left(3, -\frac{1}{2}\mathrm{e}^3\right)$ a local minimum. Note that both lie below the x-axis, and the minimum is only slightly lower than the maximum, so the dip between them is shallow.

Other features:

- As $x \to -\infty$, $y \to 0$; for $x < 0$ both $x - \frac{9}{2}$ and x are negative, so $y > 0$. On $x < 0$ the curve is increasing (as $q > 0$ there), from just above the axis up towards the asymptote.
- As $x \to 0^-$, $y \to +\infty$; as $x \to 0^+$, $y \to -\infty$ (the numerator tends to $-\frac{9}{2}$).
- The only intercept is $\left(\frac{9}{2}, 0\right)$; $y < 0$ on $\left(0, \frac{9}{2}\right)$ and $y > 0$ for $x > \frac{9}{2}$, with $y \to \infty$ as $x \to \infty$.



Here the y-axis itself is the vertical asymptote.

Checks

- A numerical derivative of y at random (a, b, x) matches the formula for $\mathrm{d}y/\mathrm{d}x$, and counting sign changes of q for random (a, b) gives two stationary points exactly when $a - b < 0$ or $a - b > 4$.
- In (i), $y(1) = 5.4366 = 2\mathrm{e}$ and $y(-\frac{1}{2}) = 0.3033$; in (ii), $y(\frac{3}{2}) = -8.9634$ and $y(3) = -10.0428$, with values just either side of each point confirming maximum or minimum.
- Sanity check on the condition: in (i) $a - b = -\frac{1}{2}$ and in (ii) $a - b = \frac{9}{2}$, both in the allowed range, as they must be.

Common slips

- Expanding the discriminant without spotting that it is $(a - b)^2 - 4(a - b)$; it factorises neatly once written in terms of $a - b$.
- Forgetting to check that a root of q can't be at the asymptote $x = b$.
- In (ii), drawing the maximum and minimum on opposite sides of the x-axis: both are below it, and the curve crosses the axis only at $x = \frac{9}{2}$.

Beyond STEP

The method here, reading off the shape of a curve from the sign of its derivative and its limits, is made rigorous in a first analysis course through the Mean Value Theorem (a positive derivative on an interval really does mean increasing) and careful definitions of limits at infinity. That e^x beats any rational function as $x \to \pm\infty$ is a standard result proved there too. Abbott's *Understanding Analysis* covers all of this beautifully.

WORKED SOLUTION

STEP I 2010, Question 3

Pure • Sum-to-Product Formulae

STEP I 2010, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q3

The key idea

The sum-to-product formulae are what turn a difference of sines (or cosines) into a product. A chord of the ellipse joining the points with parameters u and v has gradient $\dfrac{b(\sin v - \sin u)}{a(\cos v - \cos u)}$. Both the top and the bottom are differences, so the formulae factor them, a common factor cancels, and the gradient depends only on the *sum* $u + v$. Comparing two chords then comes down to comparing two cotangents.

Solution

The trigonometric identities

Using the compound-angle formulae,

$$\begin{aligned}\sin(x+y) - \sin(x-y) &= (\sin x \cos y + \cos x \sin y) - (\sin x \cos y - \cos x \sin y)\\ &= 2\cos x \sin y,\end{aligned}$$

as required. Now, given any A and B, put $x = \frac{1}{2}(A+B)$ and $y = \frac{1}{2}(A-B)$, so that $x + y = A$ and $x - y = B$. The identity becomes

$$\sin A - \sin B = 2\cos \tfrac{1}{2}(A+B)\ \sin \tfrac{1}{2}(A-B).$$

In the same way,

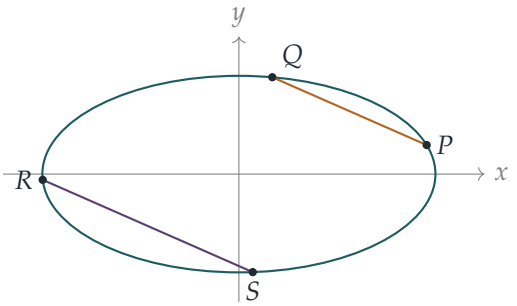
$$\begin{aligned}\cos(x+y) - \cos(x-y) &= (\cos x \cos y - \sin x \sin y) - (\cos x \cos y + \sin x \sin y)\\ &= -2\sin x \sin y,\end{aligned}$$

and the same substitution gives

$$\cos A - \cos B = -2\sin \tfrac{1}{2}(A+B)\ \sin \tfrac{1}{2}(A-B).$$

The parallel chords

All four points lie on the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$, taken anticlockwise in the order P, Q, R, S. The figure shows a parallel pair with $a=2$, $b=1$.



Gradient of a chord. Take two parameters $u<v$ in $[0,2\pi)$ such that the chord through $(a\cos u, b\sin u)$ and $(a\cos v, b\sin v)$ is not vertical, that is, $\cos v\neq\cos u$. By the two identities with $A=v$, $B=u$, its gradient is

$$\frac{b(\sin v-\sin u)}{a(\cos v-\cos u)}=\frac{2b\cos\frac{1}{2}(u+v)\,\sin\frac{1}{2}(v-u)}{-2a\sin\frac{1}{2}(u+v)\,\sin\frac{1}{2}(v-u)}=-\frac{b}{a}\cot\tfrac{1}{2}(u+v).$$

The cancellation is legitimate: $0<v-u<2\pi$, so $\frac{1}{2}(v-u)$ lies strictly between 0 and π and $\sin\frac{1}{2}(v-u)\neq 0$. Since the original denominator is non-zero, $\sin\frac{1}{2}(u+v)\neq 0$ too, so the cotangent is defined.

Write $\alpha=\frac{1}{2}(p+q)$ and $\beta=\frac{1}{2}(r+s)$. Neither PQ nor SR is vertical, so

$$\text{gradient of } PQ=-\frac{b}{a}\cot\alpha,\qquad \text{gradient of } SR=-\frac{b}{a}\cot\beta,$$

with $\sin\alpha\neq 0$ and $\sin\beta\neq 0$. Two non-vertical lines are parallel exactly when their gradients are equal, and since $b/a\neq 0$,

$$PQ\parallel SR\iff\cot\alpha=\cot\beta.$$

Ask yourself Why can't we just "cancel the cot" and conclude $\alpha=\beta$?

Because cot is not one-to-one: it repeats with period π. To keep track of this, write the difference as a single sine:

$$\cot\alpha-\cot\beta=\frac{\cos\alpha\sin\beta-\sin\alpha\cos\beta}{\sin\alpha\sin\beta}=\frac{\sin(\beta-\alpha)}{\sin\alpha\sin\beta}.$$

The denominator is non-zero, so

$$PQ\parallel SR\iff\sin(\beta-\alpha)=0\iff\beta-\alpha=k\pi\text{ for some integer }k.$$

Pinning down k. We have

$$\beta - \alpha = \tfrac{1}{2}\big[(r-p)+(s-q)\big].$$

Since $p < r$ and $q < s$, both brackets are positive, so $\beta - \alpha > 0$. Since all four parameters lie in $[0, 2\pi)$, each bracket is less than 2π, so $\beta - \alpha < 2\pi$. The only multiple of π strictly between 0 and 2π is π itself, so $k = 1$. Therefore

$$PQ \parallel SR \iff \tfrac{1}{2}(r+s-p-q) = \pi \iff r+s-p-q = 2\pi.$$

Every step is an equivalence, so this proves both directions, as required.

(The two lines can never coincide: a straight line meets an ellipse in at most two points, and P, Q, R, S are four distinct points. So "parallel" here means genuinely parallel and distinct.)

Checks

- Take $p = 0$, $q = \frac{\pi}{2}$, $r = \pi$, $s = \frac{3\pi}{2}$, so that $r+s-p-q = 2\pi$. Then $P = (a, 0)$, $Q = (0, b)$, $R = (-a, 0)$, $S = (0, -b)$, and both PQ and SR have gradient $-b/a$. With $a = 2$, $b = 1$ both gradients come out as -0.5.
- Symmetry: the condition says $\frac{1}{2}(r+s) = \frac{1}{2}(p+q) + \pi$. The point $(x, y) \mapsto (-x, -y)$ adds π to a parameter, and rotating the chord PQ through a half-turn gives a parallel chord, which fits.
- Numerically, with random a, b and random $p < q < r < s$, the gradients agreed exactly in the cases with $r+s-p-q = 2\pi$ (about 5000 constructed examples) and never otherwise.

Common slips

- Going from $\cot\alpha = \cot\beta$ straight to $\alpha = \beta$. That gives $r+s = p+q$, which is impossible here. The period of cot is the whole point.
- Forgetting to say why $\sin\frac{1}{2}(v-u)$ can be cancelled, or not using the ordering $p < q < r < s$ to rule out $k = 0$ and $k = 2$.

Beyond STEP

Writing the ellipse as $(a\cos t, b\sin t)$ shows it as the image of the unit circle under the stretch $(x, y) \mapsto (ax, by)$. A linear map like this sends parallel lines to parallel lines, so the question is really about the circle. There, the chord from angle p to angle q is perpendicular to the radius at angle $\frac{1}{2}(p+q)$, and the result becomes almost obvious. Ideas like this one, where properties preserved by a map are used to simplify a problem, run through linear algebra and geometry. Strang's *Introduction to Linear Algebra* is a good place to see linear maps at work. Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with trigonometric identities.

WORKED SOLUTION

STEP I 2010, Question 4

Pure • Integration by Substitution

STEP I 2010, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q4

The key idea

The substitution is chosen so that x and $x+1$ both become simple fractions with the same denominator t^2-1, which makes the square root of their product come out exactly. The integral then turns into the standard one in the note. For the volume, squaring y produces two easy logarithms and a cross term, and the cross term is precisely the integral we have just found.

Solution

The integral

Take $x>0$. The substitution $x=\dfrac{1}{t^2-1}$ with $t>1$ is legitimate: solving for t gives

$$t=\sqrt{1+\frac{1}{x}}=\frac{\sqrt{x+1}}{\sqrt{x}},$$

and as t runs over $(1,\infty)$, x runs over $(0,\infty)$ (decreasingly), one value of x for each t.

Now

$$x+1=\frac{1}{t^2-1}+1=\frac{t^2}{t^2-1}, \qquad \text{so} \qquad x(x+1)=\frac{t^2}{(t^2-1)^2}.$$

Since $t>1$, both t and t^2-1 are positive, so

$$\sqrt{x(x+1)}=\frac{t}{t^2-1}.$$

? **Ask yourself** Why did we need $t>1$ when taking the square root?

A square root is never negative, so $\sqrt{t^2/(t^2-1)^2}=|t|/|t^2-1|$. The condition $t>1$ lets us drop both modulus signs. It also explains the choice: $t^2-1>0$ is exactly what makes $x>0$.

Differentiating, $\dfrac{\mathrm{d}x}{\mathrm{d}t} = -\dfrac{2t}{(t^2-1)^2}$. Therefore

$$\int \frac{1}{\sqrt{x(x+1)}}\,\mathrm{d}x = \int \frac{t^2-1}{t}\cdot\left(-\frac{2t}{(t^2-1)^2}\right)\mathrm{d}t = -2\int \frac{1}{t^2-1}\,\mathrm{d}t$$
$$= -2\cdot\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right| + c = \ln\left(\frac{t+1}{t-1}\right) + c,$$

using the note with $a = 1$, and the fact that $t > 1$ makes $(t-1)/(t+1)$ positive.

Now return to x using $t = \sqrt{x+1}/\sqrt{x}$. Multiplying top and bottom by $\sqrt{x}$,

$$\frac{t+1}{t-1} = \frac{\sqrt{x+1}+\sqrt{x}}{\sqrt{x+1}-\sqrt{x}} = \frac{\left(\sqrt{x+1}+\sqrt{x}\right)^2}{(x+1)-x} = \left(\sqrt{x}+\sqrt{x+1}\right)^2,$$

where we multiplied top and bottom by $\sqrt{x+1}+\sqrt{x}$ and used the difference of two squares. Hence

$$\int \frac{1}{\sqrt{x(x+1)}}\,\mathrm{d}x = 2\ln\left(\sqrt{x}+\sqrt{x+1}\right) + c,$$

as required.

The volume

The curve has $y > 0$ for $x > 0$, since $\sqrt{x} < \sqrt{x+1}$ gives $1/\sqrt{x} > 1/\sqrt{x+1}$. So the volume swept out by the region between the curve and the x-axis is $V = \pi\int y^2\,\mathrm{d}x$. Expanding,

$$y^2 = \frac{1}{x} + \frac{1}{x+1} - \frac{2}{\sqrt{x}\sqrt{x+1}} = \frac{1}{x} + \frac{1}{x+1} - \frac{2}{\sqrt{x(x+1)}},$$

and the last term is twice the integrand from the first part. So

$$V = \pi\Big[\ln x + \ln(x+1)\Big]_{1/8}^{9/16} - 2\pi\Big[2\ln\left(\sqrt{x}+\sqrt{x+1}\right)\Big]_{1/8}^{9/16}.$$

The limits are chosen so that the square roots are tidy:

- at $x = \frac{9}{16}$: $\sqrt{x} = \frac{3}{4}$, $\sqrt{x+1} = \frac{5}{4}$, sum 2, and $x(x+1) = \frac{225}{256}$;
- at $x = \frac{1}{8}$: $\sqrt{x} = \frac{1}{2\sqrt{2}}$, $\sqrt{x+1} = \frac{3}{2\sqrt{2}}$, sum $\frac{4}{2\sqrt{2}} = \sqrt{2}$, and $x(x+1) = \frac{9}{64}$.

The first bracket is

$$\ln\frac{225}{256} - \ln\frac{9}{64} = \ln\left(\frac{225}{256}\cdot\frac{64}{9}\right) = \ln\frac{25}{4},$$

and the second is

$$2\ln 2 - 2\ln\sqrt{2} = 2\ln 2 - \ln 2 = \ln 2.$$

Therefore

$$V = \pi\ln\frac{25}{4} - 2\pi\ln 2 = \pi\ln\frac{25}{4} - \pi\ln 4 = \pi\ln\frac{25}{16} = 2\pi\ln\frac{5}{4},$$

which is the result we wanted.

Checks

- Differentiate the answer:

$$\frac{\mathrm{d}}{\mathrm{d}x}\,2\ln\left(\sqrt{x}+\sqrt{x+1}\right)=\frac{2}{\sqrt{x}+\sqrt{x+1}}\left(\frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{x+1}}\right)=\frac{1}{\sqrt{x}\sqrt{x+1}},$$

  since the bracket equals $\dfrac{\sqrt{x+1}+\sqrt{x}}{2\sqrt{x}\sqrt{x+1}}$.

- Numerically, Simpson's rule gives $\pi\int_{1/8}^{9/16}y^2\,\mathrm{d}x\approx 1.402052$, matching $2\pi\ln\frac{5}{4}\approx 1.402052$. Simpson's rule also confirms $\int_{1/8}^{9/16}\frac{\mathrm{d}x}{\sqrt{x(x+1)}}=\ln 2\approx 0.693147$.

Common slips

- Forgetting the minus sign in $\mathrm{d}x/\mathrm{d}t$. It is what turns $\ln\frac{t-1}{t+1}$ into $\ln\frac{t+1}{t-1}$.
- Expanding y^2 as $\frac{1}{x}-\frac{1}{x+1}$: the cross term $-2/\sqrt{x(x+1)}$ is the whole point of the first part.
- Forgetting the factor π, or using 2π instead of π, in the volume formula.

Beyond STEP

The answer $2\ln\left(\sqrt{x}+\sqrt{x+1}\right)$ is really $2\operatorname{arsinh}\sqrt{x}$, because $\operatorname{arsinh}u=\ln\left(u+\sqrt{u^2+1}\right)$. The substitution $x=\sinh^2\theta$ gives the same result in one line. Choosing the substitution that rationalises a square root is a systematic technique (the Euler substitutions), and it lies behind the result that every integral of a rational function of x and $\sqrt{ax^2+bx+c}$ can be done in elementary functions. Siklos, *Advanced Problems in Mathematics*, has many more integrals in this spirit.

WORKED SOLUTION

STEP I 2010, Question 5

Pure • Binomial Coefficient Identities

STEP I 2010, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q5

The key idea

Every one of these sums is the binomial expansion of $(1+x)^n$ in disguise, evaluated at $x = 1$ after we have done something to it. A factor of k in front of $\binom{n}{k}$ is what differentiating x^k produces; a factor $\frac{1}{k+1}$ is what integrating x^k produces. So we differentiate or integrate the identity

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$$

and then put $x = 1$. For k^2 we need to differentiate twice, with a multiplication by x in between.

Solution

Throughout, n is a positive integer and

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n \qquad (*)$$

holds for every real x. Because both sides are polynomials, we may differentiate or integrate them term by term and the identity survives.

Part (i)

Put $x = 1$ in $(*)$. The left-hand side is 2^n and each x^k becomes 1, so

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n,$$

as required.

Part (ii)

Differentiate $(*)$ with respect to x. The constant term disappears and $\binom{n}{k}x^k$ becomes $k\binom{n}{k}x^{k-1}$:

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \cdots + n\binom{n}{n}x^{n-1}. \qquad (**)$$

Putting $x = 1$ gives

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} = n\,2^{n-1},$$

as required.

Part (iii)

This time we want $\binom{n}{k}$ divided by $k+1$, which is what we get from $\int_0^1 x^k\,\mathrm{d}x = \frac{1}{k+1}$. Integrating $(*)$ from 0 to 1, the left-hand side is

$$\int_0^1 (1+x)^n\,\mathrm{d}x = \left[\frac{(1+x)^{n+1}}{n+1}\right]_0^1 = \frac{2^{n+1}-1}{n+1},$$

and the right-hand side is

$$\sum_{k=0}^{n}\binom{n}{k}\int_0^1 x^k\,\mathrm{d}x = \sum_{k=0}^{n}\frac{1}{k+1}\binom{n}{k}.$$

Hence

$$\binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \cdots + \frac{1}{n+1}\binom{n}{n} = \frac{1}{n+1}\left(2^{n+1}-1\right),$$

as required. Note that the lower limit matters: integrating from 0 is what produces the -1.

Part (iv)

Ask yourself If we simply differentiate $(**)$ again, we get coefficients $k(k-1)$, not k^2. How can we get k^2 instead?

Before differentiating a second time, multiply $(**)$ by x. That restores the power x^k alongside $k\binom{n}{k}$, so the next derivative brings down another factor of exactly k:

$$nx(1+x)^{n-1} = \sum_{k=1}^{n} k\binom{n}{k}x^k.$$

Differentiating, using the product rule on the left,

$$n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2} = \sum_{k=1}^{n} k^2\binom{n}{k}x^{k-1}.$$

(When $n = 1$ the second term on the left has the factor $n - 1 = 0$, so it is simply absent; the identity still holds.) Now put $x = 1$:

$$\begin{aligned}\binom{n}{1} + 2^2\binom{n}{2} + 3^2\binom{n}{3} + \cdots + n^2\binom{n}{n} &= n\,2^{n-1} + n(n-1)2^{n-2} \\ &= n\,2^{n-2}\big(2 + (n-1)\big) \\ &= n(n+1)2^{n-2},\end{aligned}$$

as required.

An alternative is to write $k^2 = k(k-1) + k$. Differentiating $(*)$ twice and putting $x = 1$ gives $\sum k(k-1)\binom{n}{k} = n(n-1)2^{n-2}$, and adding the result of part (ii) gives the same answer.

Checks

Take $n = 3$, where the coefficients are $1, 3, 3, 1$.

- (i) $1 + 3 + 3 + 1 = 8 = 2^3$.
- (ii) $3 + 6 + 3 = 12 = 3 \cdot 2^2$.
- (iii) $1 + \frac{3}{2} + 1 + \frac{1}{4} = \frac{15}{4} = \frac{1}{4}(2^4 - 1)$.
- (iv) $3 + 12 + 9 = 24 = 3 \cdot 4 \cdot 2$.

We also checked all four identities exactly, with fractions, for every n from 1 to 40.

Common slips

- In (iii), forgetting the contribution from the lower limit and getting $\frac{2^{n+1}}{n+1}$.
- In (iv), differentiating $(**)$ directly, which gives $\sum k(k-1)\binom{n}{k}$ rather than $\sum k^2\binom{n}{k}$.

Beyond STEP

These identities have probabilistic meanings. If X counts the heads in n tosses of a fair coin, then $\mathrm{P}(X = k) = \binom{n}{k}/2^n$, so part (ii) says $\mathrm{E}(X) = \frac{n}{2}$ and part (iv) says $\mathrm{E}(X^2) = \frac{n(n+1)}{4}$, giving $\mathrm{Var}(X) = \frac{n}{4}$. Differentiating a generating function to extract moments is a standard first-year technique. Grimmett and Welsh, *Probability: An Introduction*, develops it properly.

WORKED SOLUTION

STEP I 2010, Question 6

Pure • Reduction of Order

STEP I 2010, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q6

The key idea

Once we know one solution of a linear second-order equation, writing the general solution as that known solution times an unknown function u makes every term in u itself (with no derivative) cancel. What is left involves only u' and u'', so it is really a *first*-order equation for $v = u'$, and this one is separable. This technique is called *reduction of order*.

Solution

Checking that $y = \mathrm{e}^x$ works

If $y = \mathrm{e}^x$ then $y' = y'' = \mathrm{e}^x$, so

$$(x-1)\mathrm{e}^x - x\mathrm{e}^x + \mathrm{e}^x = (x - 1 - x + 1)\mathrm{e}^x = 0,$$

and $y = \mathrm{e}^x$ satisfies $(*)$.

The equation for u

Let $y = u\mathrm{e}^x$. By the product rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (u' + u)\,\mathrm{e}^x,$$
$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = (u'' + u')\,\mathrm{e}^x + (u' + u)\,\mathrm{e}^x = (u'' + 2u' + u)\,\mathrm{e}^x.$$

Substituting into the left-hand side of $(*)$ and taking out the factor e^x:

$$\begin{aligned}&\mathrm{e}^x\left[(x-1)(u'' + 2u' + u) - x(u' + u) + u\right]\\ &\quad= \mathrm{e}^x\left[(x-1)u'' + (2x - 2 - x)u' + (x - 1 - x + 1)u\right]\\ &\quad= \mathrm{e}^x\left[(x-1)u'' + (x-2)u'\right].\end{aligned}$$

The terms in u cancel, exactly as they did when we checked $y = \mathrm{e}^x$ (that check is the case $u = 1$). Since e^x is never zero, $y = u\mathrm{e}^x$ satisfies $(*)$ if and only if

$$(x-1)\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + (x-2)\frac{\mathrm{d}u}{\mathrm{d}x} = 0,$$

which is $(**)$, as required.

Solving for v and then u

With $v = \dfrac{\mathrm{d}u}{\mathrm{d}x}$, $(**)$ becomes the first-order equation

$$(x-1)\frac{\mathrm{d}v}{\mathrm{d}x} = -(x-2)\,v.$$

One solution is $v = 0$. Otherwise, working on an interval where $x \neq 1$ and $v \neq 0$, we separate the variables:

$$\int \frac{1}{v}\,\mathrm{d}v = -\int \frac{x-2}{x-1}\,\mathrm{d}x.$$

Ask yourself How do we integrate $\dfrac{x-2}{x-1}$?

Write the numerator in terms of the denominator: $x - 2 = (x-1) - 1$, so

$$\frac{x-2}{x-1} = 1 - \frac{1}{x-1}.$$

Hence

$$\ln|v| = -x + \ln|x-1| + c, \qquad \text{so} \qquad v = C(x-1)\mathrm{e}^{-x},$$

where $C = \pm\mathrm{e}^c$. Allowing $C = 0$ includes the solution $v = 0$ as well, so $v = C(x-1)\mathrm{e}^{-x}$ for an arbitrary constant C. (It is easy to check directly that this satisfies $(x-1)v' + (x-2)v = 0$ for all x, including $x = 1$.)

Now integrate to find u. Integrating by parts, with $(x-1)$ differentiated and e^{-x} integrated,

$$\begin{aligned}\int (x-1)\mathrm{e}^{-x}\,\mathrm{d}x &= -(x-1)\mathrm{e}^{-x} + \int \mathrm{e}^{-x}\,\mathrm{d}x \\ &= -(x-1)\mathrm{e}^{-x} - \mathrm{e}^{-x} = -x\mathrm{e}^{-x}.\end{aligned}$$

So $u = -Cx\mathrm{e}^{-x} + B$. Renaming $-C$ as A,

$$u = Ax\mathrm{e}^{-x} + B,$$

where A and B are arbitrary constants.

The final part

We showed above that $y = ue^x$ satisfies $(*)$ whenever u satisfies $(**)$. Our $u = Axe^{-x} + B$ satisfies $(**)$, because its derivative is $v = -A(x-1)e^{-x}$, which solves the equation for v. Therefore

$$y = ue^x = \big(Axe^{-x} + B\big)e^x = Ax + Be^x$$

satisfies $(*)$ for any constants A and B, as required.

Checks

Check $y = x$ directly: $y' = 1$, $y'' = 0$, so the left-hand side of $(*)$ is $0 - x + x = 0$. Since $(*)$ is linear, any combination $Ax + Be^x$ of the two solutions x and e^x also works. As a numerical check, with $A = 2$, $B = -1$ and $x = 0.5$ we have $y = 1 - e^{0.5} \approx -0.64872$, $y' = 2 - e^{0.5} \approx 0.35128$ and $y'' = -e^{0.5} \approx -1.64872$, giving $(-0.5)(-1.64872) - 0.5(0.35128) - 0.64872 = 0.82436 - 0.17564 - 0.64872 = 0$.

Common slips

- Forgetting that the product rule gives $2u'$ (not u') in y''.
- Integrating $\frac{x-2}{x-1}$ as a logarithm straight away, without first splitting off the 1.
- Dropping the constant when integrating v to get u. That constant B is what brings back the original solution e^x.

Beyond STEP

Reduction of order works for any linear equation $y'' + p(x)y' + q(x)y = 0$: if y_1 is one solution, putting $y = uy_1$ always gives a first-order equation for u'. Together with Abel's formula for the Wronskian, it shows that the solutions form a two-dimensional vector space, so $Ax + Be^x$ is in fact the general solution here, on each interval not containing the singular point $x = 1$. Boyce and DiPrima, *Elementary Differential Equations*, covers both ideas.

WORKED SOLUTION

STEP I 2010, Question 7

Pure • Vectors and Intersecting Lines

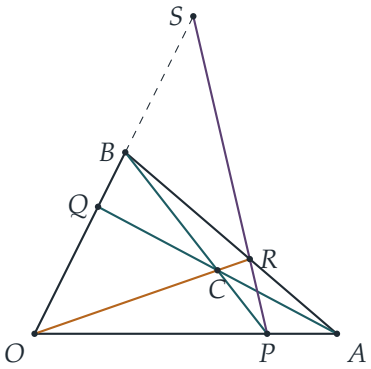
STEP I 2010, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q7

The key idea

Because O, A and B are not collinear, every point of the plane can be written as $\lambda\mathbf{a} + \mu\mathbf{b}$ in exactly one way. So to find where two lines meet, we write a general point of each line in this form and compare coefficients of $\mathbf{a}$ and $\mathbf{b}$. A point lies on the line OB exactly when its $\mathbf{a}$-coefficient is zero, and on AB exactly when its two coefficients add up to 1. Those two facts do almost all the work.



The picture uses $\alpha = \frac{1}{2}$, $\beta = \frac{7}{20}$. Note that S lies on the line OB but outside the segment OB, while Q lies between O and B.

Solution

Finding p and q

A general point of the line BC is

$$\mathbf{b} + \mu(\mathbf{c} - \mathbf{b}) = \mu\alpha\,\mathbf{a} + \big(1 - \mu(1 - \beta)\big)\mathbf{b}.$$

A point of the line OA has the form $\lambda\mathbf{a}$, with no $\mathbf{b}$ component. Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel (because O, A, B are not collinear), we may compare coefficients. The

b-coefficient must vanish, so $\mu = \dfrac{1}{1-\beta}$; this is allowed because $\beta < 1-\alpha < 1$, so $1-\beta \neq 0$. Then

$$\mathbf{p} = \mu\alpha\,\mathbf{a} = \frac{\alpha\mathbf{a}}{1-\beta},$$

as required.

The set-up is symmetric under swapping $A \leftrightarrow B$ and $\alpha \leftrightarrow \beta$ (this swaps $P \leftrightarrow Q$), so we can write down

$$\mathbf{q} = \frac{\beta\mathbf{b}}{1-\alpha}.$$

The point R

First, $\mathbf{r} = \dfrac{1}{\alpha+\beta}\,\mathbf{c}$ is a scalar multiple of $\mathbf{c}$, so R lies on the line OC.

Second, write $\mathbf{r} = t\mathbf{a} + (1-t)\mathbf{b}$ with $t = \dfrac{\alpha}{\alpha+\beta}$. Indeed $1-t = \dfrac{\beta}{\alpha+\beta}$, which is the coefficient of $\mathbf{b}$ in $\mathbf{r}$. Points of the form $t\mathbf{a} + (1-t)\mathbf{b} = \mathbf{b} + t(\mathbf{a}-\mathbf{b})$ are exactly the points of the line AB, so R lies on AB too.

The point S

A general point of the line PR is $\mathbf{p} + t(\mathbf{r}-\mathbf{p})$, which equals

$$\left[\frac{\alpha}{1-\beta} + t\left(\frac{\alpha}{\alpha+\beta} - \frac{\alpha}{1-\beta}\right)\right]\mathbf{a} \;+\; \frac{t\beta}{\alpha+\beta}\,\mathbf{b}.$$

For this point to lie on OB, the **a**-coefficient must be zero. Dividing by $\alpha \neq 0$ gives

$$\frac{1}{1-\beta} = t\cdot\frac{(\alpha+\beta)-(1-\beta)}{(1-\beta)(\alpha+\beta)}, \qquad \text{so} \qquad t = \frac{\alpha+\beta}{\alpha+2\beta-1}.$$

(The question tells us the lines meet, which forces $\alpha + 2\beta \neq 1$. If $\alpha + 2\beta = 1$ then PR is parallel to OB.) Hence

$$\mathbf{s} = \frac{t\beta}{\alpha+\beta}\,\mathbf{b} = \frac{\beta}{\alpha+2\beta-1}\,\mathbf{b}.$$

> **Ask yourself** How do we turn a position vector $k\mathbf{b}$ into the ratio of the distances from O and from B?

If X has position vector $k\mathbf{b}$, then $\overrightarrow{OX} = k\mathbf{b}$ and $\overrightarrow{BX} = (k-1)\mathbf{b}$, so

$$\frac{OX}{BX} = \frac{|k|}{|1-k|}.$$

The absolute values matter, because k may be negative or larger than 1, as it is for S in the picture.

The ratios

For Q we have $k = \dfrac{\beta}{1-\alpha}$, and $0 < k < 1$ because $0 < \beta < 1-\alpha$. Then $1-k = \dfrac{1-\alpha-\beta}{1-\alpha}$, so

$$\frac{OQ}{BQ} = \frac{\beta}{1-\alpha-\beta}.$$

For S we have $k = \dfrac{\beta}{\alpha+2\beta-1}$, so

$$1-k = \frac{\alpha+2\beta-1-\beta}{\alpha+2\beta-1} = \frac{\alpha+\beta-1}{\alpha+2\beta-1}, \qquad \frac{k}{1-k} = \frac{\beta}{\alpha+\beta-1} = -\frac{\beta}{1-\alpha-\beta}.$$

Taking absolute values, and using $\beta > 0$ and $1-\alpha-\beta > 0$,

$$\frac{OS}{BS} = \left|\frac{k}{1-k}\right| = \frac{\beta}{1-\alpha-\beta} = \frac{OQ}{BQ}.$$

This completes the proof. The minus sign before we took absolute values says that Q divides OB internally and S divides it externally, in the same ratio.

Checks

- Try $\alpha = \frac{1}{2}$, $\beta = \frac{7}{20}$ (the picture). Then $\mathbf{q} = 0.7\mathbf{b}$ and $\mathbf{s} = 1.75\mathbf{b}$, so $OQ/BQ = 0.7/0.3 = \frac{7}{3}$ and $OS/BS = 1.75/0.75 = \frac{7}{3}$. The formula gives $\beta/(1-\alpha-\beta) = 0.35/0.15 = \frac{7}{3}$.
- For random $\mathbf{a}$, $\mathbf{b}$, α, β, we computed all four intersections directly by solving pairs of line equations and compared them with the formulas, and checked that the two ratios agree. They matched in over 8000 cases.
- Symmetry: swapping $\alpha \leftrightarrow \beta$ and $\mathbf{a} \leftrightarrow \mathbf{b}$ turns the formula for $\mathbf{p}$ into the formula for $\mathbf{q}$, as it should.

Common slips

- Forgetting the absolute values. $\mathbf{s}$ may have a negative coefficient, or one bigger than 1, and a careless $k/(1-k)$ then gives a negative "ratio of lengths".
- Comparing coefficients of $\mathbf{a}$ and $\mathbf{b}$ without saying why it is allowed. It works because $\mathbf{a}$ and $\mathbf{b}$ are not parallel.

Beyond STEP

The points Q and S divide OB internally and externally in the same ratio, so they are *harmonic conjugates* with respect to O and B. The figure here, four lines and their intersections, is the classical *complete quadrilateral*. Harmonic division is preserved by projections, and it is one of the central ideas of projective geometry. The habit of writing

points in terms of two independent vectors and comparing coefficients is the start of linear algebra, in which it becomes the idea of a basis. Strang's *Introduction to Linear Algebra* develops that side.

WORKED SOLUTION

STEP I 2010, Question 8

Pure • Infinite Descent

STEP I 2010, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q8

The key idea

Both parts use the same trick, known as *infinite descent*. First we look at the equation modulo a prime (3 in part (i), 5 in part (ii)) and find that the prime is forced to divide every unknown. Dividing it out then gives a smaller solution of *exactly the same equation*, and a non-zero integer can't be divided by 3 (or 5) for ever. The only real work is spotting which residues are possible, and in part (ii) the final digit of a fourth power does that for us.

Solution

Part (i)

Rearranging,

$$a^3 = 9c^3 - 3b^3 = 3\big(3c^3 - b^3\big),$$

so 3 divides a^3. Since 3 is prime, if it divides a product $a \cdot a \cdot a$ then it divides one of the factors, so $3 \mid a$.

Write $a = 3a_1$ with a_1 an integer. Substituting and dividing by 3:

$$27a_1^3 + 3b^3 = 9c^3 \quad \Longrightarrow \quad 9a_1^3 + b^3 = 3c^3,$$

so $b^3 = 3\big(c^3 - 3a_1^3\big)$ is divisible by 3, and as before $3 \mid b$. Write $b = 3b_1$; then

$$9a_1^3 + 27b_1^3 = 3c^3 \quad \Longrightarrow \quad c^3 = 3\big(a_1^3 + 3b_1^3\big),$$

so $3 \mid c$ too. This shows that a, b and c are all divisible by 3.

Now write $c = 3c_1$ as well. The last equation becomes $27c_1^3 = 3a_1^3 + 9b_1^3$, that is,

$$a_1^3 + 3b_1^3 = 9c_1^3.$$

Ask yourself We have shown 3 divides everything. Why does that force everything to be zero?

Because $(a_1, b_1, c_1) = (a/3,\ b/3,\ c/3)$ is again an integer solution of the *original* equation, so everything we have just proved applies to it: a_1, b_1, c_1 are divisible by 3, i.e. a, b, c are divisible by 9. Repeating the argument k times shows that 3^k divides each of a, b and c, for every $k \geq 1$. But a non-zero integer n is not divisible by 3^k once $3^k > |n|$. Hence $a = b = c = 0$, and this clearly is a solution, so it is the only one.

(Equivalently: if there were a solution other than $(0,0,0)$, choose one with $|a| + |b| + |c|$ as small as possible but positive; then $(a/3, b/3, c/3)$ is a solution with a smaller positive value, a contradiction.)

Part (ii)

Since $p^4 = |p|^4$, the final digit of p^4 depends only on the final digit of $|p|$:

last digit of n	0	1	2	3	4	5	6	7	8	9
last digit of n^4	0	1	6	1	6	5	6	1	6	1

(For example $2^4 = 16$, $3^4 = 81$, $7^4 = 2401$.) So a fourth power ends in 0, 1, 5 or 6, and it ends in 0 or 5 exactly when $5 \mid n$.

Now look at the final digit of each term.

- $5r^4$ is a multiple of 5, so it ends in 0 or 5.
- $2q^4$ ends in 0 (if q^4 ends in 0 or 5) or in 2 (if q^4 ends in 1 or 6).
- p^4 ends in 0, 1, 5 or 6.

Adding a final digit from $\{0,1,5,6\}$ to one from $\{0,2\}$ gives one of $0,1,5,6,2,3,7,8$. The only ways to get 0 or 5 are $0+0$ and $5+0$. So $2q^4$ ends in 0, which means q^4 ends in 0 or 5, i.e. $5 \mid q$; and p^4 ends in 0 or 5, i.e. $5 \mid p$. This shows that p and q are divisible by 5.

(Or otherwise: by Fermat's little theorem $n^4 \equiv 1 \pmod 5$ when $5 \nmid n$, so $p^4 + 2q^4$ is congruent to $0+0$, $1+0$, $0+2$ or $1+2$ modulo 5, and only the first is 0.)

Write $p = 5p_1$ and $q = 5q_1$. Then

$$625(p_1^4 + 2q_1^4) = 5r^4 \quad \Longrightarrow \quad r^4 = 125(p_1^4 + 2q_1^4),$$

so $5 \mid r^4$, and since 5 is prime, $5 \mid r$. With $r = 5r_1$,

$$625r_1^4 = 125(p_1^4 + 2q_1^4) \quad \Longrightarrow \quad p_1^4 + 2q_1^4 = 5r_1^4.$$

So $(p/5,\ q/5,\ r/5)$ is again an integer solution of the same equation.

Exactly as in part (i), repeating this shows that 5^k divides each of p, q and r for every k, which is impossible for a non-zero integer. Hence the only integer solution is $p = q = r = 0$.

Checks

A computer search finds no solution other than $(0,0,0)$ of $a^3+3b^3=9c^3$ with $|a|,|b|,|c|\leq 60$, nor of $p^4+2q^4=5r^4$ with $|p|,|q|,|r|\leq 150$. Running through all 100 pairs of final digits for p and q confirms that p^4+2q^4 ends in 0 or 5 only when both final digits are 0 or 5. A quick hand check: $p=q=1$ gives $p^4+2q^4=3$, not a multiple of 5, as the table predicts.

Common slips

- Stopping after showing 3 (or 5) divides each unknown. That alone doesn't give zero; the point is that dividing out gives a solution of the *same* equation, so the argument repeats.
- Going from "$3 \mid a^3$" to "$3 \mid a$" without a reason. It works because 3 is prime; it would fail for, say, $9 \mid a^3$ implying $9 \mid a$ (try $a=3$).
- Forgetting negative integers when listing final digits. Fourth powers are even powers, so $p^4=|p|^4$ and the table covers every case.

Beyond STEP

Infinite descent goes back to Fermat, who used it to prove that $x^4+y^4=z^2$ has no solution in positive integers (and so the $n=4$ case of his Last Theorem). The idea used here, that an equation with no non-trivial solutions modulo a power of a prime can have no non-trivial integer solutions, grows into the study of p-adic numbers and the Hasse principle. Davenport's *The Higher Arithmetic* is a beautiful introduction to congruences and descent.

WORKED SOLUTION

STEP I 2010, Question 9

Mechanics • Friction and Equilibrium

STEP I 2010, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q9

The key idea

Put the corner on the floor at a known place and walk to the centre of mass along the two perpendicular edges: that gives the distance from the wall. For the equilibrium, "limiting at both contacts" fixes each friction force as μ times its normal reaction, so the two resolutions give both reactions in terms of the weight. One moment equation then finishes the job, and writing $\mu = \tan\lambda$ turns the algebra into compound-angle formulae.

Solution

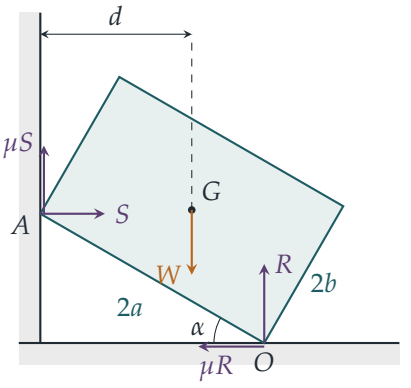
The model

The lamina is a rigid uniform rectangle of weight W, so its centre of mass G is its centre. Call O the corner on the floor and A the corner against the wall, so that OA is an edge of length $2a$ making angle α with the floor, and the edge of length $2b$ from O points up and away from the wall. The forces are:

- at O: normal reaction R upwards and friction F horizontally;
- at A: normal reaction S horizontally away from the wall and friction P vertically;
- the weight W downwards at G.

The lamina is on the point of slipping with O sliding away from the wall and A sliding down it, so friction opposes this: F acts towards the wall and P acts upwards. Since friction is limiting at both contacts,

$$F = \mu R, \qquad P = \mu S.$$



Distance of the centre of mass from the wall

Since OA has length $2a$ and makes angle α with the horizontal, O is a horizontal distance $2a\cos\alpha$ from the wall.

From O, reach G by going half-way along each of the two edges at O. Going a along OA (towards the wall) moves us $a\cos\alpha$ towards the wall. The edge of length $2b$ is perpendicular to OA, so it makes angle $\frac{1}{2}\pi - \alpha$ with the horizontal and leans away from the wall; going b along it moves us $b\cos(\frac{1}{2}\pi - \alpha) = b\sin\alpha$ away from the wall. Hence the distance of G from the wall is

$$d = 2a\cos\alpha - a\cos\alpha + b\sin\alpha = a\cos\alpha + b\sin\alpha,$$

as required. In the same way, G is a height $a\sin\alpha + b\cos\alpha$ above the floor, and A is a height $2a\sin\alpha$ above it.

The equilibrium condition

Resolving horizontally and vertically,

$$S = \mu R,$$
$$R + \mu S = W.$$

Substituting the first into the second gives $R(1+\mu^2) = W$, so

$$R = \frac{W}{1+\mu^2}, \qquad S = \frac{\mu W}{1+\mu^2}.$$

Now take moments about O, which removes R and μR. Horizontally, G is $2a\cos\alpha - d = a\cos\alpha - b\sin\alpha$ nearer the wall than O, so the weight has moment $W(a\cos\alpha - b\sin\alpha)$, tending to turn the lamina towards the wall. At A, the force S acts along a line a height $2a\sin\alpha$ above O, and μS acts along the wall, a horizontal distance $2a\cos\alpha$ from O; both tend to turn the lamina away from the wall. So

$$W(a\cos\alpha - b\sin\alpha) = S\cdot 2a\sin\alpha + \mu S\cdot 2a\cos\alpha = \frac{2a\mu W(\sin\alpha + \mu\cos\alpha)}{1+\mu^2}. \qquad (*)$$

Ask yourself How does $\mu = \tan\lambda$ tidy up the right-hand side?

With $\mu = \tan\lambda$ we have $1 + \mu^2 = \sec^2\lambda$, so

$$\frac{\mu}{1+\mu^2} = \tan\lambda\cos^2\lambda = \sin\lambda\cos\lambda,$$
$$\sin\alpha + \mu\cos\alpha = \frac{\sin\alpha\cos\lambda + \cos\alpha\sin\lambda}{\cos\lambda} = \frac{\sin(\alpha+\lambda)}{\cos\lambda}.$$

So the right-hand side of $(*)$ is $2aW\sin\lambda\sin(\alpha+\lambda)$. Cancelling W and using $2\sin P\sin Q = \cos(P-Q) - \cos(P+Q)$ with $P = \alpha+\lambda$, $Q = \lambda$,

$$\begin{aligned} b\sin\alpha &= a\cos\alpha - 2a\sin\lambda\sin(\alpha+\lambda) \\ &= a\cos\alpha - a\big(\cos\alpha - \cos(\alpha+2\lambda)\big) \\ &= a\cos(2\lambda+\alpha), \end{aligned}$$

which is the required result.

The square lamina

If the lamina is square then $b = a$, and dividing by $a \neq 0$,

$$\cos(2\lambda+\alpha) = \sin\alpha = \cos\left(\tfrac{1}{2}\pi - \alpha\right).$$

So $2\lambda + \alpha = \pm\left(\frac{1}{2}\pi - \alpha\right) + 2k\pi$ for some integer k. We need to pick the right branch. Here $0 < \alpha < \frac{1}{2}\pi$ (the lamina leans), and $0 < \lambda < \frac{1}{2}\pi$ since $\mu > 0$, so $0 < 2\lambda + \alpha < \frac{3}{2}\pi$.

- The $+$ sign gives $2\lambda + \alpha = \frac{1}{2}\pi - \alpha + 2k\pi$; only $k = 0$ lands in $(0, \frac{3}{2}\pi)$ (since $\frac{1}{2}\pi - \alpha$ lies in $(0, \frac{1}{2}\pi)$), giving $\lambda = \frac{1}{4}\pi - \alpha$.
- The $-$ sign gives $2\lambda = -\frac{1}{2}\pi + 2k\pi$, i.e. $\lambda = -\frac{1}{4}\pi + k\pi$, and no such value lies in $(0, \frac{1}{2}\pi)$.

Hence $\lambda = \frac{1}{4}\pi - \alpha$, as required. (In particular equilibrium of this limiting kind needs $\alpha < \frac{1}{4}\pi$, which matches the need for G to lie on the wall side of O: $a\cos\alpha > a\sin\alpha$.)

Checks

- If $\mu = 0$ ($\lambda = 0$) the condition becomes $a\cos\alpha = b\sin\alpha$: G is directly above O, which is exactly when a lamina could balance on its corner with no help from the wall.
- For a square at $\alpha = 15°$ the result gives $\lambda = 30°$, so $\mu = 1/\sqrt{3} \approx 0.577$. Putting $a = b = 1$, $\alpha = \frac{\pi}{12}$, $\mu = 1/\sqrt{3}$ into $(*)$, both sides equal $W(\cos 15° - \sin 15°) \approx 0.707W$.
- Numerically, for 2000 random rectangles and angles we found μ from the force and moment equations by bisection and confirmed $a\cos(2\lambda+\alpha) = b\sin\alpha$ to within 10^{-14}.

Common slips

- Getting the directions of friction wrong: decide first which way each contact would slide, then oppose it.
- Using the distance $a\cos\alpha + b\sin\alpha$ from the *wall* as the moment arm about O; the arm about O is $a\cos\alpha - b\sin\alpha$.
- Forgetting that $\lambda = \frac{3}{4}\pi$ also solves the trigonometric equation; it is ruled out because $\mu > 0$ forces $0 < \lambda < \frac{1}{2}\pi$.

Beyond STEP

The angle λ with $\tan\lambda = \mu$ is the *angle of friction*: at limiting friction the total contact force makes angle λ with the normal. A neat alternative approach uses the fact that a rigid body in equilibrium under three forces needs them to be concurrent: the two total contact forces and the weight must meet in a point, which turns the problem into pure geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops statics of rigid bodies and friction alongside the rest of Newtonian mechanics.

WORKED SOLUTION

STEP I 2010, Question 10

Mechanics • Vector Kinematics

STEP I 2010, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q10

The key idea

Everything follows from differentiating $\mathbf{r}$ and taking dot products: the angle between two vectors comes from $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$, and perpendicularity means a zero dot product. For the second particle, the useful observation is that Q at time t is exactly where P was at time $t-T$. Writing the position as a complex number, $z = \mathrm{e}^{(1+\mathrm{i})t}$, turns the last part into a one-line calculation.

Solution

Velocity and acceleration

Write $\mathbf{r} = \mathrm{e}^t(\cos t,\ \sin t)$. By the product rule,

$$\begin{aligned}\mathbf{v} = \dot{\mathbf{r}} &= \mathrm{e}^t\big(\cos t - \sin t,\ \sin t + \cos t\big),\\ \mathbf{a} = \dot{\mathbf{v}} &= \mathrm{e}^t\big((\cos t - \sin t) + (-\sin t - \cos t),\ (\sin t + \cos t) + (\cos t - \sin t)\big)\\ &= \mathrm{e}^t\big(-2\sin t,\ 2\cos t\big).\end{aligned}$$

The angle between velocity and displacement

The magnitudes are $|\mathbf{r}| = \mathrm{e}^t$ and

$$|\mathbf{v}|^2 = \mathrm{e}^{2t}\big[(\cos t - \sin t)^2 + (\sin t + \cos t)^2\big] = \mathrm{e}^{2t}\cdot 2,$$

since the cross terms $\mp 2\sin t\cos t$ cancel. So $|\mathbf{v}| = \sqrt{2}\,\mathrm{e}^t$. The dot product is

$$\mathbf{r}\cdot\mathbf{v} = \mathrm{e}^{2t}\big[\cos t(\cos t - \sin t) + \sin t(\sin t + \cos t)\big] = \mathrm{e}^{2t}(\cos^2 t + \sin^2 t) = \mathrm{e}^{2t}.$$

Hence, if θ is the angle between $\mathbf{r}$ and $\mathbf{v}$,

$$\cos\theta = \frac{\mathbf{r}\cdot\mathbf{v}}{|\mathbf{r}||\mathbf{v}|} = \frac{\mathrm{e}^{2t}}{\mathrm{e}^t\cdot\sqrt{2}\,\mathrm{e}^t} = \frac{1}{\sqrt{2}},$$

so $\theta = \frac{\pi}{4}$ for every t, as required. (Neither vector is ever zero, since $\mathrm{e}^t > 0$, so the angle is always defined.)

The acceleration is perpendicular to the displacement

$$\mathbf{r}\cdot\mathbf{a} = e^{2t}\left[-2\sin t\cos t + 2\sin t\cos t\right] = 0,$$

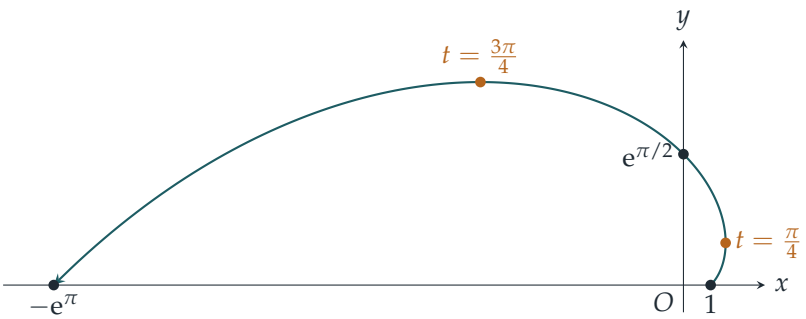
and both vectors are non-zero, so $\mathbf{a}$ is always perpendicular to $\mathbf{r}$. Notice also that $\mathbf{a} = 2e^t(-\sin t, \cos t)$ is $\mathbf{r}$ rotated by $\frac{\pi}{2}$ and doubled.

The path for $0 \le t \le \pi$

In polar coordinates the path is $r = e^\theta$ with $\theta = t$: the distance from O grows exponentially as the particle turns anticlockwise, an *equiangular* (logarithmic) spiral. Useful landmarks:

- $t = 0$: the point $(1, 0)$, moving in direction $(1, 1)$, at $\frac{\pi}{4}$ to the x-axis.
- $t = \frac{\pi}{4}$: $\dot{x} = e^t(\cos t - \sin t) = 0$, so x is greatest, at the point $\left(\frac{1}{\sqrt{2}}e^{\pi/4}, \frac{1}{\sqrt{2}}e^{\pi/4}\right) \approx (1.55, 1.55)$.
- $t = \frac{\pi}{2}$: the point $(0, e^{\pi/2}) \approx (0, 4.81)$.
- $t = \frac{3\pi}{4}$: $\dot{y} = e^t(\sin t + \cos t) = 0$, so y is greatest, at $\left(-\frac{1}{\sqrt{2}}e^{3\pi/4}, \frac{1}{\sqrt{2}}e^{3\pi/4}\right) \approx (-7.46, 7.46)$.
- $t = \pi$: the point $(-e^\pi, 0) \approx (-23.1, 0)$, crossing the axis at $\frac{\pi}{4}$, heading down and to the left.

At every point the path crosses the radius from O at $\frac{\pi}{4}$, which fixes its shape.



The second particle

Ask yourself Where is Q at time t, in terms of P's motion?

Q reaches each point T later than P, so at time t it is where P was at time $t - T$:

$$\mathbf{r}_Q(t) = \mathbf{r}(t-T) = e^{t-T}\big(\cos(t-T),\ \sin(t-T)\big).$$

Then

$$\begin{aligned}|PQ|^2 &= |\mathbf{r}(t)|^2 + |\mathbf{r}(t-T)|^2 - 2\,\mathbf{r}(t)\cdot\mathbf{r}(t-T)\\ &= \mathrm{e}^{2t} + \mathrm{e}^{2t-2T} - 2\mathrm{e}^{2t-T}\big(\cos t\cos(t-T) + \sin t\sin(t-T)\big)\\ &= \mathrm{e}^{2t}\big(1 - 2\mathrm{e}^{-T}\cos T + \mathrm{e}^{-2T}\big),\end{aligned}$$

using $\cos t\cos(t-T) + \sin t\sin(t-T) = \cos\big(t-(t-T)\big) = \cos T$. Therefore

$$|PQ| = k\,\mathrm{e}^{t}, \qquad k = \sqrt{1 - 2\mathrm{e}^{-T}\cos T + \mathrm{e}^{-2T}},$$

and k depends only on T, not on t. So the distance between P and Q is proportional to e^t, as required. (The bracket is $|1-\mathrm{e}^{-T}\mathrm{e}^{\mathrm{i}T}|^2$, which is positive for $T \neq 0$, since $\mathrm{e}^{-T} \neq 1$; so $k > 0$.)

Complex-number version. With $z = x + \mathrm{i}y$, P is at $z_P = \mathrm{e}^t\mathrm{e}^{\mathrm{i}t} = \mathrm{e}^{(1+\mathrm{i})t}$ and Q at $z_Q = \mathrm{e}^{(1+\mathrm{i})(t-T)}$. So

$$z_P - z_Q = \mathrm{e}^{(1+\mathrm{i})t}\big(1 - \mathrm{e}^{-(1+\mathrm{i})T}\big), \qquad |z_P - z_Q| = \mathrm{e}^t\,\big|1 - \mathrm{e}^{-(1+\mathrm{i})T}\big|.$$

This also shows that $\overrightarrow{QP}$ is a fixed complex multiple of z_P, so the segment PQ makes a constant angle with OP: the triangle OQP keeps the same shape and just grows and turns.

Checks

- Differentiating $z = \mathrm{e}^{(1+\mathrm{i})t}$ gives $\dot z = (1+\mathrm{i})z$ and $\ddot z = (1+\mathrm{i})^2 z = 2\mathrm{i}z$. Multiplying by $1+\mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$ rotates by $\frac{\pi}{4}$, and multiplying by $2\mathrm{i}$ rotates by $\frac{\pi}{2}$: both of the first results at a glance.
- Numerically, for random t the angle between $\mathbf{r}$ and $\mathbf{v}$ came out as $0.785398\ldots = \frac{\pi}{4}$ and $\mathbf{r}\cdot\mathbf{a} = 0$. With $T = 1$, $|PQ|/\mathrm{e}^t = 0.85895\ldots$ at every t tested, matching $\sqrt{1 - 2\mathrm{e}^{-1}\cos 1 + \mathrm{e}^{-2}}$.
- As $T \to 0$, $k \approx T\sqrt{2}$, so $|PQ| \approx \sqrt{2}\,\mathrm{e}^t T = |\mathbf{v}|T$, which is the distance travelled in time T, as it should be.

Common slips

- Taking Q's position as $\mathbf{r}(t+T)$: Q is *behind* P, so it is at $\mathbf{r}(t-T)$. (The answer is still proportional to e^t, but with a different constant.)
- Sketching a spiral that turns clockwise, or showing the path through the origin. The distance from O is $\mathrm{e}^t \geq 1$ and increases as the particle moves anticlockwise.

Beyond STEP

The curve $r = \mathrm{e}^{\theta}$ is the logarithmic spiral: every scaling of it about O is a rotation of it, which is why the crossing angle is constant and why OQP keeps its shape. In first-year

courses, motion in a plane is often handled with polar unit vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}$, in which the radial acceleration $\ddot{r} - r\dot{\theta}^2$ vanishes here, giving the perpendicularity directly. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops polar coordinates for exactly this kind of problem.

WORKED SOLUTION

STEP I 2010, Question 11

Mechanics • Collisions on a Circle

STEP I 2010, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

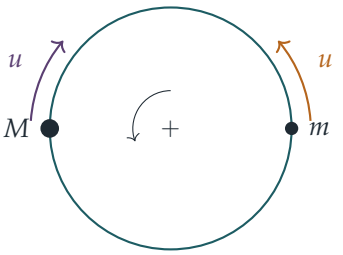
step.maths.org/questions/10-s1-q11

The key idea

Although the particles move on a circle, each collision is an ordinary one-dimensional collision along the common tangent: the smooth groove pushes only perpendicular to the motion, so it cannot change speeds or affect the tangential impulses. So we use conservation of momentum and Newton's experimental law with *signed* velocities measured round the circle. After that, two facts carry all the information: the total signed momentum never changes, and the relative speed is multiplied by e at every collision.

Solution

We model the particles as points in a smooth horizontal groove. Take anticlockwise as positive, with m initially moving anticlockwise at u and M clockwise at u, so their signed velocities are u and $-u$.



Between collisions the only horizontal force on each particle is the normal reaction of the smooth groove, which is perpendicular to the velocity. It does no work, so each particle keeps a constant speed. At a collision the impulses the particles exert on each other act along the common tangent, are equal and opposite, and the groove supplies no tangential impulse. So in each collision the total signed momentum round the circle is unchanged, and Newton's law applies to the signed velocities.

The first collision

Let the signed velocities after the first collision be p (for m) and q (for M). Conservation of momentum and Newton's law give

$$\begin{aligned} mp + Mq &= mu - Mu, \\ q - p &= e\big(u - (-u)\big) = 2eu. \end{aligned}$$

Substituting $q = p + 2eu$ into the first equation, $(m+M)p = (m-M)u - 2eMu$, so

$$p = -\frac{u\,(M - m + 2eM)}{M+m}, \qquad q = p + 2eu = \frac{u\,\big(2em - (M-m)\big)}{M+m}.$$

The speeds after the first collision are therefore

$$\text{speed of } m = \frac{u\,(M - m + 2eM)}{M+m}, \qquad \text{speed of } M = \frac{u\,|2em - (M-m)|}{M+m}.$$

Ask yourself Which signs mean that a particle has changed direction?

The particle m was moving in the positive sense, so it reverses exactly when $p < 0$. Since $M > m$ and $e \geq 0$, the bracket $M - m + 2eM$ is positive, so m *always* reverses. The particle M was moving in the negative sense, so it reverses exactly when $q > 0$, that is, when

$$2em > M - m.$$

So under this condition both particles change direction, as required.

After a further $2n$ collisions

Why $mv - MV = u(M-m)$. The total signed momentum is $mu - Mu = -u(M-m)$ at the start. Collisions conserve it, and between collisions the speeds don't change, so it keeps this value for ever.

Since both particles reverse at every collision, after the first collision m moves clockwise and M anticlockwise; each later collision swaps this round. After a further $2n$ collisions, $2n+1$ collisions have taken place in all, an odd number, so m is again moving clockwise and M anticlockwise. Their signed velocities are $-v$ and V, and

$$m(-v) + MV = -u(M-m) \quad \Longrightarrow \quad mv - MV = u(M-m).$$

The relative speed. After each collision the particles move apart in opposite directions round the circle with constant speeds, and they next meet on the far side, approaching each other at the same relative speed with which they separated. Newton's law then says that the speed of separation after a collision is e times the speed of approach before it. The initial speed of approach is $2u$, so after $2n+1$ collisions the speed of separation is $2ue^{2n+1}$. As the particles are moving in opposite directions, this is $v + V$:

$$v + V = 2ue^{2n+1}.$$

Solving. Multiply the second equation by M and add it to the first:

$$(m+M)v = u(M-m) + 2Mue^{2n+1},$$

and multiply the second by m and subtract the first:

$$(m+M)V = 2mue^{2n+1} - u(M-m).$$

Hence

$$v = \frac{u\big(M-m+2Me^{2n+1}\big)}{M+m}, \qquad V = \frac{u\big(2me^{2n+1}-(M-m)\big)}{M+m}.$$

Checks

- Putting $n = 0$ gives back the speeds after the first collision found above.
- With $m = 2$, $M = 3$, $e = 0.8$, $u = 1$, the formulae give $(v, V) = (1.16, 0.44)$ for $n = 0$ and $(0.8144, 0.2096)$ for $n = 1$. A step-by-step simulation of the particles going round the circle and colliding gives exactly these values after the 1st and 3rd collisions.
- Kinetic energy should not increase. With $e = 1$, $v = u\,\frac{3M-m}{M+m}$ and $V = u\,\frac{3m-M}{M+m}$, and you can check that $\frac{1}{2}mv^2 + \frac{1}{2}MV^2 = \frac{1}{2}(m+M)u^2$.

Common slips

- Mixing speeds and velocities: Newton's law and conservation of momentum must be written with signed velocities. Only at the end do you convert to speeds.
- Forgetting that $2n$ further collisions means $2n+1$ in total, which gives the power e^{2n+1} rather than e^{2n}.
- Worrying about the centripetal force from the groove. It is perpendicular to the motion, so it neither changes the speeds nor enters the collision equations.

Beyond STEP

The formula for V shows that the given assumption can't last for ever when $e < 1$: once $2me^{2n+1}$ falls below $M-m$, the heavier particle stops reversing. Momentum is conserved here only in a rotational sense. The groove's force always points through the centre, so the conserved quantity is really the *angular momentum* about the centre, and the circle's radius cancels. Conservation laws that come from symmetries like this are a central theme of first-year mechanics; Kleppner and Kolenkow, *An Introduction to Mechanics*, treats angular momentum and collisions carefully.

WORKED SOLUTION

STEP I 2010, Question 12

Probability and Statistics • Expectation via Tail Sums

STEP I 2010, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q12

The key idea

The tail-sum formula comes from writing $n\,\mathrm{P}(X = n)$ as $\mathrm{P}(X = n)$ added up n times, and then collecting the terms in a different order. For the penguins, the event $X \geq n$ is much easier to describe than $X = n$: you still need to open more boxes after $n - 1$ of them exactly when those $n - 1$ boxes all contain the same kind of penguin. So the tail probabilities are simple, and the formula we have just proved turns $\mathrm{E}(X)$ into two geometric series.

Solution

The definition and the tail-sum formula

Since X takes only the values $1, 2, 3, \ldots$,

$$\mathrm{E}(X) = \sum_{m=1}^{\infty} m\,\mathrm{P}(X = m),$$

provided this series converges.

For each $n \geq 1$, the event $X \geq n$ is the disjoint union of the events $X = m$ for $m \geq n$, so $\mathrm{P}(X \geq n) = \sum_{m \geq n} \mathrm{P}(X = m)$. Hence

$$\begin{aligned}\sum_{n=1}^{\infty} \mathrm{P}(X \geq n) &= \sum_{n=1}^{\infty} \sum_{m=n}^{\infty} \mathrm{P}(X = m) \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{m} \mathrm{P}(X = m) \\ &= \sum_{m=1}^{\infty} m\,\mathrm{P}(X = m) = \mathrm{E}(X).\end{aligned}$$

Ask yourself Why are we allowed to swap the order of the two sums?

The double sum adds $\mathrm{P}(X = m)$ over all pairs (n, m) with $1 \le n \le m$. In the first line we fix n and sum over m; in the second we fix m and sum over the m values $n = 1, \ldots, m$. Every term is non-negative, and for a series of non-negative terms the total does not depend on the order of summation (both sides may be infinite, in which case both are). It helps to picture the terms laid out in a triangle with row m containing m copies of $\mathrm{P}(X = m)$: summing by rows gives $\mathrm{E}(X)$, summing by columns gives the tail sum. This proves the formula.

The tail probabilities

Assume, as is intended, that the contents of different boxes are independent. Let D_k be the event that box k contains a daddy penguin, so $\mathrm{P}(D_k) = p$ and $\mathrm{P}(D_k') = q$.

Take $n \ge 2$. We have $X \ge n$ exactly when the first $n-1$ boxes do *not* yet contain one of each kind, that is, when they are all daddies or all mummies. These two cases are disjoint, so by independence

$$\mathrm{P}(X \ge n) = p^{n-1} + q^{n-1} \qquad (n \ge 2).$$

In particular, with $n = 4$,

$$\mathrm{P}(X \ge 4) = p^3 + q^3,$$

as required. Also $X \ge 2$ always, since one box can't hold both kinds, so $\mathrm{P}(X \ge 1) = \mathrm{P}(X \ge 2) = 1$. (The formula above gives $p + q = 1$ at $n = 2$, which agrees, but it gives 2 at $n = 1$, so the $n = 1$ term must be treated separately.)

The expectation

By the tail-sum formula,

$$\begin{aligned}\mathrm{E}(X) &= \mathrm{P}(X \ge 1) + \sum_{n=2}^{\infty} \left(p^{n-1} + q^{n-1}\right) \\ &= 1 + \sum_{k=1}^{\infty} p^k + \sum_{k=1}^{\infty} q^k \\ &= 1 + \frac{p}{1-p} + \frac{q}{1-q},\end{aligned}$$

where both geometric series converge because $0 < p < 1$ and $0 < q < 1$ (this is where $p \ne 0$, $q \ne 0$ is used). Since $1 - p = q$ and $1 - q = p$,

$$\mathrm{E}(X) = 1 + \frac{p}{q} + \frac{q}{p} = 1 + \frac{p^2 + q^2}{pq}.$$

Now $p^2 + q^2 = (p+q)^2 - 2pq = 1 - 2pq$, so

$$\mathrm{E}(X) = 1 + \frac{1 - 2pq}{pq} = 1 + \frac{1}{pq} - 2 = \frac{1}{pq} - 1,$$

as required.

The final part

By the AM–GM inequality for the positive numbers p and q,

$$\sqrt{pq} \le \frac{p+q}{2} = \frac{1}{2}, \qquad \text{so} \qquad 0 < pq \le \frac{1}{4}.$$

Ask yourself Which way does the inequality go when we take reciprocals?

Since $pq > 0$, taking reciprocals reverses the inequality: $\dfrac{1}{pq} \ge 4$. Therefore

$$\mathrm{E}(X) = \frac{1}{pq} - 1 \ge 4 - 1 = 3,$$

with equality exactly when $p = q = \frac{1}{2}$. (Equivalently, $pq = p(1-p) = \frac{1}{4} - (p - \frac{1}{2})^2 \le \frac{1}{4}$.)

Checks

- With $p = q = \frac{1}{2}$: $X - 1$ is the wait for the first box differing from box 1, a geometric variable with mean 2, so $\mathrm{E}(X) = 3$, matching $\frac{1}{1/4} - 1$.
- As $p \to 0$, $\mathrm{E}(X) \to \infty$: daddy penguins are rare, so you wait a long time. That matches $\frac{1}{pq} - 1$ blowing up.
- Directly, $\mathrm{P}(X = n) = p^{n-1}q + q^{n-1}p$ for $n \ge 2$. Summing $n\,\mathrm{P}(X = n)$ numerically for $p = \frac{1}{3}$ gives 3.5, and a simulation of 200 000 collectors with $p = 0.3$ gave a mean of 3.761 against the exact $\frac{1}{0.21} - 1 \approx 3.762$.

Common slips

- Using $p^{n-1} + q^{n-1}$ for $n = 1$ too, which gives a "probability" of 2 and an expectation that is 1 too large.
- Writing $\mathrm{P}(X \ge 4) = p^4 + q^4$: only the first *three* boxes need to match for a fourth box to be needed.
- Forgetting that reciprocals reverse inequalities between positive numbers in the final part.

Beyond STEP

The tail-sum formula has a continuous twin: for a non-negative random variable, $\mathrm{E}(X) = \int_0^\infty \mathrm{P}(X > x)\,\mathrm{d}x$, proved by exactly the same swap of order (Tonelli's theorem justifies it in general). The penguin problem is the two-type case of the *coupon collector's problem*; with N equally likely types the expected number of boxes is $N(1 + \frac{1}{2} + \cdots + \frac{1}{N})$,

roughly $N \ln N$. Both ideas appear in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 2010, Question 13

Probability and Statistics • Poisson Waiting Times

STEP I 2010, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q13

The key idea

"The first text arrives between 1 and 2 hours" is the same as "no text in the first hour, but not no text in the first two hours". Each of those is a Poisson probability of zero events, which is a single exponential. Substituting $x = \mathrm{e}^{\lambda}$ then turns everything into a quadratic, and Vieta's formulas (sum and product of roots) do the rest of the work, including for Mildred.

Solution

We take the usual Poisson model: the number of texts George receives in any interval of length t hours is Poisson with mean λt, and counts in non-overlapping intervals are independent. Time is measured from the start of the morning. Since time is continuous, the probability of the first text arriving at exactly 1 or exactly 2 hours is zero, so it doesn't matter whether the endpoints are included.

George's equation

Let T be the time in hours until George's first text. Then $T > t$ exactly when there are no texts in $[0, t]$, so

$$\mathrm{P}(T > t) = \mathrm{P}(\text{Poisson}(\lambda t) = 0) = \mathrm{e}^{-\lambda t}.$$

The event $\{1 < T \leq 2\}$ is $\{T > 1\}$ with $\{T > 2\}$ removed, and $\{T > 2\} \subseteq \{T > 1\}$, so

$$p = \mathrm{P}(1 < T \leq 2) = \mathrm{P}(T > 1) - \mathrm{P}(T > 2) = \mathrm{e}^{-\lambda} - \mathrm{e}^{-2\lambda}.$$

Multiplying through by $\mathrm{e}^{2\lambda}$ gives $p\mathrm{e}^{2\lambda} = \mathrm{e}^{\lambda} - 1$, that is,

$$p\mathrm{e}^{2\lambda} - \mathrm{e}^{\lambda} + 1 = 0,$$

as required.

Two positive values of λ

Put $x = e^{\lambda}$. Since $\lambda \mapsto e^{\lambda}$ is a strictly increasing bijection from $(0, \infty)$ to $(1, \infty)$, positive solutions λ correspond exactly to solutions $x > 1$ of

$$px^2 - x + 1 = 0.$$

Here $p > 0$ (it is the probability of an event that happens whenever, say, exactly one text arrives in $[1, 2]$ and none before), so this really is a quadratic.

Ask yourself It is easy to show the roots are real. How do we know both of them exceed 1?

Real and distinct. The discriminant is $1 - 4p > 0$ because $4p < 1$, so there are two distinct real roots $x_1 < x_2$.

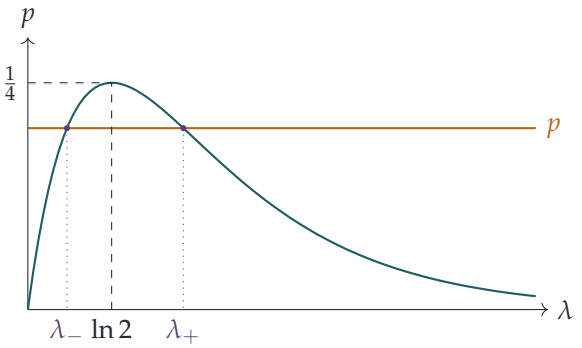
Both exceed 1. By Vieta's formulas, $x_1 + x_2 = \frac{1}{p}$ and $x_1 x_2 = \frac{1}{p}$. Then

$$(x_1 - 1)(x_2 - 1) = x_1 x_2 - (x_1 + x_2) + 1 = \tfrac{1}{p} - \tfrac{1}{p} + 1 = 1 > 0,$$

so $x_1 - 1$ and $x_2 - 1$ have the same sign. Their sum is $x_1 + x_2 - 2 = \frac{1}{p} - 2 > 0$, because $p < \frac{1}{4}$ gives $\frac{1}{p} > 4$. So both are positive: $x_1 > 1$ and $x_2 > 1$.

Hence $\lambda_{\pm} = \ln\left(\dfrac{1 \pm \sqrt{1 - 4p}}{2p}\right)$ are two distinct positive values of λ satisfying the equation.

The picture makes sense of this. As a function of λ, $p = e^{-\lambda} - e^{-2\lambda}$ is 0 at $\lambda = 0$, tends to 0 as $\lambda \to \infty$, and (writing $u = e^{-\lambda}$, $p = u - u^2$) has maximum $\frac{1}{4}$ at $u = \frac{1}{2}$, i.e. $\lambda = \ln 2$. A horizontal line at height $p < \frac{1}{4}$ cuts it twice: a quiet phone (texts rarely arrive at all) and a busy one (texts usually arrive within the first hour) can give the same p.



Mildred: $\lambda_1 + \lambda_2$

Each phone on its own is exactly George's situation, so e^{λ_1} and e^{λ_2} both satisfy $px^2 - x + 1 = 0$. That quadratic has only two roots, and $\lambda_1 \neq \lambda_2$, so e^{λ_1} and e^{λ_2} are its two

roots x_1 and x_2. (This needs $p < \frac{1}{4}$, which holds because two distinct values exist: at $p = \frac{1}{4}$ there is a repeated root.) By the product of roots,

$$e^{\lambda_1+\lambda_2} = e^{\lambda_1}e^{\lambda_2} = x_1x_2 = \frac{1}{p}, \qquad \text{so} \qquad \lambda_1 + \lambda_2 = \ln\frac{1}{p} = -\ln p.$$

Mildred's first text

> **Ask yourself** What is the distribution of the total number of texts Mildred receives on the two phones together?

The two counts are independent Poisson variables, so their sum is Poisson with the sum of the means: in t hours she receives $\text{Poisson}\big((\lambda_1 + \lambda_2)t\big)$ texts in total. Equivalently, let T_1, T_2 be the waiting times on the two phones; they are independent, and her first text comes at $T = \min(T_1, T_2)$ with

$$\mathrm{P}(T > t) = \mathrm{P}(T_1 > t)\,\mathrm{P}(T_2 > t) = e^{-(\lambda_1+\lambda_2)t}.$$

Either way, writing $\mu = \lambda_1 + \lambda_2$, so that $e^{-\mu} = p$, the same argument as for George gives

$$\mathrm{P}(1 < T \le 2) = e^{-\mu} - e^{-2\mu} = p - p^2 = p(1-p).$$

Checks

- With $\lambda = 0.8$, $p = e^{-0.8} - e^{-1.6} \approx 0.2474$; a simulation of 2×10^5 exponential waiting times gave 0.2462.
- With $p = 0.2$: the roots are $\lambda_1 \approx 0.3235$ and $\lambda_2 \approx 1.2859$, both positive, with sum $1.6094 = \ln 5 = -\ln 0.2$. Simulating $\min(T_1, T_2)$ gave 0.1601 for the final probability, against $p - p^2 = 0.16$.
- Sense check: $p(1-p) < p$, as it should be, since with two phones the first text tends to come sooner, making "no text for an hour" less likely.

Common slips

- Showing the roots in x are real and positive but forgetting that $\lambda > 0$ needs $x > 1$, not just $x > 0$.
- Adding the probabilities for the two phones, or multiplying them (p^2), for the final part. The right event is "first text on either phone", governed by the combined rate.

Beyond STEP

The waiting time to the first event of a Poisson process is exponential, and the merging fact used here (independent Poisson processes with rates λ_1, λ_2 combine into one

with rate $\lambda_1 + \lambda_2$) is one of the basic properties of Poisson processes, alongside "thinning" (keeping each event independently with probability q). These ideas underpin queueing theory and much of applied probability. Grimmett and Welsh, *Probability: An Introduction*, treats Poisson processes in a way that follows on naturally from this question.

WORKED SOLUTION

STEP II 2010, Question 1

Pure • Curvature and Osculating Circles

STEP II 2010, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q1

The key idea

"Touches" means the circle and the curve share the point P and the gradient there; "the rate of change of the gradient" means $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$. So we need a circle whose y, $\frac{\mathrm{d}y}{\mathrm{d}x}$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ all agree with the curve's at P. Differentiating the equation of a general circle implicitly, twice, turns this into simple equations for the centre.

Solution

The curve at P

For $y = 1 - x + \tan x$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -1 + \sec^2 x = \tan^2 x, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2\tan x \sec^2 x.$$

At $x = \frac{1}{4}\pi$ we have $\tan x = 1$ and $\sec^2 x = 2$, so

$$y = 2 - \tfrac{1}{4}\pi, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = 1, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4.$$

So $P = \left(\frac{1}{4}\pi,\ 2 - \frac{1}{4}\pi\right)$.

The circle

Let the circle be $(x-a)^2 + (y-b)^2 = r^2$. Differentiating implicitly with respect to x,

$$(x-a) + (y-b)\frac{\mathrm{d}y}{\mathrm{d}x} = 0,$$

and differentiating again (using the product rule on the second term),

$$1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + (y-b)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0.$$

Ask yourself Why can we put the curve's values straight into the circle's equations?

These two equations hold at every point of the circle, with y, $\mathrm{d}y/\mathrm{d}x$ and $\mathrm{d}^2y/\mathrm{d}x^2$ being the circle's own values. The two conditions say that, at P, the circle passes through P with the same gradient as C and the same second derivative. So at P we may use $x = \frac{1}{4}\pi$, $y = 2 - \frac{1}{4}\pi$, $\mathrm{d}y/\mathrm{d}x = 1$ and $\mathrm{d}^2y/\mathrm{d}x^2 = 4$.

The second equation gives

$$1 + 1 + 4(y - b) = 0 \quad \Longrightarrow \quad y - b = -\tfrac{1}{2} \quad \Longrightarrow \quad b = 2 - \tfrac{1}{4}\pi + \tfrac{1}{2} = \tfrac{5}{2} - \tfrac{1}{4}\pi.$$

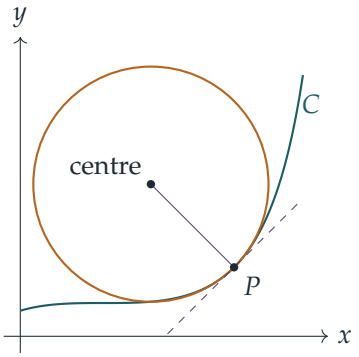
The first then gives

$$x - a = -(y - b) \cdot 1 = \tfrac{1}{2} \quad \Longrightarrow \quad a = \tfrac{1}{4}\pi - \tfrac{1}{2}.$$

Finally, $r^2 = (x - a)^2 + (y - b)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. So the osculating circle has

$$\text{centre } \left(\tfrac{1}{4}\pi - \tfrac{1}{2},\ \tfrac{5}{2} - \tfrac{1}{4}\pi\right) \qquad \text{and radius} \qquad \frac{1}{\sqrt{2}}.$$

It makes sense geometrically. The centre lies a distance $\frac{1}{\sqrt{2}}$ from P along the direction $(-1, 1)$, which is perpendicular to the tangent of gradient 1. It lies on the side where $y > 2 - \frac{1}{4}\pi$, that is, above the curve, which is right because $\mathrm{d}^2y/\mathrm{d}x^2 > 0$ and the curve is concave upwards there.



Checks

- The general radius-of-curvature formula $\rho = \left(1 + y'^2\right)^{3/2} / |y''|$ gives $2^{3/2}/4 = 1/\sqrt{2}$, as found.
- Numerically, the centre is about $(0.2854,\ 1.7146)$ and P is about $(0.7854,\ 1.2146)$; their distance is 0.70711. Taking the lower half of the circle, $y = b - \sqrt{r^2 - (x - a)^2}$, and differentiating numerically at $x = 0.7854$ gives gradient 1.0000 and second derivative 4.000, matching the curve.

Common slips

- Placing the centre on the wrong side of the tangent: the sign of $y - b$ is fixed by the second equation, so let the algebra decide.
- Forgetting that differentiating $(y - b)\,\mathrm{d}y/\mathrm{d}x$ needs the product rule, which is where the $(\mathrm{d}y/\mathrm{d}x)^2$ term comes from.

Beyond STEP

The radius of the osculating circle is the radius of curvature, and its reciprocal $\kappa = |y''|/(1 + y'^2)^{3/2}$ is the curvature, the central quantity in the differential geometry of curves. For a curve in space, curvature joins torsion in the Frenet–Serret formulae, which describe how a curve bends and twists. Siklos, *Advanced Problems in Mathematics*, has more STEP questions in this spirit.

WORKED SOLUTION

STEP II 2010, Question 2

Pure • Trigonometric Integration

STEP II 2010, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q2

The key idea

The cubic $7\sin x - 8\sin^3 x$ looks awkward to integrate until we notice that it is built from the triple-angle formula: twice $\sin 3x$ plus a bit left over. The triple-angle formula for cosine then turns the answer back into a polynomial in $c = \cos\alpha$. For Eustace's integral, $\sin^2\alpha = 1 - c^2$ puts his (wrong) answer into the same variable, and comparing the two gives a quartic in c whose roots we have been handed a hint about.

Solution

The triple-angle formulae

Using the addition formulae and $\sin^2 x = 1 - \cos^2 x$,

$$\begin{aligned}\cos 3x = \cos(2x + x) &= \cos 2x \cos x - \sin 2x \sin x \\ &= (2\cos^2 x - 1)\cos x - 2\sin^2 x \cos x \\ &= 2\cos^3 x - \cos x - 2(1 - \cos^2 x)\cos x \\ &= 4\cos^3 x - 3\cos x,\end{aligned}$$

as required. In the same way, using $\cos^2 x = 1 - \sin^2 x$,

$$\begin{aligned}\sin 3x = \sin(2x + x) &= \sin 2x \cos x + \cos 2x \sin x \\ &= 2\sin x \cos^2 x + (1 - 2\sin^2 x)\sin x \\ &= 2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x \\ &= 3\sin x - 4\sin^3 x.\end{aligned}$$

Part (i)

From the sine formula, $8\sin^3 x = 6\sin x - 2\sin 3x$, so

$$7\sin x - 8\sin^3 x = \sin x + 2\sin 3x.$$

Now the integral is easy:

$$\mathrm{I}(\alpha) = \Big[-\cos x - \tfrac{2}{3}\cos 3x \Big]_0^{\alpha} = -\cos\alpha - \tfrac{2}{3}\cos 3\alpha + 1 + \tfrac{2}{3}.$$

By the cosine formula, $\cos 3\alpha = 4c^3 - 3c$, so

$$\mathrm{I}(\alpha) = -c - \tfrac{2}{3}(4c^3 - 3c) + \tfrac{5}{3} = -\tfrac{8}{3}c^3 + c + \tfrac{5}{3},$$

which is the result we wanted.

For a value of c giving $\mathrm{I}(\alpha) = 0$: taking $\alpha = 0$ makes the integral zero trivially, so $c = 1$ works (check: $-\frac{8}{3} + 1 + \frac{5}{3} = 0$). In fact it is the only one, because $8c^3 - 3c - 5 = (c-1)(8c^2 + 8c + 5)$ and the quadratic has discriminant $64 - 160 < 0$.

Part (ii)

Eustace's rule gives $\int \sin x \,\mathrm{d}x = \frac{1}{2}\sin^2 x$ and $\int \sin^3 x \,\mathrm{d}x = \frac{1}{4}\sin^4 x$, so his value of the definite integral is

$$\mathrm{E}(\alpha) = \Big[\tfrac{7}{2}\sin^2 x - 2\sin^4 x \Big]_0^{\alpha} = \tfrac{7}{2}s^2 - 2s^4, \qquad s = \sin\alpha.$$

To compare with part (i), write $s^2 = 1 - c^2$:

$$\mathrm{E}(\alpha) = \tfrac{7}{2}(1-c^2) - 2(1-c^2)^2 = \tfrac{3}{2} + \tfrac{1}{2}c^2 - 2c^4.$$

The value $\cos\beta = -\frac{1}{6}$. With $c = -\frac{1}{6}$,

$$\mathrm{I}(\beta) = -\tfrac{8}{3}\cdot\big(-\tfrac{1}{216}\big) - \tfrac{1}{6} + \tfrac{5}{3} = \tfrac{1}{81} + \tfrac{3}{2} = \tfrac{245}{162},$$
$$\mathrm{E}(\beta) = \tfrac{3}{2} + \tfrac{1}{2}\cdot\tfrac{1}{36} - 2\cdot\tfrac{1}{1296} = \tfrac{3}{2} + \tfrac{1}{72} - \tfrac{1}{648} = \tfrac{3}{2} + \tfrac{8}{648} = \tfrac{245}{162}.$$

The two agree, so Eustace gets the right value of $\mathrm{I}(\beta)$ by luck.

All values of α. Eustace is right exactly when $\mathrm{E}(\alpha) = \mathrm{I}(\alpha)$, that is (multiplying by 6 and collecting terms)

$$12c^4 - 16c^3 - 3c^2 + 6c + 1 = 0.$$

Ask yourself How do we factorise a quartic like this without a formula?

We already know two roots. The value $c = -\frac{1}{6}$ works, by the calculation just done, so $6c + 1$ is a factor. And $c = 1$ (that is, $\alpha = 0$) must work, because both I and E are 0 there. Dividing out $6c + 1$ (compare coefficients of c^3 and c),

$$12c^4 - 16c^3 - 3c^2 + 6c + 1 = (6c+1)(2c^3 - 3c^2 + 1),$$

and then, since $c = 1$ is a root of the cubic,

$$2c^3 - 3c^2 + 1 = (c-1)(2c^2 - c - 1) = (c-1)^2(2c+1).$$

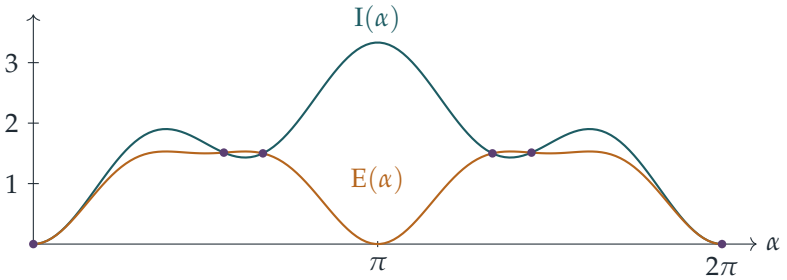
So the condition is

$$(6c+1)(2c+1)(c-1)^2 = 0, \qquad \text{that is} \qquad c = 1,\ -\tfrac{1}{2} \text{ or } -\tfrac{1}{6}.$$

Since $c = \cos\alpha$ and every value of c in $[-1,1]$ comes from some α, Eustace gets the correct value of $\mathrm{I}(\alpha)$ exactly when, for some integer n,

$$\alpha = 2n\pi, \qquad \alpha = 2n\pi \pm \tfrac{2\pi}{3}, \qquad \text{or} \qquad \alpha = 2n\pi \pm \arccos\left(-\tfrac{1}{6}\right).$$

(Negative α is included: both I and E depend on α only through $\cos\alpha$.) The graph shows the true integral (teal) and Eustace's answer (amber) for $0 \le \alpha \le 2\pi$; they meet at $\alpha = 0$, $\frac{2\pi}{3}$, $\arccos(-\frac{1}{6}) \approx 1.738$, and the mirror images of these in $\alpha = \pi$.



Checks

- At $\alpha = \pi$ ($c = -1$) the formula gives $\mathrm{I}(\pi) = \frac{8}{3} - 1 + \frac{5}{3} = \frac{10}{3}$, which matches $\int_0^{\pi} (\sin x + 2\sin 3x)\,\mathrm{d}x = 2 + \frac{4}{3}$.
- Simpson's rule on the original integrand gives $\mathrm{I}(\beta) \approx 1.5123457$, matching $\frac{245}{162} = 1.5123456\ldots$; at $\alpha = \frac{2\pi}{3}$ both the integral and Eustace's answer equal 1.5.

Common slips

- Forgetting the lower limit: $-\cos 0 - \frac{2}{3}\cos 0 = -\frac{5}{3}$, which supplies the $+\frac{5}{3}$.
- Missing that $c = 1$ is a *double* root of the quartic, or forgetting it altogether and reporting only $c = -\frac{1}{2}$ and $c = -\frac{1}{6}$.
- Giving only α in $[0, \pi]$; the question asks for all values.

Beyond STEP

Writing powers like $\sin^3 x$ as combinations of $\sin x$ and $\sin 3x$ is the start of Fourier analysis: every power $\sin^n x$ is a finite sum of $\sin kx$ or $\cos kx$, which you can find quickly using $\sin x = (\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x})/2\mathrm{i}$. The polynomial expressing $\cos nx$ in terms of $\cos x$ is the Chebyshev polynomial T_n, with $T_3(c) = 4c^3 - 3c$. Needham's *Visual Complex Analysis* shows why de Moivre's theorem makes all these identities natural.

WORKED SOLUTION

STEP II 2010, Question 3

Pure • Fibonacci Numbers and Recurrences

STEP II 2010, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q3

The key idea

Four given terms and four unknowns: substitute $n = 0, 1, 2, 3$ into $(*)$ and solve. The trick is to divide one equation by another, which cancels a and turns the conditions into statements about $\lambda + \mu$ and $\lambda\mu$, the symmetric functions that make λ and μ the roots of a quadratic. Once the formula is known, the sum in (iii) splits into two geometric series.

Solution

Part (i)

Putting $n = 0, 1, 2, 3$ into $(*)$ gives

$$\begin{aligned} a + b &= 0, & a\lambda + b\mu &= 1, \\ a\lambda^2 + b\mu^2 &= 1, & a\lambda^3 + b\mu^3 &= 2. \end{aligned}$$

The first gives $b = -a$, so the other three become

$$a(\lambda - \mu) = 1, \qquad a(\lambda^2 - \mu^2) = 1, \qquad a(\lambda^3 - \mu^3) = 2.$$

In particular $a(\lambda - \mu) = 1$ tells us that $a \neq 0$ and $\lambda \neq \mu$. Factorising,

$$\lambda^2 - \mu^2 = (\lambda - \mu)(\lambda + \mu), \qquad \lambda^3 - \mu^3 = (\lambda - \mu)(\lambda^2 + \lambda\mu + \mu^2),$$

so dividing the second and third equations by $a(\lambda - \mu) = 1$ (allowed, as it is non-zero) gives

$$\lambda + \mu = 1 \qquad \text{and} \qquad \lambda^2 + \lambda\mu + \mu^2 = 2,$$

the second of which is the required result.

? **Ask yourself** How do we get λ and μ themselves from these two equations?

Write the second in terms of the sum and product: $\lambda^2+\lambda\mu+\mu^2=(\lambda+\mu)^2-\lambda\mu$, so $1-\lambda\mu=2$ and $\lambda\mu=-1$. Numbers with sum 1 and product -1 are the roots of

$$t^2-t-1=0, \qquad \text{so} \qquad t=\frac{1\pm\sqrt{5}}{2}.$$

Which root is λ? From $a(\lambda-\mu)=1$ and $a>0$ we need $\lambda-\mu>0$, so λ is the larger root. Then $\lambda-\mu=\sqrt{5}$, and

$$\lambda=\frac{1+\sqrt{5}}{2}, \qquad \mu=\frac{1-\sqrt{5}}{2}, \qquad a=\frac{1}{\sqrt{5}}, \qquad b=-\frac{1}{\sqrt{5}}.$$

(The other labelling would force $a=-1/\sqrt{5}$, which is ruled out.) Conversely, these values satisfy all four equations, since every step above can be reversed.

Part (ii)

We have $F_6=(\lambda^6-\mu^6)/\sqrt{5}$. Rather than expanding $\left(\frac{1+\sqrt{5}}{2}\right)^6$ directly, use the fact that λ is a root of $t^2=t+1$ to reduce powers:

$$\begin{aligned}\lambda^2&=\lambda+1,\\ \lambda^3&=\lambda^2+\lambda=2\lambda+1,\\ \lambda^6&=(2\lambda+1)^2=4\lambda^2+4\lambda+1=4(\lambda+1)+4\lambda+1=8\lambda+5.\end{aligned}$$

Exactly the same holds for μ, which satisfies the same quadratic, so $\mu^6=8\mu+5$. Hence

$$F_6=\frac{(8\lambda+5)-(8\mu+5)}{\sqrt{5}}=\frac{8(\lambda-\mu)}{\sqrt{5}}=\frac{8\sqrt{5}}{\sqrt{5}}=8.$$

Part (iii)

Using $(*)$,

$$\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}}=\frac{1}{2\sqrt{5}}\sum_{n=0}^{\infty}\left[\left(\frac{\lambda}{2}\right)^n-\left(\frac{\mu}{2}\right)^n\right].$$

Since $\lambda\approx 1.618$ and $\mu\approx -0.618$, both common ratios $\lambda/2$ and $\mu/2$ have modulus less than 1. So both geometric series converge, and we may split the sum into the difference of the two:

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}}&=\frac{1}{2\sqrt{5}}\left[\frac{1}{1-\lambda/2}-\frac{1}{1-\mu/2}\right]=\frac{1}{\sqrt{5}}\left[\frac{1}{2-\lambda}-\frac{1}{2-\mu}\right]\\ &=\frac{1}{\sqrt{5}}\cdot\frac{\lambda-\mu}{(2-\lambda)(2-\mu)}=\frac{1}{4-2(\lambda+\mu)+\lambda\mu},\end{aligned}$$

using $\lambda-\mu=\sqrt{5}$. With $\lambda+\mu=1$ and $\lambda\mu=-1$ the denominator is $4-2-1=1$, so

$$\sum_{n=0}^{\infty}\frac{F_n}{2^{n+1}}=1.$$

Checks

The formula should give integers: $F_4 = (\lambda^4 - \mu^4)/\sqrt{5}$ with $\lambda^4 = 3\lambda + 2$ gives $F_4 = 3$, and $F_5 = 5$, $F_6 = 8$ fit the pattern $F_{n+2} = F_{n+1} + F_n$ seen in the first four terms. For (iii), the partial sums 0, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{19}{32}$, $\ldots$ creep up towards 1; adding the first 300 terms numerically gives 1.0000000 to machine precision. The terms decay only like $(0.809)^n$, so convergence is slow.

Common slips

- Forgetting to use $a > 0$ to decide which root is λ: without it the answer isn't determined.
- Dividing by $\lambda - \mu$ without noting why it is non-zero (it is, because $a(\lambda - \mu) = 1$).
- Splitting the infinite sum into two series without checking that each converges.

Beyond STEP

The formula here is Binet's formula for the Fibonacci numbers, and the method is the general one for linear recurrences: λ and μ are the roots of the characteristic equation $t^2 = t + 1$ of $F_{n+2} = F_{n+1} + F_n$. Part (iii) evaluates the generating function $\sum F_n x^n = x/(1 - x - x^2)$ at $x = \frac{1}{2}$ (and halves it), which gives 1 again. Generating functions are a major tool in first-year combinatorics and probability; for the golden ratio λ and the Fibonacci numbers in the theory of continued fractions, see Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 2010, Question 4

Pure • Integration by Substitution

STEP II 2010, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q4

The key idea

The substitution $u = a - x$ swaps the ends of the interval and turns $f(x)$ into $f(a-x)$ and back again. So the integral equals its "partner", in which the numerator is the other half of the denominator. Adding the two versions makes the numerator equal the denominator, and the integrand becomes 1. The rest of the question is about spotting this structure in disguise. Part (ii) uses the same trick with the substitution $x \mapsto 1/x$, which swaps the ends of $[\frac{1}{2}, 2]$.

Solution

Part (i)

We assume throughout that $f(x) + f(a-x) \neq 0$ on $[0, a]$, so the integrand is defined. In I, substitute $u = a - x$, so $\mathrm{d}x = -\mathrm{d}u$. When $x = 0$, $u = a$, and when $x = a$, $u = 0$. Hence

$$I = \int_a^0 \frac{f(a-u)}{f(a-u)+f(u)}\,(-\mathrm{d}u) = \int_0^a \frac{f(a-u)}{f(u)+f(a-u)}\,\mathrm{d}u.$$

Since u is a dummy variable, we can rename it x, which gives the required second form of I.

Now add the two expressions for I:

$$2I = \int_0^a \frac{f(x)+f(a-x)}{f(x)+f(a-x)}\,\mathrm{d}x = \int_0^a 1\,\mathrm{d}x = a, \qquad \text{so} \qquad I = \frac{a}{2}.$$

The first integral.

? **Ask yourself** Where is the $f(x) + f(a-x)$ hiding in $\ln(2 + x - x^2)$?

The quadratic factorises: $2 + x - x^2 = (1 + x)(2 - x)$. So

$$\ln(2 + x - x^2) = \ln(1 + x) + \ln(2 - x),$$

which is valid because both factors are positive on $[0, 1]$. Take $a = 1$ and $f(x) = \ln(1 + x)$. Then $f(1 - x) = \ln(2 - x)$, and the integrand is exactly $\dfrac{f(x)}{f(x) + f(1 - x)}$. The denominator is never zero, since $2 + x - x^2 \geq 2$ on $[0, 1]$. Therefore

$$\int_0^1 \frac{\ln(x + 1)}{\ln(2 + x - x^2)}\,\mathrm{d}x = \frac{1}{2}.$$

The second integral. Expand the denominator with the compound-angle formula:

$$\sin\left(x + \tfrac{\pi}{4}\right) = \sin x \cos\tfrac{\pi}{4} + \cos x \sin\tfrac{\pi}{4} = \frac{\sin x + \cos x}{\sqrt{2}}.$$

So the integrand is $\sqrt{2}\,\dfrac{\sin x}{\sin x + \cos x}$. With $a = \frac{\pi}{2}$ and $f = \sin$, we have $f(\frac{\pi}{2} - x) = \cos x$. The denominator $\sin x + \cos x$ is positive on $[0, \frac{\pi}{2}]$. By the result above,

$$\int_0^{\pi/2} \frac{\sin x}{\sin(x + \frac{\pi}{4})}\,\mathrm{d}x = \sqrt{2} \cdot \frac{\pi/2}{2} = \frac{\sqrt{2}\,\pi}{4}.$$

Part (ii)

Call the integral J. Its integrand is $\dfrac{1}{x} \cdot \dfrac{\sin x}{\sin x + \sin(1/x)}$. The denominator is safe: for $x \in [\frac{1}{2}, 2]$, both x and $1/x$ lie in $[\frac{1}{2}, 2] \subset (0, \pi)$, so $\sin x > 0$ and $\sin(1/x) > 0$.

Ask yourself Which substitution swaps $\sin x$ and $\sin\frac{1}{x}$ and keeps the limits $\frac{1}{2}$ and 2?

The answer is $x = 1/u$. Then $\mathrm{d}x = -\mathrm{d}u/u^2$. When $x = \frac{1}{2}$, $u = 2$, and when $x = 2$, $u = \frac{1}{2}$. The factor $\frac{1}{x}$ becomes u, so

$$J = \int_2^{1/2} u \cdot \frac{\sin(1/u)}{\sin(1/u) + \sin u} \cdot \left(-\frac{\mathrm{d}u}{u^2}\right) = \int_{1/2}^2 \frac{\sin(1/u)}{u\left(\sin u + \sin\frac{1}{u}\right)}\,\mathrm{d}u.$$

Renaming u as x and adding this to the original form of J:

$$2J = \int_{1/2}^2 \frac{\sin x + \sin\frac{1}{x}}{x\left(\sin x + \sin\frac{1}{x}\right)}\,\mathrm{d}x = \int_{1/2}^2 \frac{1}{x}\,\mathrm{d}x = \ln 2 - \ln\tfrac{1}{2} = 2\ln 2.$$

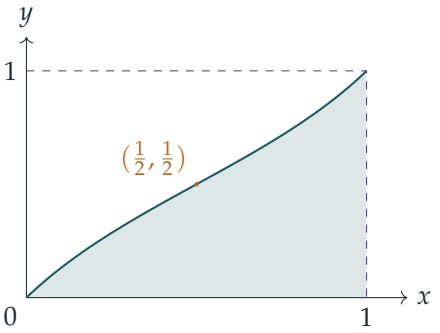
Hence

$$\int_{1/2}^2 \frac{\sin x}{x\left(\sin x + \sin\frac{1}{x}\right)}\,\mathrm{d}x = \ln 2.$$

The factor $\frac{1}{x}$ is essential here. The measure $\frac{\mathrm{d}x}{x}$ is unchanged by $x \mapsto 1/x$ (apart from the orientation), in the same way that $\mathrm{d}x$ is unchanged by $x \mapsto a - x$.

Checks

- The integrand $g(x) = \frac{f(x)}{f(x)+f(a-x)}$ satisfies $g(x) + g(a-x) = 1$. Its graph is therefore symmetric under a half-turn about $(\frac{a}{2}, \frac{1}{2})$, so the area under it is half the area of the $a \times 1$ rectangle. The picture shows this for the first integral.



- Numerically, Simpson's rule gives 0.500000 for the first integral, 1.110721 for the second (and $\frac{\sqrt{2}\pi}{4} \approx 1.110721$), and 0.693147 for part (ii) (and $\ln 2 \approx 0.693147$).

Common slips

- Forgetting the factor $\sqrt{2}$ when rewriting $\sin(x + \frac{\pi}{4})$, which gives $\frac{\pi}{4}$ instead of $\frac{\sqrt{2}\pi}{4}$.
- In part (ii), losing track of the $\frac{1}{x}$: it turns into u, and combines with $-\frac{1}{u^2}$ to leave $-\frac{1}{u}$.

Beyond STEP

Both tricks exploit a symmetry of the interval that leaves the integral's "measure" unchanged: reflection $x \mapsto a - x$ preserves $\mathrm{d}x$, and inversion $x \mapsto 1/x$ preserves $\frac{\mathrm{d}x}{x}$. At university this becomes the idea of an invariant measure. For example, $\frac{\mathrm{d}x}{x}$ is the natural measure on the positive reals under multiplication, and it underlies the Mellin transform. For the real-analysis foundations of substitution, see Abbott, *Understanding Analysis*. For many more STEP-style integration tricks, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2010, Question 5

Pure • Vector Angles and Cones

STEP II 2010, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q5

The key idea

Everything here comes down to one fact: the angle θ between a line with direction $\mathbf{d}$ and a vector $\mathbf{a}$ satisfies $\cos\theta = \dfrac{\mathbf{d}\cdot\mathbf{a}}{|\mathbf{d}|\,|\mathbf{a}|}$. "Inclined equally" and "inclined at angle α" turn straight into equations in the components of $\mathbf{d}$. For the final part, notice that the Cartesian equation has the same shape as the relationship you have just found. A set of lines through O, all making the same angle with a fixed line, is a cone.

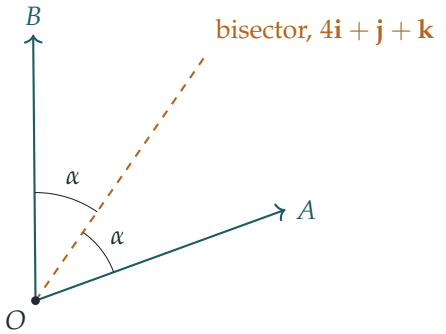
Solution

Write $\mathbf{a} = \mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i}-\mathbf{j}-\mathbf{k}$, so that $|\mathbf{a}| = \sqrt{3}$ and $|\mathbf{b}| = \sqrt{27} = 3\sqrt{3}$. Also $\mathbf{a}\cdot\mathbf{b} = 5-1-1 = 3$. Hence

$$\cos 2\alpha = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}|\,|\mathbf{b}|} = \frac{3}{\sqrt{3}\cdot 3\sqrt{3}} = \frac{1}{3}.$$

Since $\cos 2\alpha > 0$, the angle 2α is acute, so α lies between 0 and $\frac{\pi}{4}$. We will need

$$\cos^2\alpha = \tfrac{1}{2}(1+\cos 2\alpha) = \tfrac{2}{3}.$$



(The picture shows the plane OAB only schematically.)

Part (i)

Let $\mathbf{d} = m\mathbf{i} + n\mathbf{j} + p\mathbf{k}$ be the direction of L_1. The cosines of the angles that $\mathbf{d}$ makes with OA and OB are

$$\frac{\mathbf{d}\cdot\mathbf{a}}{|\mathbf{d}|\sqrt{3}} = \frac{m+n+p}{\sqrt{3}\,|\mathbf{d}|} \qquad \text{and} \qquad \frac{\mathbf{d}\cdot\mathbf{b}}{|\mathbf{d}|\,3\sqrt{3}} = \frac{5m-n-p}{3\sqrt{3}\,|\mathbf{d}|}.$$

Equal angles means equal cosines, since cos is one-to-one on $[0, \pi]$. Multiplying through by $3\sqrt{3}\,|\mathbf{d}|$ gives

$$3(m+n+p) = 5m - n - p \quad \Longleftrightarrow \quad m = 2(n+p).$$

Ask yourself A line has no preferred direction. What if the line makes angle θ with OA but $\pi - \theta$ with OB?

If we take "the angle between two lines" to mean the acute one, then the cosines need only be equal up to sign. The other sign gives $3(m+n+p) = -(5m-n-p)$, that is, $4m + n + p = 0$. These lines lie in the plane through O perpendicular to the bisector, which contains the external bisector of $\angle AOB$. The intended answer is $m = 2(n+p)$, but it's worth knowing that the second family exists.

For the bisector of $\angle AOB$, we add the *unit* vectors along OA and OB. The sum of two unit vectors bisects the angle between them because they span a rhombus. So

$$\frac{\mathbf{a}}{\sqrt{3}} + \frac{\mathbf{b}}{3\sqrt{3}} = \frac{3\mathbf{a}+\mathbf{b}}{3\sqrt{3}} = \frac{8\mathbf{i}+2\mathbf{j}+2\mathbf{k}}{3\sqrt{3}},$$

and we can take $m = 4$, $n = 1$, $p = 1$ (or any non-zero multiple). As a check, $4 = 2(1+1)$, so this direction satisfies the relation above.

Part (ii)

Let $\mathbf{e} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$ be the direction of L_2. The line makes angle α with OA when

$$\frac{u+v+w}{\sqrt{3}\,\sqrt{u^2+v^2+w^2}} = \pm\cos\alpha.$$

The $\pm$ allows for either sense along the line. Squaring removes the sign, and using $\cos^2\alpha = \frac{2}{3}$ we get

$$(u+v+w)^2 = 2(u^2+v^2+w^2).$$

Expanding the left-hand side as $u^2+v^2+w^2+2(vw+wu+uv)$ and rearranging gives the relationship

$$u^2+v^2+w^2 = 2(vw+wu+uv).$$

Ask yourself How does a condition on a *direction* turn into a *surface*?

Take any point $(x, y, z) \neq (0, 0, 0)$. It lies on exactly one line through O, namely the line with direction $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. By part (ii), that point satisfies

$$x^2 + y^2 + z^2 = 2(yz + zx + xy)$$

exactly when its line through O makes angle α with OA. The origin also satisfies the equation. So the surface is the union of all lines through O inclined at angle α to the line OA.

This is a **(double) right circular cone** with vertex at the origin and axis the line $x = y = z$ (the line OA). Its semi-vertical angle α satisfies

$$\cos\alpha = \sqrt{\tfrac{2}{3}}, \qquad \tan\alpha = \tfrac{1}{\sqrt{2}}, \qquad \alpha \approx 35.3^\circ,$$

which is half of $\angle AOB$. It has two nappes because μ runs over all real numbers. The bisector from part (i) lies on the cone: for $(4, 1, 1)$ we get $16 + 1 + 1 = 18 = 2(1 + 4 + 4)$.

Checks

- The point $(1, 1, 0)$ satisfies $2 = 2(0 + 0 + 1)$. Its angle with $\mathbf{i} + \mathbf{j} + \mathbf{k}$ has cosine $2/(\sqrt{2}\sqrt{3}) = \sqrt{2/3}$, as it should.
- Numerically, $\arccos(1/3)/2 = 0.61548$ rad, and the bisector $4\mathbf{i} + \mathbf{j} + \mathbf{k}$ makes angle 0.61548 rad with both OA and OB.
- The identity $x^2 + y^2 + z^2 - 2(yz + zx + xy) = (x^2 + y^2 + z^2)(2 - 3c^2)$ holds, where c is the cosine of the angle with $\mathbf{i} + \mathbf{j} + \mathbf{k}$. So the left-hand side vanishes exactly when $c^2 = \frac{2}{3}$.

Common slips

- Bisecting with $\mathbf{a} + \mathbf{b}$ instead of the sum of *unit* vectors. Since $|\mathbf{b}| \neq |\mathbf{a}|$, $\mathbf{a} + \mathbf{b} = 6\mathbf{i}$ is not the bisector.
- Using $\cos\alpha = \frac{1}{2}\cos 2\alpha$ or similar. You need the double-angle formula, $\cos^2\alpha = \frac{1}{2}(1 + \cos 2\alpha)$.
- Describing the surface just as "a cone" without giving its vertex, axis and semi-vertical angle.

Beyond STEP

The equation of the cone is a *quadratic form*, $\mathbf{x}^{\mathsf{T}} M\mathbf{x} = 0$ with M a symmetric matrix. In first-year linear algebra you classify such surfaces by the eigenvalues of M. Here M has diagonal entries 1 and off-diagonal entries -1, so its eigenvalues are -1 (along $\mathbf{i} + \mathbf{j} + \mathbf{k}$) and 2 (twice). One negative and two equal positive eigenvalues are exactly the signature of a circular cone about the eigenvector of the odd one out. Strang's

Introduction to Linear Algebra develops this in its treatment of symmetric matrices and quadratic forms.

WORKED SOLUTION

STEP II 2010, Question 6

Pure • Regular Tetrahedron Geometry

STEP II 2010, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

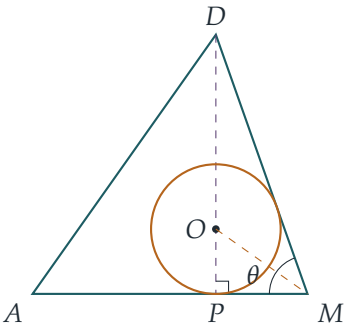
step.maths.org/questions/10-s2-q6

The key idea

A regular tetrahedron is so symmetric that almost everything happens in one vertical cross-section: the plane through an edge-midpoint, the opposite vertex and the apex. In that plane every 3D question becomes a right-angled triangle. For the sphere, symmetry tells us where the centre must be, and the angle from part (ii) tells us how far it sits from each face.

Solution

Throughout, let M be the midpoint of BC. Every face is an equilateral triangle of side 1, so each median (such as AM or DM) has length $\sqrt{1-\frac{1}{4}} = \frac{\sqrt{3}}{2}$.



Part (i)

Since D is equidistant from A, B and C, and DP is perpendicular to the plane ABC, Pythagoras gives $PA^2 = DA^2 - DP^2 = DB^2 - DP^2 = PB^2$, and likewise $PB = PC$. So P is the circumcentre of the equilateral triangle ABC, which is also its centroid. The

centroid lies two-thirds of the way along each median, so

$$PA = \tfrac{2}{3} \cdot \tfrac{\sqrt{3}}{2} = \tfrac{1}{\sqrt{3}}, \qquad PM = \tfrac{1}{3} \cdot \tfrac{\sqrt{3}}{2} = \tfrac{1}{2\sqrt{3}}.$$

Then, from the right angle at P in triangle DPA,

$$PD^2 = DA^2 - PA^2 = 1 - \tfrac{1}{3} = \tfrac{2}{3}, \qquad \text{so} \qquad PD = \sqrt{\tfrac{2}{3}} = \frac{\sqrt{6}}{3}.$$

Part (ii)

All six edges look the same (any edge can be moved onto any other by a symmetry of the tetrahedron), so it is enough to find the angle θ between faces ABC and DBC, which meet along BC.

Ask yourself Which angle in the diagram actually *is* the angle between the two faces?

The angle between two planes meeting in a line is measured using two lines, one in each plane, both perpendicular to the common line at the same point. In the equilateral triangle ABC the median AM is perpendicular to BC; in the equilateral triangle DBC the median DM is perpendicular to BC. So $\theta = \angle AMD$, and since P lies on AM between A and M, also $\theta = \angle PMD$.

Triangle DPM has a right angle at P, with $PM = \frac{1}{2\sqrt{3}}$ and hypotenuse $DM = \frac{\sqrt{3}}{2}$. Hence

$$\cos\theta = \frac{PM}{DM} = \frac{1/(2\sqrt{3})}{\sqrt{3}/2} = \frac{1}{3},$$

as required. (As a check, $PD^2 = DM^2 - PM^2 = \frac{3}{4} - \frac{1}{12} = \frac{2}{3}$, agreeing with part (i).)

Part (iii)

The best sphere. Let O be the point of DP with $OP = r$, where r is to be found so that O is also at distance r from the three sloping faces. By symmetry, it is enough to deal with the face DBC.

The plane AMD is perpendicular to BC (it contains AM and DM, both perpendicular to BC), so it is perpendicular to the face DBC, and the perpendicular from O to that face lies in this cross-section: it is the perpendicular from O to the line MD. Since O is at distance OP from line MP and at distance r from line MD and we want these equal, O must lie on the bisector of angle θ at M. So

$$r = PM \tan\tfrac{\theta}{2}.$$

From part (ii), using $\tan^2\frac{\theta}{2} = \dfrac{1-\cos\theta}{1+\cos\theta}$,

$$\tan^2\tfrac{\theta}{2} = \frac{1-\frac{1}{3}}{1+\frac{1}{3}} = \frac{1}{2}, \qquad \tan\tfrac{\theta}{2} = \frac{1}{\sqrt{2}}$$

(positive, since θ is acute). Therefore

$$r = \frac{1}{2\sqrt{3}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2\sqrt{6}} = \frac{\sqrt{6}}{12}.$$

The sphere with centre O and radius $\frac{\sqrt{6}}{12}$ touches all four faces, and lies inside the tetrahedron because O is an interior point at distance at least r from every face.

Ask yourself How do we know no sphere, placed anywhere, can be bigger than this one?

Nothing larger fits. Take any point X inside the tetrahedron, at distances d_1, d_2, d_3, d_4 from the four faces. Joining X to the vertices splits the tetrahedron into four pyramids, one on each face, each with base area $\frac{\sqrt{3}}{4}$. Adding their volumes,

$$V = \tfrac{1}{3} \cdot \tfrac{\sqrt{3}}{4} (d_1 + d_2 + d_3 + d_4).$$

But taking $X = D$ (where three distances are 0 and one is PD) shows $V = \frac{1}{3} \cdot \frac{\sqrt{3}}{4} PD$. Hence for every interior point

$$d_1 + d_2 + d_3 + d_4 = PD = \frac{\sqrt{6}}{3}.$$

A sphere of radius ρ centred at X fits inside only if every $d_i \geq \rho$, so $4\rho \leq \frac{\sqrt{6}}{3}$, that is, $\rho \leq \frac{\sqrt{6}}{12}$. So the largest sphere has radius

$$r = \frac{\sqrt{6}}{12} = \tfrac{1}{4}\, PD,$$

with equality exactly when all four distances are equal (at O).

Checks

- Coordinates: $A = (0,0,0)$, $B = (1,0,0)$, $C = (\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$, $D = (\frac{1}{2}, \frac{\sqrt{3}}{6}, \sqrt{2/3})$ give all six edges of length 1, and $PD \approx 0.8165$.
- The outward unit normals of any two faces have dot product $-\frac{1}{3}$, confirming $\cos\theta = \frac{1}{3}$ (about $70.5°$).
- The point $(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{12})$ is at distance $0.20412\ldots = \frac{\sqrt{6}}{12}$ from all four faces. Random interior points always gave a distance sum of $0.8165\ldots$, and their smallest distance never exceeded 0.2041.
- The centre is a quarter of the way up PD. This matches the fact that the incentre of a regular tetrahedron is its centroid, which lies a quarter of the way from a face to the opposite vertex.

Common slips

- Using $\angle DAM$ or $\angle DBA$ as the angle between faces. The lines must both be perpendicular to the common edge, which DM and AM are.
- Taking PA to be half a median instead of two-thirds of one.
- Finding the sphere touching the four faces without saying why no sphere elsewhere can be larger. The distance-sum identity settles this in two lines.

Beyond STEP

The constant sum of distances to the faces is the 3D version of Viviani's theorem for an equilateral triangle. It comes from barycentric coordinates, which describe any point of a simplex as a weighted average of its vertices. The angle $\arccos\frac{1}{3}$ appears in chemistry too: the angle between bonds in methane is $\arccos(-\frac{1}{3}) \approx 109.5^\circ$, the angle subtended at the centre by two vertices of this same tetrahedron. For more of this kind of 3D reasoning, see Siklos, *Advanced Problems in Mathematics*, or Strang, *Introduction to Linear Algebra* for the vector approach with dot and cross products.

WORKED SOLUTION

STEP II 2010, Question 7

Pure • Cubic Equations

STEP II 2010, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q7

The key idea

A cubic with positive leading coefficient crosses the axis once only exactly when its two turning points lie on the same side of the axis. Part (ii) is the classical trick for solving cubics with no x^2 term: the substitution $x = u + q/u$ is chosen so that the cross terms in x^3 cancel against $-3qx$, which leaves a quadratic in u^3. Part (iii) then turns out to be the same cubic in disguise, with p in place of x.

Solution

Part (i)

Write $f(x) = x^3 - 3qx - q(1+q)$. Then $f'(x) = 3x^2 - 3q$, which vanishes at $x = \pm\sqrt{q}$ (real, since $q > 0$). Since $f''(x) = 6x$, the point at $x = -\sqrt{q}$ is a local maximum and the point at $x = \sqrt{q}$ is a local minimum.

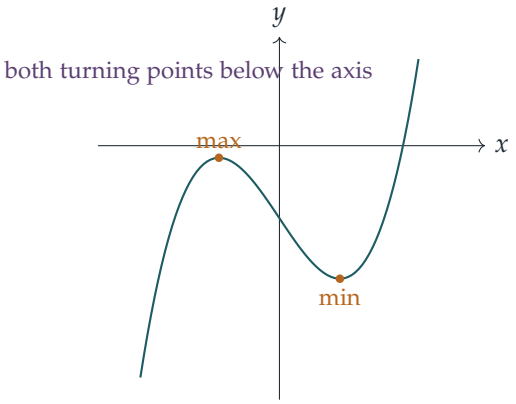
At the local maximum,

$$\begin{aligned} f(-\sqrt{q}) &= -q\sqrt{q} + 3q\sqrt{q} - q - q^2 \\ &= -q\left(q - 2\sqrt{q} + 1\right) \\ &= -q\left(\sqrt{q} - 1\right)^2. \end{aligned}$$

Since $q > 0$ and $q \neq 1$ (so $\sqrt{q} \neq 1$), this is strictly negative. The local minimum is lower still:

$$f(\sqrt{q}) = q\sqrt{q} - 3q\sqrt{q} - q - q^2 = -2q\sqrt{q} - q - q^2 < 0.$$

So both turning points lie below the x-axis. For $x < \sqrt{q}$ the curve never rises above its local maximum value, which is negative, so it does not meet the axis there. For $x > \sqrt{q}$ the function is strictly increasing, from a negative value to $+\infty$, so it crosses the axis exactly once. Hence the curve crosses the x-axis once only.



Part (ii)

Cubing $x = u + q/u$ with the binomial theorem,

$$\begin{aligned} x^3 &= u^3 + 3u^2 \cdot \frac{q}{u} + 3u \cdot \frac{q^2}{u^2} + \frac{q^3}{u^3} \\ &= u^3 + \frac{q^3}{u^3} + 3q\left(u + \frac{q}{u}\right) = u^3 + \frac{q^3}{u^3} + 3qx. \end{aligned}$$

So $x^3 - 3qx = u^3 + q^3/u^3$, and the cubic becomes

$$u^3 + \frac{q^3}{u^3} = q(1+q).$$

Multiplying by u^3 (note $u \neq 0$ for the substitution to make sense), the quadratic equation satisfied by u^3 is

$$\left(u^3\right)^2 - q(1+q)\,u^3 + q^3 = 0.$$

? **Ask yourself** How does this quadratic factorise?

The two roots must have sum $q + q^2$ and product $q^3 = q \cdot q^2$, so they are q and q^2:

$$\left(u^3 - q\right)\left(u^3 - q^2\right) = 0.$$

Taking real cube roots, $u = q^{1/3}$ or $u = q^{2/3}$. Both give the same x:

$$q^{1/3} + \frac{q}{q^{1/3}} = q^{1/3} + q^{2/3}, \qquad q^{2/3} + \frac{q}{q^{2/3}} = q^{2/3} + q^{1/3}.$$

As a direct check, with $s = q^{1/3}$ and $x = s + s^2$ we have $x^3 = s^3 + 3s^4 + 3s^5 + s^6$ and $3qx = 3s^4 + 3s^5$, so $x^3 - 3qx = s^3 + s^6 = q(1+q)$. By part (i) the cubic has only one real root when $q > 0$, $q \neq 1$, so the real root is

$$x = q^{1/3} + q^{2/3}.$$

Part (iii)

Since $\alpha + \beta = p$ and $\alpha\beta = q$,

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = p^3 - 3qp,$$

as required.

If one root is the square of the other, then either $\alpha^2 = \beta$ or $\beta^2 = \alpha$, so in either case

$$(\alpha^2 - \beta)(\beta^2 - \alpha) = 0.$$

This expression is symmetric in α and β, so we can write it in terms of p and q. Expanding,

$$\begin{aligned}(\alpha^2 - \beta)(\beta^2 - \alpha) &= \alpha^2\beta^2 - (\alpha^3 + \beta^3) + \alpha\beta \\ &= q^2 - (p^3 - 3qp) + q.\end{aligned}$$

Setting this to zero gives the relationship

$$p^3 - 3qp - q(1 + q) = 0.$$

This is exactly the cubic of part (ii) with p in place of x. When $q > 0$, $q \neq 1$ and p is real, p must be its only real root, so

$$p = q^{1/3} + q^{2/3}.$$

This makes sense: the roots are then $\alpha = q^{1/3}$ and $\beta = q^{2/3} = \alpha^2$, whose sum is p and product is q.

Checks

Take $q = 8$: part (ii) gives $x = 2 + 4 = 6$, and indeed $216 - 144 - 72 = 0$. In part (iii), $p = 6$ gives $t^2 - 6t + 8 = (t - 2)(t - 4)$, with $4 = 2^2$. A computer confirmed both turning-point values in (i) for thousands of random q, the identity $x^3 - 3qx = u^3 + q^3/u^3$ for random complex u, and that $p = q^{1/3} + q^{2/3}$ always yields a pair of roots with one the square of the other.

Common slips

- In (i), checking only that the local minimum is negative. That alone allows three crossings; it is the local maximum being negative that matters, and that is where $q \neq 1$ is used.
- In (ii), worrying that the two values of u^3 give two different roots. They give the same x, as they must, because u and q/u play symmetric roles.

Beyond STEP

The substitution in (ii) is the heart of Cardano's formula for the general cubic, and it is where complex numbers first became unavoidable in mathematics: when a cubic has three real roots, the quadratic for u^3 has complex roots. The excluded case $q = 1$ is a nice illustration. Then $p = -1$ is also allowed, and $t^2 + t + 1 = 0$ has roots ω and ω^2, the complex cube roots of unity, each the square of the other. The full story of when polynomial equations can be solved by radicals is Galois theory, told well in Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2010, Question 8

Pure • Areas Between Curves

STEP II 2010, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q8

The key idea

Since $\sin x \leq 1$, the curve $y = \mathrm{e}^{-x}\sin x$ can never rise above $y = \mathrm{e}^{-x}$; it meets it exactly where $\sin x = 1$, and there it just touches. So each area is a single integral of $\mathrm{e}^{-x}(1 - \sin x)$ between consecutive touching points, with no sign changes to worry about. The integral of $\mathrm{e}^{-x}\sin x$ is a standard "integrate by parts twice" job, and because the limits are all points where $\sin x = 1$ and $\cos x = 0$, the antiderivative collapses to something very simple there.

Solution

The sketch

C_1 is the familiar decaying exponential, starting at $(0, 1)$ and decreasing towards 0. C_2 is a damped sine wave: it starts at the origin, is zero at every multiple of π, and is trapped between $y = \mathrm{e}^{-x}$ and $y = -\mathrm{e}^{-x}$ because $-1 \leq \sin x \leq 1$. The oscillations shrink very fast; already the first negative dip has depth only about 0.014 (at $x = \frac{5\pi}{4}$).

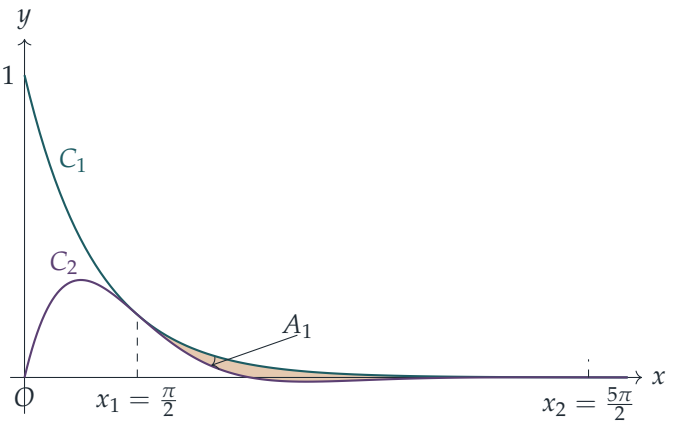
Where do the curves meet? Since $\mathrm{e}^{-x} > 0$,

$$\mathrm{e}^{-x}\sin x = \mathrm{e}^{-x} \iff \sin x = 1 \iff x = \tfrac{\pi}{2} + 2k\pi, \quad k = 0, 1, 2, \ldots$$

At these points the curves touch rather than cross: C_2 lies on or below C_1 everywhere. Indeed the gradients agree there too, since

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{-x}\sin x\right) = \mathrm{e}^{-x}(\cos x - \sin x) = -\mathrm{e}^{-x} \quad \text{when } \sin x = 1,\ \cos x = 0.$$

Notice that the first maximum of C_2 is at $x = \frac{\pi}{4}$ (where $\cos x = \sin x$), *before* the first contact at $x = \frac{\pi}{2}$: the sketch should show C_2 peaking and then coming down to meet C_1 tangentially.



The areas A_n

The nth contact point is the nth solution of $\sin x = 1$ with $x > 0$:

$$x_n = \tfrac{\pi}{2} + 2(n-1)\pi = \tfrac{(4n-3)\pi}{2}, \qquad x_{n+1} = x_n + 2\pi.$$

Between x_n and x_{n+1} we have $\mathrm{e}^{-x} - \mathrm{e}^{-x}\sin x = \mathrm{e}^{-x}(1-\sin x) \geq 0$, so C_1 is the upper boundary throughout and

$$A_n = \int_{x_n}^{x_{n+1}} \mathrm{e}^{-x}(1-\sin x)\,\mathrm{d}x.$$

? **Ask yourself** How do we integrate $\mathrm{e}^{-x}\sin x$?

Integrate by parts twice and you get back to the integral you started with. Let $I = \int \mathrm{e}^{-x}\sin x\,\mathrm{d}x$. Integrating by parts with $u = \sin x$, $\mathrm{d}v = \mathrm{e}^{-x}\,\mathrm{d}x$, and then again with $u = \cos x$,

$$\begin{aligned} I &= -\mathrm{e}^{-x}\sin x + \int \mathrm{e}^{-x}\cos x\,\mathrm{d}x \\ &= -\mathrm{e}^{-x}\sin x - \mathrm{e}^{-x}\cos x - \int \mathrm{e}^{-x}\sin x\,\mathrm{d}x, \end{aligned}$$

so $2I = -\mathrm{e}^{-x}(\sin x + \cos x)$ and $I = -\frac{1}{2}\mathrm{e}^{-x}(\sin x + \cos x) + c$. (You can confirm by differentiating.)

Hence an antiderivative of $\mathrm{e}^{-x}(1-\sin x)$ is

$$F(x) = -\mathrm{e}^{-x} + \tfrac{1}{2}\mathrm{e}^{-x}(\sin x + \cos x).$$

Now the neat part: at every contact point $\sin x = 1$ and $\cos x = 0$, so there

$$F(x) = -\mathrm{e}^{-x} + \tfrac{1}{2}\mathrm{e}^{-x} = -\tfrac{1}{2}\mathrm{e}^{-x}.$$

Therefore, using $x_{n+1} = x_n + 2\pi$,

$$\begin{aligned} A_n = F(x_{n+1}) - F(x_n) &= \tfrac{1}{2}\mathrm{e}^{-x_n} - \tfrac{1}{2}\mathrm{e}^{-x_n - 2\pi} \\ &= \tfrac{1}{2}\mathrm{e}^{-x_n - 2\pi}\left(\mathrm{e}^{2\pi} - 1\right). \end{aligned}$$

Finally $x_n + 2\pi = \frac{(4n-3)\pi}{2} + \frac{4\pi}{2} = \frac{(4n+1)\pi}{2}$, so

$$A_n = \tfrac{1}{2}\left(\mathrm{e}^{2\pi} - 1\right)\mathrm{e}^{-(4n+1)\pi/2},$$

as required.

The infinite sum

The formula shows that $A_{n+1} = \mathrm{e}^{-2\pi}A_n$, so the areas form a geometric series with first term $A_1 = \frac{1}{2}(\mathrm{e}^{2\pi} - 1)\mathrm{e}^{-5\pi/2}$ and common ratio $r = \mathrm{e}^{-2\pi}$. Since $0 < r < 1$ the series converges, and

$$\sum_{n=1}^{\infty} A_n = \frac{A_1}{1 - \mathrm{e}^{-2\pi}} = \frac{\frac{1}{2}(\mathrm{e}^{2\pi} - 1)\mathrm{e}^{-5\pi/2}}{\mathrm{e}^{-2\pi}(\mathrm{e}^{2\pi} - 1)} = \tfrac{1}{2}\mathrm{e}^{-\pi/2}.$$

Checks

- The total is the whole area between the curves to the right of x_1, so it should equal $\int_{\pi/2}^{\infty} \mathrm{e}^{-x}(1 - \sin x)\,\mathrm{d}x = \lim_{X\to\infty} F(X) - F(\frac{\pi}{2}) = 0 + \frac{1}{2}\mathrm{e}^{-\pi/2}$. It does, with no series needed.
- Numerically, $A_1 \approx 0.10375$ and $\frac{1}{2}\mathrm{e}^{-\pi/2} \approx 0.10394$: almost all the area is in the first region, as the sketch suggests, because $\mathrm{e}^{-2\pi} \approx 0.0019$. Simpson's rule on $\mathrm{e}^{-x}(1 - \sin x)$ over $[\frac{\pi}{2}, \frac{5\pi}{2}]$ gives the same 0.103746.

Common slips

- Taking the contact points to be where C_2 has its maxima. The maxima are at $x = \frac{\pi}{4} + k\pi$; the curves touch at $x = \frac{\pi}{2} + 2k\pi$.
- Counting every solution of $\sin x = \pm 1$, or of $\sin x = 1$ spaced by π: consecutive contacts are 2π apart.
- A sign error in $\int \mathrm{e}^{-x}\sin x\,\mathrm{d}x$. Differentiate your answer to check it.

Beyond STEP

The curve C_2 is the shape of a damped oscillation, the motion of a spring with friction, and the envelope $\pm\mathrm{e}^{-x}$ governs how fast it dies away. Integrals like $\int \mathrm{e}^{-x}\sin x\,\mathrm{d}x$ are much quicker as the imaginary part of $\int \mathrm{e}^{(-1+\mathrm{i})x}\,\mathrm{d}x$, a trick used constantly with differential equations and Laplace transforms in first-year courses. Boyce and DiPrima, *Elementary Differential Equations*, treats damped oscillations in detail.

WORKED SOLUTION

STEP II 2010, Question 9

Mechanics • Colliding Projectiles

STEP II 2010, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

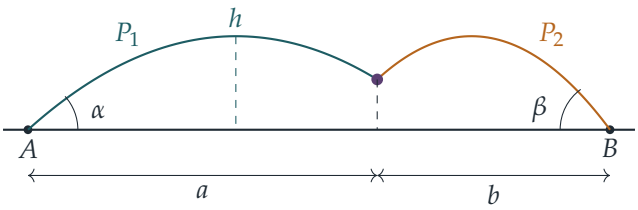
step.maths.org/questions/10-s2-q9

The key idea

Two particles projected at the same moment fall with the same acceleration g, so the only way their heights can agree at the moment of collision is for their initial vertical velocities to be equal. Then their vertical motions are identical: same greatest height h, reached at the same time, and both would land at the same time. The collision happens between the top of the flight and the landing, and the horizontal distance each particle has covered is proportional to the time, which pins it between two simple bounds.

Solution

We model both particles as projectiles moving freely under gravity g, with no air resistance. Let P_1 have speed u and P_2 have speed v, take $t = 0$ at projection and let the collision happen at time $t_c > 0$.



Equal vertical motions

At time t the heights of P_1 and P_2 are

$$y_1 = ut\sin\alpha - \tfrac{1}{2}gt^2, \qquad y_2 = vt\sin\beta - \tfrac{1}{2}gt^2.$$

At the collision $y_1 = y_2$, so $(u\sin\alpha - v\sin\beta)\,t_c = 0$, and since $t_c > 0$,

$$u\sin\alpha = v\sin\beta.$$

So the two particles have exactly the same vertical motion. In particular both reach the same greatest height h at the same time T, where

$$T = \frac{u\sin\alpha}{g} = \frac{v\sin\beta}{g}, \qquad h = \frac{u^2\sin^2\alpha}{2g} = \frac{v^2\sin^2\beta}{2g},$$

and both would return to the ground at time $2T$.

Ask yourself What do "in the air" and "after P_1 has attained its maximum height" tell us about t_c?

P_1 is at its highest at time T, so "after" gives $t_c > T$. Both particles are above the ground only for $0 < t < 2T$, so a collision in the air needs $t_c < 2T$. Hence

$$T < t_c < 2T.$$

The bound on b

Horizontally, P_2 moves with constant speed $v\cos\beta$, so $b = v\cos\beta\, t_c$. At time T it would have covered

$$v\cos\beta\, T = \frac{v^2\sin\beta\cos\beta}{g} = \frac{v^2\sin^2\beta}{2g}\cdot\frac{2\cos\beta}{\sin\beta} = 2h\cot\beta.$$

Therefore

$$b = v\cos\beta\, T\cdot\frac{t_c}{T} = 2h\cot\beta\cdot\frac{t_c}{T},$$

and since $1 < t_c/T < 2$ and $2h\cot\beta > 0$ (as β is acute),

$$2h\cot\beta < b < 4h\cot\beta,$$

as required.

The bound on a

In exactly the same way, $a = u\cos\alpha\, t_c$ and

$$u\cos\alpha\, T = \frac{u^2\sin\alpha\cos\alpha}{g} = 2h\cot\alpha,$$

so $a = 2h\cot\alpha\cdot t_c/T$, and $1 < t_c/T < 2$ gives

$$2h\cot\alpha < a < 4h\cot\alpha.$$

This uses the same h: h is the greatest height of P_1 by definition, and it is also the greatest height of P_2 because the vertical motions are identical.

Checks

- Boundary cases: a collision exactly at the top ($t_c = T$) gives $a = 2h\cot\alpha$, the familiar half-range $\frac{u^2\sin 2\alpha}{2g}$; a collision at the moment of landing ($t_c = 2T$) gives $a = 4h\cot\alpha$, the full range $\frac{u^2\sin 2\alpha}{g}$. The question's strict inequalities sit between these.

- Numerically: with $g = 9.8$, $u = 20$, $\alpha = 0.7$, $\beta = 0.9$ (radians), $v = u \sin\alpha / \sin\beta$ and $t_c = 1.5T$, we get $h \approx 8.470$ and $a \approx 30.17$, which lies between $2h\cot\alpha \approx 20.11$ and $4h\cot\alpha \approx 40.22$ (indeed it is exactly 1.5×20.11). We tested 20 000 random set-ups this way, stepping P_1's motion to confirm h and checking that the particles really coincide.

Common slips

- Treating the times to reach the collision point as different for the two particles: they are projected simultaneously, so it is the same t_c.
- Forgetting to justify the upper bound: it comes from the collision being in the air, that is, before the common landing time $2T$.

Beyond STEP

The key step is a change of frame: in a frame falling freely with acceleration g, both particles move in straight lines at constant velocity, so they collide only if they were aimed at each other in that frame, which is the classic "monkey and hunter" idea. Thinking in accelerating frames, and the fictitious forces they bring when the acceleration is not uniform, is a central theme of first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles, relative motion and non-inertial frames with many good problems.

WORKED SOLUTION

STEP II 2010, Question 10

Mechanics • Elastic Collisions

STEP II 2010, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q10

This question has a full guided treatment in *Beyond STEP*, Section 10.2, *The Stacked-Ball Effect*: https://beyondstep.keithmcnulty.org/ch10/stacked-balls.html

The key idea

Everything rests on one collision: a particle hitting a stationary particle elastically. Conservation of momentum plus Newton's experimental law (with $e = 1$) give the speed passed on, and it depends only on the *ratio* of the masses. In part (ii) every collision has the same mass ratio λ, so each one multiplies the speed by the same factor, and a geometric progression does the rest.

Solution

Model the particles as point masses moving along a straight line on a smooth table, with all velocities measured in the direction of B's initial motion.

Part (i)

Let B and A have velocities v_B and v_A just after the collision. Conservation of momentum gives

$$bmu = bm\,v_B + m\,v_A,$$

and because the collision is perfectly elastic ($e = 1$), Newton's experimental law gives speed of separation equal to speed of approach:

$$v_A - v_B = u.$$

Substituting $v_B = v_A - u$ into the momentum equation and cancelling m,

$$bu = b(v_A - u) + v_A \quad \Longrightarrow \quad v_A = \frac{2bu}{b+1}.$$

So the speed of A is $\dfrac{2b}{b+1}u$.

This formula only used the ratio b of the masses, and the derivation works for any $b > 0$. Since $b+1 > b$ and $u > 0$,

$$v_A = 2u \cdot \frac{b}{b+1} < 2u,$$

whatever the value of b. So A can never be made to move at twice B's initial speed, though by taking b large it can get as close to $2u$ as we like.

Part (ii)

The masses, in order along the line, are

$$B_1 : \ \lambda^n m, \quad B_2 : \ \lambda^{n-1} m, \quad \ldots, \quad B_n : \ \lambda m, \quad A : \ m.$$

Each particle is λ times as heavy as the next one along. So whenever a particle moving with speed v hits the next (stationary) one, part (i) applies with $b = \lambda$, and the struck particle moves off with speed kv, where

$$k = \frac{2\lambda}{\lambda+1}.$$

From the equation $v_B = v_A - u$ in part (i), the striking particle carries on forwards with speed

$$kv - v = \frac{\lambda - 1}{\lambda + 1} v,$$

which is positive since $\lambda > 1$.

So B_1 gives B_2 speed ku, then B_2 gives B_3 speed k^2u, and so on, until B_n, moving at $k^{n-1}u$, gives A speed

$$v_A = k^n u = \left(\frac{2\lambda}{\lambda+1}\right)^n u.$$

Ask yourself Could there be extra collisions, with a B_i catching up with the one in front after it has passed its speed on?

After all n collisions, B_i moves forward with speed $\dfrac{\lambda-1}{\lambda+1}k^{i-1}u$ for $i = 1, \ldots, n$, and A moves with speed $k^n u$. Since $\lambda > 1$ we have $k > 1$ (because $2\lambda > \lambda + 1$), so these speeds increase strictly along the line: each particle moves faster than the one behind it, and they separate. There are no further collisions, and A's final speed is $k^n u$.

Because $k > 1$, $k^n \to \infty$ as $n \to \infty$. So for any target speed V, choosing n large enough that $k^n u > V$ makes A move faster than V: there is no upper limit on A's speed.

The case $\lambda = 4$

Now $k = \frac{8}{5}$, and we need the least n with $\left(\frac{8}{5}\right)^n > 20$. Taking logarithms to base 10 (an increasing function, so the inequality is preserved),

$$n \log_{10} \tfrac{8}{5} > \log_{10} 20.$$

Writing $\frac{8}{5} = \frac{2^4}{10}$ and $20 = 2 \times 10$,

$$\log_{10} \tfrac{8}{5} = 4 \log_{10} 2 - 1 \approx 0.20412,$$
$$\log_{10} 20 = \log_{10} 2 + 1 \approx 1.30103.$$

Since $\log_{10} \frac{8}{5} > 0$ we may divide by it, giving

$$n > \frac{1.30103}{0.20412} \approx 6.37.$$

The least such integer is $n = 7$. (The margin is comfortable: 6.37 is far from an integer, so the approximation for $\log_{10} 2$ cannot change the answer.)

Checks

- In part (i), $b = 1$ gives $v_A = u$ and $v_B = 0$: equal masses swap velocities, as they should. As $b \to \infty$, $v_A \to 2u$.
- Kinetic energy: with $v_A = \frac{2b}{b+1}u$ and $v_B = \frac{b-1}{b+1}u$, we get $bv_B^2 + v_A^2 = \frac{b(b-1)^2+4b^2}{(b+1)^2}u^2 = bu^2$, so energy is conserved.
- Directly: $1.6^6 \approx 16.78 < 20$ and $1.6^7 \approx 26.84 > 20$, confirming $n = 7$. A step-by-step simulation of the chain with exact fractions also gave A's speed as exactly $k^n u$ after exactly n collisions.

Common slips

- Getting the masses the wrong way round: B_1 is the *heaviest* ($\lambda^n m$) and B_n the lightest of the Bs (λm), with A lighter still.
- Counting $n - 1$ collisions instead of n: the speed is passed along $B_1 \to B_2 \to \cdots \to B_n \to A$, which is n steps.
- Forgetting to rule out later collisions, or giving the non-integer 6.37 instead of the least integer $n = 7$.

Beyond STEP

A single elastic collision can at most double the speed, but a chain of collisions has no such limit. This is the idea behind the "stacked ball" demonstration, where a light ball

dropped on top of a heavy one rebounds far higher than it fell. Analysing collisions through momentum and energy (and, more neatly, in the centre-of-mass frame, where an elastic collision just reverses each velocity) is a central theme of first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats it in depth.

WORKED SOLUTION

STEP II 2010, Question 11

Mechanics • Three-Force Equilibrium

STEP II 2010, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q11

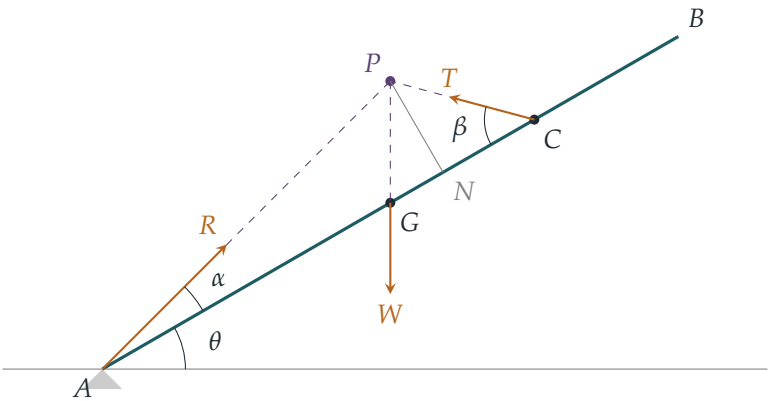
The key idea

Only three forces act on the rod: its weight, the reaction at A and the tension at C. When three non-parallel forces keep a body in equilibrium, their lines of action must pass through a single point. So the line of the reaction from A, the line of the string from C and the vertical through the midpoint of the rod all meet. Measuring distances along and perpendicular to the rod from that common point turns the whole question into a little right-angled trigonometry, with no need to find the forces themselves.

Solution

The model and the diagram

We model the rod as a uniform rigid rod, so its weight W acts at its midpoint G, with $AG = 2L$. The string is light, so it pulls on the rod at C along its own length, where $AC = 3L$. The support exerts a single force R at A (we are not told whether it is smooth or rough, and we don't need to be). The diagram shows the case $\theta = 30°$, $\beta = 45°$ to scale.



Showing the formula

The weight acts vertically through G; the string's line of action passes through C and the reaction's through A. These three lines are not parallel, so all three pass through one point, which we call P.

Ask yourself Why must the three lines of action meet at a point?

Let the lines of action of R and T meet at P. Taking moments about P, both R and T have zero moment, so for the rod to be in rotational equilibrium the weight must also have zero moment about P. That means the vertical line through G passes through P too.

Let N be the foot of the perpendicular from P to the rod, and let $h = PN$. In the right-angled triangle PNA the angle at A is α, so $AN = h\cot\alpha$. In the right-angled triangle PNC the angle at C is β, so $NC = h\cot\beta$. Since $AN + NC = AC$,

$$h\cot\alpha + h\cot\beta = 3L. \tag{1}$$

(If α or β is obtuse, N lies beyond A or C, but the signed lengths $h\cot\alpha$ and $h\cot\beta$ still add up to AC, so (1) holds in every case.)

Now use the third line of action. The rod makes angle θ with the horizontal, so the perpendicular PN makes angle θ with the vertical PG. In the right-angled triangle PNG the angle at P is θ, so $GN = h\tan\theta$, and N lies on the upper side of G (further from A), as in the diagram. Hence

$$h\cot\alpha = AN = AG + GN = 2L + h\tan\theta. \tag{2}$$

Subtracting (2) from (1) gives $h\cot\beta = L - h\tan\theta$, that is,

$$L = h(\cot\beta + \tan\theta).$$

Substituting this value of L into (2):

$$h\cot\alpha = 2h(\cot\beta + \tan\theta) + h\tan\theta = h(3\tan\theta + 2\cot\beta).$$

Since $h \neq 0$ (otherwise P would lie on the rod at G, and the string, which is attached at C, would lie along the rod, so it could not balance the moment of the weight about A), we may divide by h:

$$\cot\alpha = 3\tan\theta + 2\cot\beta,$$

as required.

The case $\theta = 30°$, $\beta = 45°$

With $\tan 30° = \frac{1}{\sqrt{3}}$ and $\cot 45° = 1$,

$$\cot\alpha = \frac{3}{\sqrt{3}} + 2 = 2 + \sqrt{3}.$$

Now we need $\cot 15^\circ$. Using $\tan(A-B)$ with $A = 45^\circ$, $B = 30^\circ$,

$$\tan 15^\circ = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{2} = 2 - \sqrt{3},$$

so

$$\cot 15^\circ = \frac{1}{2-\sqrt{3}} = \frac{2+\sqrt{3}}{(2-\sqrt{3})(2+\sqrt{3})} = 2 + \sqrt{3}.$$

Hence $\cot\alpha = \cot 15^\circ$. The angle α lies between 0° and 180°, and cot is strictly decreasing on that interval, so it takes each value exactly once. Therefore $\alpha = 15^\circ$.

Checks

- *An independent route.* Taking moments about A gives $T \cdot 3L\sin\beta = W \cdot 2L\cos\theta$. Resolving along and perpendicular to the rod then gives the components of R, and $\cot\alpha$ is their ratio; this reproduces the formula. We checked it numerically this way for 10^5 random pairs (θ, β), with agreement to about 10^{-15}.
- *Special values.* For $\theta = 30^\circ$, $\beta = 45^\circ$ the numerical equilibrium gives $\alpha = 15.000^\circ$, and $\cot 15^\circ = 3.7320\ldots = 2 + \sqrt{3}$. The diagram is drawn to scale: P lies directly above G, at height $\sqrt{3}L$ above A.
- *Limiting case.* If $\theta \to 0$ and $\beta = 90^\circ$ (horizontal rod, vertical string), the formula gives $\cot\alpha = 0$, so $\alpha = 90^\circ$: the reaction is vertical too, as it must be when every other force is vertical.

Common slips

- Putting the weight at C, or taking $AG = 3L/2$: the weight acts at the midpoint of the whole rod, $2L$ from A.
- Getting the sign of GN wrong. Because the rod slopes upwards from A, the vertical through G meets the perpendicular from P on the side of G away from A, so $AN = 2L + h\tan\theta$, not $2L - h\tan\theta$.
- Stopping at $\cot\alpha = 2 + \sqrt{3}$ without showing that this is $\cot 15^\circ$, and without saying why it fixes α uniquely.

Beyond STEP

The three-force theorem is a special case of a general fact in statics. Any system of forces on a rigid body reduces to a single force through a chosen point plus a couple, and equilibrium means both are zero. At university this becomes the vector condition that the total force and the total moment $\sum \mathbf{r}_i \times \mathbf{F}_i$ both vanish, which then underpins rigid-body dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops torque and equilibrium in this vector language.

WORKED SOLUTION

STEP II 2010, Question 12

Probability and Statistics • Mean and Median

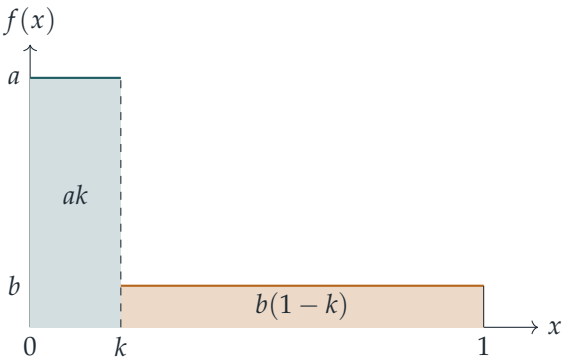
STEP II 2010, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q12

The key idea

The density is a two-step function, so every probability is an area of a rectangle and the only real condition is that the total area is 1. That condition fixes k in terms of a and b, and once k is eliminated everything is algebra. For the median, the question is simply whether the tall first rectangle already holds half of the probability.



Solution

The two rectangles in the sketch have areas ak and $b(1-k)$, so the density integrates to 1 exactly when

$$ak + b(1-k) = 1, \qquad \text{that is,} \qquad k = \frac{1-b}{a-b}.$$

(We may divide by $a-b$ because $a > b$.) We use this value of k throughout.

Why $a > 1$ and $b < 1$. The equation $ak + b(1-k) = 1$ says that 1 is a weighted average of a and b, with weights k and $1-k$, both strictly between 0 and 1. A weighted average with positive weights lies strictly between the two values, so $b < 1 < a$. To put it without words: $1 - b = k(a-b) > 0$ and $a - 1 = (1-k)(a-b) > 0$.

Part (i)

Integrating $xf(x)$ over each piece,

$$\mathrm{E}(X) = \int_0^k ax\,\mathrm{d}x + \int_k^1 bx\,\mathrm{d}x = \tfrac{1}{2}ak^2 + \tfrac{1}{2}b(1-k^2) = \tfrac{1}{2}\big(b + (a-b)k^2\big).$$

Now $(a-b)k^2 = \dfrac{(1-b)^2}{a-b}$, so

$$\mathrm{E}(X) = \frac{b(a-b)+(1-b)^2}{2(a-b)} = \frac{ab - b^2 + 1 - 2b + b^2}{2(a-b)} = \frac{1-2b+ab}{2(a-b)},$$

as required.

Part (ii)

The cumulative distribution function is

$$\mathrm{F}(x) = \begin{cases} ax & 0 \le x < k, \\ ak + b(x-k) & k \le x \le 1. \end{cases}$$

It is continuous and strictly increasing on $[0,1]$, so the median M is the unique solution of $\mathrm{F}(M) = \frac{1}{2}$. Which formula applies depends on whether the first rectangle holds at least half of the probability.

> **Ask yourself** How do we turn "the first rectangle has area at least $\frac{1}{2}$" into a condition on a and b?

Substitute for k and clear the positive denominator $a-b$:

$$ak \ge \tfrac{1}{2} \iff 2a(1-b) \ge a-b \iff 2a - 2ab \ge a - b \iff a + b \ge 2ab.$$

Case $a+b \ge 2ab$. Then $\mathrm{F}(k) = ak \ge \frac{1}{2}$, so the median lies in $[0,k]$ and solves $aM = \frac{1}{2}$:

$$M = \frac{1}{2a}.$$

Case $a+b \le 2ab$. Then $ak \le \frac{1}{2}$, so the median lies in $[k,1]$. It is easiest to work from the right-hand end: the probability to the right of M is $b(1-M)$, and this must equal $\frac{1}{2}$. Hence

$$M = 1 - \frac{1}{2b} = \frac{2b-1}{2b}.$$

When $a+b = 2ab$ both formulae apply and agree, since then $1 - \frac{1}{2b} = \frac{2ab-a}{2ab} = \frac{b}{2ab} = \frac{1}{2a}$.

Part (iii)

We compute $\mathrm{E}(X) - M$ in each case and show it is positive.

Case $a+b \geq 2ab$. Over the common denominator $2a(a-b)$,

$$\mathrm{E}(X) - \frac{1}{2a} = \frac{a(1-2b+ab)-(a-b)}{2a(a-b)} = \frac{b-2ab+a^2b}{2a(a-b)} = \frac{b(a-1)^2}{2a(a-b)}.$$

Case $a+b \leq 2ab$. Over the common denominator $2b(a-b)$,

$$\begin{aligned}\mathrm{E}(X) - \frac{2b-1}{2b} &= \frac{b(1-2b+ab)-(2b-1)(a-b)}{2b(a-b)} \\ &= \frac{(b-2b^2+ab^2)-(2ab-2b^2-a+b)}{2b(a-b)} = \frac{a(1-b)^2}{2b(a-b)}.\end{aligned}$$

Ask yourself Why are these strictly positive, and not merely non-negative?

In both cases a, b and $a-b$ are positive, and the squares are strictly positive because we showed at the start that $a > 1$ and $b < 1$. So $\mathrm{E}(X) - M > 0$ in every case, and $M < \mathrm{E}(X)$.

Checks

- Take $a = 3$, $b = \frac{1}{2}$: then $k = \frac{1}{5}$, $a+b = \frac{7}{2} \geq 3 = 2ab$, and $\mathrm{E}(X) = \frac{1-1+\frac{3}{2}}{5} = 0.3$ while $M = \frac{1}{6}$. The difference $0.1\overline{3}$ matches $\frac{b(a-1)^2}{2a(a-b)} = \frac{2}{15}$.
- Take $a = \frac{5}{4}$, $b = \frac{9}{10}$: now $a+b = 2.15 \leq 2.25 = 2ab$, so $M = 1 - \frac{5}{9} = \frac{4}{9} \approx 0.444$, and $\mathrm{E}(X) = \frac{13}{28} \approx 0.464$.
- As $a \to 1^+$ (so $b \to 1^-$) the density tends to the uniform one, and both differences tend to 0: mean and median coincide at $\frac{1}{2}$, as they should by symmetry.

We checked numerically, for 300 random pairs (a, b), that the median found by bisection on F matches the case formula, that the mean by midpoint quadrature matches part (i), and that $\mathrm{E}(X) - M$ equals the factorised expressions above.

Common slips

- Forgetting to eliminate k: the answers are wanted in terms of a and b only.
- Writing the second median as the solution of $b(M-k) = \frac{1}{2}$, forgetting the probability ak already accumulated on $[0, k)$.
- Concluding only $\mathrm{E}(X) \geq M$: the strict inequality needs $a \neq 1$ and $b \neq 1$, which is exactly what the opening part provides.

Beyond STEP

This density is decreasing, so it is "skewed to the right": its long tail pulls the mean beyond the median. That pattern is common but not universal, and a first course in probability shows how mean, median and mode relate for unimodal distributions and where the familiar rules of thumb fail. Step functions like this one are also the building blocks of histograms, and of the simple functions used to define the integral in measure theory. For more on continuous distributions, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP II 2010, Question 13

Probability and Statistics • Independent Events

STEP II 2010, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q13

The key idea

Rosalind passes exactly when she wins at least one game against Pardeep *and* at least one against Quentin. "At least one" is always easiest through the complement: the chance of winning at least one of n games is 1 minus the chance of losing them all. With independent games, each strategy's success probability is then a product of two such factors, and comparing strategies means factorising a difference. Writing $a = 1 - p$ and $b = 1 - q$ for the losing probabilities keeps the algebra tidy.

Solution

We assume that the results of different games are independent, and that each game against Pardeep (or Quentin) is won with probability p (or q) whatever has happened before. Order doesn't matter: whether or not play stops once she has qualified, she qualifies in exactly the same outcomes.

If Rosalind plays Pardeep m times and Quentin n times, let A be the event that she wins at least one game against Pardeep and B the event that she wins at least one against Quentin. Then $\mathrm{P}(A) = 1 - (1-p)^m$, since A fails only if she loses all m games, and similarly $\mathrm{P}(B) = 1 - (1-q)^n$. The two events involve different games, so they are independent, and

$$\mathrm{P}(\text{accepted}) = \big(1 - (1-p)^m\big)\big(1 - (1-q)^n\big).$$

Throughout, write $a = 1 - p$ and $b = 1 - q$. Since $0 < p < q < 1$, we have $0 < b < a < 1$.

Part (i)

Playing Pardeep twice and Quentin once gives success probability $(1 - a^2)(1 - b)$; playing Quentin twice and Pardeep once gives $(1 - a)(1 - b^2)$. Factorising each difference

of squares,

$$\begin{aligned}(1-a^2)(1-b)-(1-a)(1-b^2) &= (1-a)(1-b)\big[(1+a)-(1+b)\big]\\ &= (1-a)(1-b)(a-b)\\ &= pq(q-p).\end{aligned}$$

This is strictly positive because $p, q > 0$ and $q > p$. So playing Pardeep twice gives the larger chance of acceptance, and that is what she should choose.

Part (ii)

The three strategies succeed with probabilities

$$S_1 = (1-a^3)(1-b), \qquad S_2 = (1-a^2)(1-b^2), \qquad S_3 = (1-a)(1-b^3).$$

Each contains the factor $(1-a)(1-b) = pq$, because $1-x^k = (1-x)(1+x+\cdots+x^{k-1})$.

Strategy 1 always beats strategy 3. We have

$$\begin{aligned}S_1 - S_3 &= pq\big[(1+a+a^2)-(1+b+b^2)\big]\\ &= pq\,(a-b)(1+a+b) > 0,\end{aligned}$$

since $a > b$. So strategy 3 is never the best choice, and only strategies 1 and 2 need comparing.

Strategy 1 against strategy 2. We have

$$S_1 - S_2 = pq\big[(1+a+a^2)-(1+a)(1+b)\big] = pq\big[a^2-b(1+a)\big].$$

Returning to p and q,

$$S_1 - S_2 = pq\,D, \qquad \text{where} \qquad D = (1-p)^2-(1-q)(2-p).$$

Since $pq > 0$, strategy 1 is better than strategy 2 exactly when $D > 0$.

Ask yourself How does the condition $q - p > \frac{1}{2}$ help to make D positive?

The only term in D involving q is $-(1-q)(2-p)$, and it increases as q increases, because $2-p > 0$. So we replace q by the smallest value the condition allows. If $q - p > \frac{1}{2}$ then $1-q < \frac{1}{2}-p$, and since $2-p > 0$,

$$(1-q)(2-p) < \big(\tfrac{1}{2}-p\big)(2-p) = 1-\tfrac{5}{2}p+p^2.$$

Therefore

$$D > (1-2p+p^2)-\big(1-\tfrac{5}{2}p+p^2\big) = \tfrac{1}{2}p > 0.$$

So $S_1 > S_2$, and together with $S_1 > S_3$ this shows that Rosalind should choose strategy 1 when $q - p > \frac{1}{2}$.

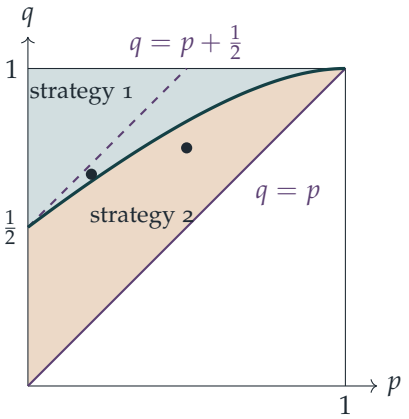
When $q - p < \frac{1}{2}$, either can win. Two examples:

- $p = \frac{1}{2}, q = \frac{3}{4}$ (so $q - p = \frac{1}{4}$). Here $D = \frac{1}{4} - \frac{1}{4} \cdot \frac{3}{2} = -\frac{1}{8} < 0$. Indeed $S_1 = \frac{7}{8} \cdot \frac{3}{4} = \frac{42}{64}$ and $S_2 = \frac{3}{4} \cdot \frac{15}{16} = \frac{45}{64}$, so **strategy 2 is better**.
- $p = \frac{1}{5}, q = \frac{2}{3}$ (so $q - p = \frac{7}{15} < \frac{1}{2}$). Here $D = \frac{16}{25} - \frac{1}{3} \cdot \frac{9}{5} = \frac{16}{25} - \frac{15}{25} = \frac{1}{25} > 0$. Indeed $S_1 = \frac{61}{125} \cdot \frac{2}{3} = \frac{122}{375}$ and $S_2 = \frac{9}{25} \cdot \frac{8}{9} = \frac{120}{375}$, so **strategy 1 is better**.

The whole picture is clear in the (p, q) plane. Setting $D = 0$ gives the boundary curve $q = 1 - \dfrac{(1-p)^2}{2-p}$. One can check that

$$(p + \tfrac{1}{2}) - \left(1 - \frac{(1-p)^2}{2-p}\right) = \frac{p}{2(2-p)} > 0,$$

so the curve lies just below the line $q = p + \frac{1}{2}$. Above the curve strategy 1 is best; between the curve and the diagonal $q = p$ strategy 2 is best.



The two dots are the examples above: $(\frac{1}{2}, \frac{3}{4})$ lies in the strategy 2 region and $(\frac{1}{5}, \frac{2}{3})$ just above the curve.

Checks

In part (i), let $q \to 1$: Quentin is a certain win, so only the Pardeep games matter and more of them is clearly better, which matches the difference $pq(q-p) \to p(1-p) > 0$. For part (ii) we enumerated all 2^4 outcomes exactly with fractions for 2000 random pairs $p < q$, confirming the formulas for S_1, S_2, S_3 and the factorisations. A simulation with $p = \frac{1}{2}, q = \frac{3}{4}$ gave $S_1 \approx 0.657$ and $S_2 \approx 0.703$, against the exact $\frac{21}{32} = 0.65625$ and $\frac{45}{64} \approx 0.7031$.

Common slips

- Computing "at least one win" as mp rather than $1 - (1-p)^m$; adding probabilities of overlapping events overcounts.

- Forgetting strategy 3: to show strategy 1 is best you must beat *both* alternatives.
- Giving examples with $q - p < \frac{1}{2}$ but not checking the condition $0 < p < q < 1$.

Beyond STEP

The intuition is "spend your effort where you are weakest", but the second part shows it isn't always right: the extra game against Pardeep has diminishing returns, and when p is not too small a second game against Quentin is worth more. Allocating a fixed budget of trials to maximise a product of "at least one success" probabilities is a small instance of the optimisation questions behind reliability theory and the design of experiments. Grimmett and Welsh, *Probability: An Introduction*, covers independence and complements carefully and has many problems in this spirit.

WORKED SOLUTION

STEP III 2010, Question 1

Pure • Mean and Variance

STEP III 2010, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q1

The key idea

A and B are the mean and variance of the first n numbers; C and D are the mean and variance once one more number is added. Variance is easiest to handle in the form "mean of the squares minus the square of the mean", which is what part (ii) gives us. With that, D comes out as a quadratic in x_{n+1}, and both inequalities are statements about that quadratic.

Solution

Throughout, write $x = x_{n+1}$ for brevity.

Part (i)

By definition $\sum_{k=1}^{n} x_k = nA$, so

$$C = \frac{1}{n+1}\left(\sum_{k=1}^{n} x_k + x\right) = \frac{nA + x}{n+1}.$$

Part (ii)

Expand the square and use $\sum_{k=1}^{n} x_k = nA$ and $\sum_{k=1}^{n} 1 = n$:

$$\begin{aligned} B &= \frac{1}{n}\sum_{k=1}^{n}\left(x_k^2 - 2Ax_k + A^2\right) = \frac{1}{n}\sum_{k=1}^{n} x_k^2 - \frac{2A}{n}\cdot nA + \frac{1}{n}\cdot nA^2 \\ &= \frac{1}{n}\sum_{k=1}^{n} x_k^2 - A^2, \end{aligned}$$

as required.

Part (iii)

Part (ii) holds for any list of numbers, so applying it to the $n+1$ numbers $x_1, \ldots, x_{n+1}$ (with mean C) gives

$$D = \frac{1}{n+1}\sum_{k=1}^{n+1} x_k^2 - C^2.$$

From part (ii) again, $\sum_{k=1}^{n} x_k^2 = n(B+A^2)$, so $\sum_{k=1}^{n+1} x_k^2 = n(B+A^2) + x^2$. Using part (i) for C,

$$D = \frac{nB + nA^2 + x^2}{n+1} - \frac{(nA+x)^2}{(n+1)^2}.$$

Multiply through by $(n+1)^2$ and expand:

$$\begin{aligned}(n+1)^2 D &= (n+1)\left(nB + nA^2 + x^2\right) - \left(n^2A^2 + 2nAx + x^2\right)\\ &= n(n+1)B + (n^2 + n - n^2)A^2 - 2nAx + (n+1-1)x^2\\ &= n(n+1)B + n\left(x^2 - 2Ax + A^2\right)\\ &= n(n+1)B + n(x-A)^2.\end{aligned}$$

Hence

$$\boxed{D = \frac{n}{n+1}B + \frac{n}{(n+1)^2}\left(x_{n+1} - A\right)^2.}$$

The first inequality. Multiplying by $n+1$,

$$(n+1)D = nB + \frac{n}{n+1}(x_{n+1} - A)^2 \;\geq\; nB,$$

since the last term is a non-negative multiple of a square. This holds for every value of x_{n+1}, with equality exactly when $x_{n+1} = A$.

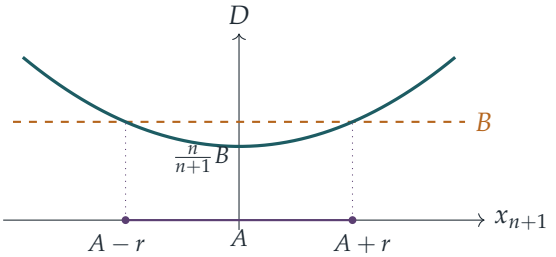
Ask yourself How do we turn $D < B$ into a condition on x_{n+1} without worrying about square roots of negative numbers?

The second inequality. Note first that $B \geq 0$, being an average of squares, so $\sqrt{(n+1)B/n}$ is a real number. Now, using the boxed formula, each step below is reversible (we only subtract the same quantity from both sides and multiply by the positive number $(n+1)^2/n$):

$$\begin{aligned}D < B &\iff \frac{n}{(n+1)^2}(x_{n+1} - A)^2 < B - \frac{n}{n+1}B = \frac{B}{n+1}\\ &\iff (x_{n+1} - A)^2 < \frac{(n+1)B}{n}\\ &\iff \left|x_{n+1} - A\right| < \sqrt{\frac{(n+1)B}{n}}\\ &\iff A - \sqrt{\frac{(n+1)B}{n}} < x_{n+1} < A + \sqrt{\frac{(n+1)B}{n}}.\end{aligned}$$

The third step takes non-negative square roots of both sides, which preserves a strict inequality between non-negative numbers. This is the required "if and only if". (If $B = 0$ the interval is empty, and indeed then $D \geq 0 = B$ always.)

The picture below shows D as a function of x_{n+1} for $n = 3$, $A = 0$, $B = 1$: a parabola with minimum $\frac{n}{n+1}B$ at $x_{n+1} = A$, dipping below the level B exactly on the interval found above.



Here $r = \sqrt{(n+1)B/n}$.

Checks

- Take $n = 1$: then $B = 0$, and the formula gives $D = \frac{1}{4}(x_2 - x_1)^2$, which is right for the two numbers x_1, x_2, whose mean is their midpoint.
- Take $x_1, x_2, x_3 = 0, 0, 3$, so $A = 1$, $B = 2$, and add $x_4 = 1 = A$. Directly, $D = \frac{1}{4}(1 + 1 + 4 + 0) = \frac{3}{2} = \frac{3}{4}B$, the minimum value.
- We checked all three results exactly (with fractions) on 20 000 random lists, including the equivalence $D < B \iff (x_{n+1} - A)^2 < (n+1)B/n$, and confirmed $D = B$ at both endpoints of the interval in 216 cases where they are rational.

Common slips

- Forgetting that $\sum_{k=1}^{n} A^2 = nA^2$, not A^2, when expanding in part (ii).
- Writing D in terms of $\sum x_k^2$ and stopping there: the question wants it in terms of B, A, x_{n+1} and n, so the sum must be replaced using part (ii).
- Square-rooting $(x_{n+1} - A)^2 < k$ to get $x_{n+1} - A < \sqrt{k}$ only, losing the lower end of the interval.

Beyond STEP

The formula for D is an *updating formula*: it lets a computer maintain the mean and variance of a data stream one value at a time without storing all the data, an idea refined in Welford's algorithm. The term $n(x_{n+1} - A)^2/(n+1)^2$ is a small instance of the decomposition of variance into "within-group" and "between-group" parts that

underlies analysis of variance in statistics. For more on this, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 2010, Question 2

Pure • Definite Integrals

STEP III 2010, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q2

The key idea

Every denominator here is a quadratic (in x, or in x^2) whose roots turn out to be $\pm e^a$ and $\pm e^{-a}$. Writing the hyperbolic coefficient in exponentials makes the factorisation visible at once, because e^a and e^{-a} have product 1 and sum $2\cosh a$ (or difference $2\sinh a$). After that, partial fractions reduce everything to logarithms or arctangents, and the work is in tidying the answer into hyperbolic form.

Solution

Part (i)

By definition,

$$\cosh a = \tfrac{1}{2}\left(e^a + e^{-a}\right).$$

So $2\cosh a = e^a + e^{-a}$ and $e^a \cdot e^{-a} = 1$, which means

$$x^2 + 2x\cosh a + 1 = (x + e^a)(x + e^{-a}).$$

Both factors are positive for $x \geq 0$, so the integrand is continuous on $[0, 1]$. Since $a > 0$, the two roots $-e^a$ and $-e^{-a}$ are distinct, and the partial fractions are

$$\frac{1}{(x+e^{-a})(x+e^a)} = \frac{1}{e^a - e^{-a}}\left(\frac{1}{x+e^{-a}} - \frac{1}{x+e^a}\right) = \frac{1}{2\sinh a}\left(\frac{1}{x+e^{-a}} - \frac{1}{x+e^a}\right),$$

as you can check by recombining: the numerator becomes $(x+e^a) - (x+e^{-a}) = e^a - e^{-a}$. Integrating,

$$\begin{aligned}\int_0^1 \frac{\mathrm{d}x}{x^2 + 2x\cosh a + 1} &= \frac{1}{2\sinh a}\left[\ln\frac{x+e^{-a}}{x+e^a}\right]_0^1 \\ &= \frac{1}{2\sinh a}\left(\ln\frac{1+e^{-a}}{1+e^a} - \ln\frac{e^{-a}}{e^a}\right).\end{aligned}$$

The first logarithm simplifies nicely: $1+e^{a}=e^{a}(e^{-a}+1)$, so $\dfrac{1+e^{-a}}{1+e^{a}}=e^{-a}$. The second is $\ln e^{-2a}=-2a$. Hence

$$\int_0^1 \frac{\mathrm{d}x}{x^2+2x\cosh a+1}=\frac{1}{2\sinh a}\left(-a+2a\right)=\frac{a}{2\sinh a},$$

which is the required result.

Part (ii)

The first integral. Now the coefficient is $2\sinh a=e^{a}-e^{-a}$ and the constant term is $-1=e^{a}\cdot(-e^{-a})$, so

$$x^2+2x\sinh a-1=(x+e^{a})(x-e^{-a}).$$

> **Ask yourself** Is the integrand well behaved on the whole range $[1,\infty)$?

The roots are $-e^{a}<0$ and e^{-a}, and $e^{-a}<1$ because $a>0$. So for $x\geq 1$ both factors are positive and the integrand is continuous. For large x it behaves like $1/x^2$, so the improper integral converges. Partial fractions as before, now with $e^{a}-(-e^{-a})=2\cosh a$:

$$\frac{1}{(x-e^{-a})(x+e^{a})}=\frac{1}{2\cosh a}\left(\frac{1}{x-e^{-a}}-\frac{1}{x+e^{a}}\right).$$

Integrating from 1 to X and letting $X\to\infty$,

$$\int_1^X \frac{\mathrm{d}x}{x^2+2x\sinh a-1}=\frac{1}{2\cosh a}\left(\ln\frac{X-e^{-a}}{X+e^{a}}-\ln\frac{1-e^{-a}}{1+e^{a}}\right).$$

As $X\to\infty$ the fraction $\frac{X-e^{-a}}{X+e^{a}}\to 1$, so the first logarithm tends to 0. For the second, take out e^{-a} from the numerator:

$$\frac{1-e^{-a}}{1+e^{a}}=e^{-a}\,\frac{e^{a}-1}{e^{a}+1}=e^{-a}\tanh\tfrac{a}{2},$$

using $\dfrac{e^{a}-1}{e^{a}+1}=\dfrac{e^{a/2}-e^{-a/2}}{e^{a/2}+e^{-a/2}}=\tanh\frac{a}{2}$ (divide top and bottom by $e^{a/2}$). So the second logarithm is $-a+\ln\tanh\frac{a}{2}$, and

$$\int_1^\infty \frac{\mathrm{d}x}{x^2+2x\sinh a-1}=\frac{a-\ln\tanh\frac{a}{2}}{2\cosh a}=\frac{a+\ln\coth\frac{a}{2}}{2\cosh a}.$$

Both terms in the numerator are positive (since $0<\tanh\frac{a}{2}<1$), as they must be for a positive integrand.

The second integral. This is the quadratic of part (i) with x replaced by x^2:

$$x^4+2x^2\cosh a+1=(x^2+e^{a})(x^2+e^{-a}),$$

and both factors are positive for all real x, so the integrand is continuous, and it is $O(1/x^4)$ at infinity, so the integral converges. Treating x^2 as the variable in the same partial fractions,

$$\frac{1}{(x^2+e^{-a})(x^2+e^{a})} = \frac{1}{2\sinh a}\left(\frac{1}{x^2+e^{-a}} - \frac{1}{x^2+e^{a}}\right).$$

Each piece is a standard arctangent integral: for $c>0$,

$$\int_0^\infty \frac{\mathrm{d}x}{x^2+c^2} = \left[\frac{1}{c}\arctan\frac{x}{c}\right]_0^\infty = \frac{\pi}{2c}.$$

With $c = e^{-a/2}$ and $c = e^{a/2}$ respectively (each piece converges on its own, so we may split the integral),

$$\int_0^\infty \frac{\mathrm{d}x}{x^4+2x^2\cosh a+1} = \frac{1}{2\sinh a}\cdot\frac{\pi}{2}\left(e^{a/2}-e^{-a/2}\right) = \frac{\pi}{2}\cdot\frac{2\sinh\frac{a}{2}}{2\sinh a}.$$

Finally, the double-angle formula $\sinh a = 2\sinh\frac{a}{2}\cosh\frac{a}{2}$ cancels the $\sinh\frac{a}{2}$:

$$\int_0^\infty \frac{\mathrm{d}x}{x^4+2x^2\cosh a+1} = \frac{\pi}{4\cosh\frac{a}{2}}.$$

Ask yourself Is there a way to see the $\cosh\frac{a}{2}$ coming?

Yes. Call the integral I. The substitution $x\mapsto 1/x$ shows $I = \int_0^\infty \frac{x^2}{x^4+2x^2\cosh a+1}\,\mathrm{d}x$ as well, so

$$2I = \int_0^\infty \frac{1+1/x^2}{x^2+1/x^2+2\cosh a}\,\mathrm{d}x.$$

With $u = x - 1/x$ we have $\mathrm{d}u = (1+1/x^2)\,\mathrm{d}x$ and $x^2+1/x^2 = u^2+2$, while u runs from $-\infty$ to ∞. The denominator becomes $u^2+2+2\cosh a = u^2+4\cosh^2\frac{a}{2}$, so $2I = \pi/(2\cosh\frac{a}{2})$, the same answer.

Checks

- As $a\to 0$, part (i) becomes $\int_0^1 (1+x)^{-2}\,\mathrm{d}x = \frac{1}{2}$, and $a/(2\sinh a)\to\frac{1}{2}$. The last integral becomes $\int_0^\infty (1+x^2)^{-2}\,\mathrm{d}x = \frac{\pi}{4}$, matching $\pi/(4\cosh 0)$.
- Numerically, with $a=1$, Simpson's rule gives 0.4254590641 for part (i), 0.5741556186 for the first integral of (ii) and 0.6965059227 for the second, agreeing with $\frac{1}{2\sinh 1}$, $\frac{1-\ln\tanh\frac{1}{2}}{2\cosh 1}$ and $\frac{\pi}{4\cosh\frac{1}{2}}$ to ten decimal places.

Common slips

- Getting the partial-fraction constant wrong: it is $1/(\text{difference of the roots})$, which is $1/(2\sinh a)$ in part (i) but $1/(2\cosh a)$ in the first integral of (ii).

- Forgetting to check that $e^{-a} < 1$, so that the root of the denominator lies outside $[1, \infty)$.
- Leaving the answers in exponentials: the question asks for hyperbolic functions, so $\frac{e^a-1}{e^a+1}$ should become $\tanh \frac{a}{2}$.

Beyond STEP

The second integral of (ii) is usually done at university by contour integration: the integrand has simple poles at $\pm ie^{\pm a/2}$, and summing residues in the upper half-plane gives $\pi/(4\cosh \frac{a}{2})$ in a few lines. The same method handles $\int_0^\infty \mathrm{d}x/(1+x^n)$ and many integrals that resist real methods. Needham's *Visual Complex Analysis* gives a geometric account of residues and why they work.

WORKED SOLUTION

STEP III 2010, Question 3

Pure • Roots of Unity

STEP III 2010, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 6.1, *Polynomials Whose Roots Are Roots of Unity*: https://beyondstep.keithmcnulty.org/ch06/cyclotomic.html

The key idea

Every root of unity a has a single *order*: the smallest positive m with $a^m = 1$. Being a primitive nth root of unity just means having order exactly n. So we find $\mathrm{C}_n(x)$ by listing the nth roots of unity $\mathrm{e}^{2\pi \mathrm{i} k/n}$, throwing away those that already return to 1 at an earlier power, and multiplying out $(x-a)$ over what is left. The fact that each number has only one order is also what settles the final part.

Solution

Throughout, the nth roots of unity are ω^k for $k = 0, 1, \ldots, n-1$, where $\omega = \mathrm{e}^{2\pi \mathrm{i}/n}$, and $(\omega^k)^m = 1$ exactly when n divides km. Since C_n is monic with no repeated roots, and its roots are exactly the primitive nth roots of unity,

$$\mathrm{C}_n(x) = \prod_{a \text{ primitive}} (x - a).$$

The 4th roots

The 4th roots of unity are $1, \mathrm{i}, -1, -\mathrm{i}$. Now $1^1 = 1$ and $(-1)^2 = 1$, so these two are not primitive. But $\mathrm{i}^1 = \mathrm{i}$, $\mathrm{i}^2 = -1$, $\mathrm{i}^3 = -\mathrm{i}$, none equal to 1, and similarly for $-\mathrm{i}$ (its powers are the conjugates). So the primitive 4th roots of unity are i and $-\mathrm{i}$, and

$$\mathrm{C}_4(x) = (x - \mathrm{i})(x + \mathrm{i}) = x^2 + 1,$$

as required.

Part (i)

$n = 1$. The only 1st root of unity is 1, and there is no m with $0 < m < 1$, so 1 is primitive: $C_1(x) = x - 1$.

$n = 2$. The square roots of unity are ± 1; 1 is excluded since $1^1 = 1$, while $(-1)^1 \neq 1$. So $C_2(x) = x + 1$.

$n = 3$ **and** $n = 5$. When n is prime, a root ω^k with $1 \leq k \leq n-1$ satisfies $(\omega^k)^m = 1$ only when $n \mid km$, and since n is prime and does not divide k, this forces $n \mid m$. So every nth root of unity other than 1 is primitive. These are the roots of

$$\frac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \cdots + x + 1,$$

which is monic, with distinct roots. Hence

$$C_3(x) = x^2 + x + 1, \qquad C_5(x) = x^4 + x^3 + x^2 + x + 1.$$

$n = 6$. With $\omega = e^{\pi i/3}$, the power ω^k has order $6/\gcd(k, 6)$. So $\omega^0 = 1$ has order 1, $\omega^3 = -1$ has order 2, ω^2, ω^4 have order 3, and only ω and $\omega^5 = \overline{\omega}$ have order 6. Their sum is $2\cos\frac{\pi}{3} = 1$ and their product is $|\omega|^2 = 1$, so

$$C_6(x) = x^2 - x + 1.$$

(As a cross-check, every 6th root of unity has order 1, 2, 3 or 6, so $x^6 - 1 = C_1C_2C_3C_6$, and indeed $(x-1)(x+1)(x^2+x+1)(x^2-x+1) = (x^3-1)(x^3+1) = x^6 - 1$.)

Part (ii)

Let a be any root of $x^4 + 1 = 0$. Then $a^4 = -1$, so $a^8 = 1$. The order d of a divides 8 (write $8 = qd + r$ with $0 \leq r < d$; then $a^r = a^8(a^d)^{-q} = 1$, so $r = 0$ by minimality of d). So $d \in \{1, 2, 4, 8\}$. If d were 1, 2 or 4 then $a^4 = 1$, contradicting $a^4 = -1$. Hence every root of $x^4 + 1$ is a primitive 8th root of unity.

Conversely, ω^k with $\omega = e^{\pi i/4}$ has order 8 exactly when k is odd (if k is even, $(\omega^k)^4 = 1$), so there are exactly four primitive 8th roots: $\omega, \omega^3, \omega^5, \omega^7$. The quartic $x^4 + 1$ has four distinct roots (they are $e^{(2j+1)\pi i/4}$, $j = 0, 1, 2, 3$), all primitive 8th roots, so they are precisely these four. Thus $C_8(x) = x^4 + 1$.

No other n works: the roots of C_n all have order n, and the roots of $x^4 + 1$ have order 8. So the answer is $n = 8$.

Part (iii)

The argument used for $n = 3$ and $n = 5$ in part (i) applies to any prime p: every pth root of unity except 1 is primitive. Therefore

$$C_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1.$$

(For $p = 2$ this gives $x + 1$, agreeing with part (i).)

Part (iv)

Suppose, for contradiction, that $C_q(x) \equiv C_r(x)C_s(x)$ for some positive integers q, r, s.

First, C_s has at least one root, since $e^{2\pi i/s}$ is a primitive sth root of unity: $(e^{2\pi i/s})^m = e^{2\pi i m/s} = 1$ only when $s \mid m$. The same goes for C_r. So C_r and C_s both have degree at least 1.

Take any root a of C_r. By the identity, $C_q(a) = C_r(a)C_s(a) = 0$, so a is a primitive qth root of unity as well as a primitive rth root of unity.

> **Ask yourself** Can one number be a primitive rth root of unity and a primitive qth root of unity with $r \neq q$?

No. If, say, $r < q$, then $a^r = 1$ with $0 < r < q$, which contradicts a being a primitive qth root. If $q < r$, swap the roles. So $r = q$, and in the same way (using a root of C_s) $s = q$.

But then the identity reads $C_q = C_q^2$. Comparing degrees, $\deg C_q = 2 \deg C_q$, so $\deg C_q = 0$, contradicting the fact that C_q has a root. (Alternatively, C_q^2 has repeated roots, which C_q does not.) Hence no such q, r, s exist.

Checks

- The degree of C_n counts the k in $1 \leq k \leq n$ with $\gcd(k, n) = 1$: degrees $1, 1, 2, 2, 4, 2$ for $n = 1, \ldots, 6$, and 4 for $n = 8$, all matching.
- Every nth root of unity has some order d dividing n, so $x^n - 1$ is the product of C_d over the divisors d of n. For $n = 4$: $(x-1)(x+1)(x^2+1) = x^4 - 1$; for $n = 8$: $(x^4-1)(x^4+1) = x^8 - 1$.
- Numerically, multiplying out $(x - a)$ over the primitive roots for $n \leq 40$ reproduces every polynomial above, gives $x^4 + 1$ only for $n = 8$, and finds no identity $C_q = C_r C_s$ with $q, r, s \leq 30$.

Common slips

- Including 1 (or -1) among the primitive roots: C_n is not $x^n - 1$.
- In part (ii), checking only that $x^4 + 1$ divides $x^8 - 1$ without ruling out smaller orders: the point is that $a^4 = -1 \neq 1$.
- In part (iv), arguing from degrees alone: $\deg C_q = \deg C_r + \deg C_s$ can happen (e.g. $4 = 2 + 2$), so the roots, not the degrees, give the contradiction.

Beyond STEP

The C_n are the *cyclotomic polynomials*. The identity $x^n - 1 = \prod_{d \mid n} C_d(x)$ shows they always have integer coefficients, and a deeper theorem says each is irreducible over the

rationals; they are central to Galois theory and to Gauss's discovery of which regular polygons can be constructed with ruler and compass. Allenby's *Rings, Fields and Groups* develops this algebra, and Needham's *Visual Complex Analysis* gives a geometric picture of roots of unity.

WORKED SOLUTION

STEP III 2010, Question 4

Pure • Common Roots of Polynomials

STEP III 2010, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q4

The key idea

If α satisfies two quadratics with the same leading coefficient, it also satisfies their difference, and the x^2 terms cancel to leave a *linear* equation. That pins α down, and putting it back into one quadratic gives the condition. In part (ii) the same trick works once we divide the cubic by the quadratic: on the roots of the quadratic, the cubic behaves exactly like its remainder, so we are back to two quadratics and can quote part (i).

Solution

Throughout, "root" means a root in the complex numbers, so every quadratic has two roots (counted with multiplicity).

Part (i)

Suppose α satisfies both equations. Subtracting the second from the first,

$$(a-c)\alpha + (b-d) = 0.$$

Since $a \neq c$ we may divide by $a-c$, giving

$$\alpha = -\frac{b-d}{a-c},$$

as required.

The condition. Write $k = a-c \neq 0$ and $m = b-d$, so the condition is $m^2 - amk + bk^2 = 0$.

($\Rightarrow$) If there is a common root, it is $\alpha = -m/k$ by the first result. Substituting into $\alpha^2 + a\alpha + b = 0$,

$$\frac{m^2}{k^2} - \frac{am}{k} + b = 0,$$

and multiplying by $k^2 \neq 0$ gives $m^2 - amk + bk^2 = 0$.

($\Leftarrow$) Suppose $m^2 - amk + bk^2 = 0$, and *define* $\alpha = -m/k$. Dividing the condition by k^2 and reading the step above backwards shows that $\alpha^2 + a\alpha + b = 0$. Also $(a-c)\alpha + (b-d) = k\alpha + m = 0$ by the choice of α. So

$$\alpha^2 + c\alpha + d = (\alpha^2 + a\alpha + b) - \big((a-c)\alpha + (b-d)\big) = 0 - 0 = 0,$$

and α is a common root. This proves the equivalence when $a \neq c$.

Ask yourself Why isn't the forward direction enough on its own?

The forward direction only says that *if* a common root exists, the condition holds. For "if and only if" we must also show that the condition produces a common root. That is why we defined α from the condition and checked that it satisfies *both* equations.

Dropping $a \neq c$. Suppose $a = c$. Then the condition reads $(b-d)^2 = 0$, that is, $b = d$.

- If $b = d$, the two equations are identical, so they certainly have a common root (both of their roots are common), and the condition holds.
- If $b \neq d$, a common root α would give $(a-c)\alpha + (b-d) = b - d = 0$ on subtracting, which is a contradiction. So there is no common root, and the condition fails.

In both cases "common root" and "condition" agree, so **yes, the result still holds** without the condition $a \neq c$.

(The one subtlety is when only real roots count. If $a = c$, $b = d$ and $a^2 < 4b$, the condition holds, yet the identical equations have no real roots. When $a \neq c$ there is no such problem, since for real coefficients the common root $-(b-d)/(a-c)$ is automatically real.)

Part (ii)

Divide the cubic by $x^2 + ax + b$. Since $(x+1)(x^2 + ax + b) = x^3 + (a+1)x^2 + (a+b)x + b$,

$$x^3 + (a+1)x^2 + qx + r = (x+1)(x^2 + ax + b) + (q - a - b)x + (r - b).$$

So if $x^2 + ax + b = 0$ at $x = \alpha$, the cubic takes the value $(q - a - b)\alpha + (r - b)$ there, and so does the quadratic

$$(x^2 + ax + b) + (q - a - b)x + (r - b) = x^2 + (q - b)x + r.$$

Hence α is a common root of the quadratic and the cubic if and only if it is a common root of

$$x^2 + ax + b = 0 \qquad \text{and} \qquad x^2 + (q-b)x + r = 0.$$

Now apply part (i) with $c = q - b$ and $d = r$. Then $a - c = a + b - q$ and $b - d = b - r$, so there is a common root if and only if

$$(b-r)^2 - a(b-r)(a+b-q) + b(a+b-q)^2 = 0.$$

Because the part (i) result holds whether or not $a = c$, no extra condition is needed here.

The example. Divide both equations by 2 to make them monic:

$$x^2 + \tfrac{5}{2}x + b = 0, \qquad x^3 + \tfrac{7}{2}x^2 + \tfrac{5}{2}x + \tfrac{1}{2} = 0.$$

These have the required form with $a = \frac{5}{2}$ (and indeed $a + 1 = \frac{7}{2}$), $q = \frac{5}{2}$ and $r = \frac{1}{2}$. Then $a + b - q = b$ and $b - r = b - \frac{1}{2}$, so the condition becomes

$$\left(b - \tfrac{1}{2}\right)^2 - \tfrac{5}{2}\left(b - \tfrac{1}{2}\right)b + b \cdot b^2 = 0.$$

Multiplying by 4:

$$\begin{aligned}(2b-1)^2 - 5b(2b-1) + 4b^3 &= 0\\ 4b^3 - 6b^2 + b + 1 &= 0.\end{aligned}$$

By inspection $b = 1$ is a root, and

$$4b^3 - 6b^2 + b + 1 = (b-1)(4b^2 - 2b - 1).$$

The quadratic factor gives $b = \dfrac{2 \pm \sqrt{4+16}}{8} = \dfrac{1 \pm \sqrt{5}}{4}$. So the equations have a common root exactly when

$$b = 1 \qquad \text{or} \qquad b = \frac{1+\sqrt{5}}{4} \qquad \text{or} \qquad b = \frac{1-\sqrt{5}}{4}.$$

Otherwise. The cubic here has no b in it, so we can just solve it. Trying $x = -\frac{1}{2}$ gives $-\frac{1}{4} + \frac{7}{4} - \frac{5}{2} + 1 = 0$, so

$$2x^3 + 7x^2 + 5x + 1 = (2x+1)(x^2 + 3x + 1),$$

with roots $-\frac{1}{2}$ and $\frac{1}{2}(-3 \pm \sqrt{5})$. A common root x forces $2b = -(2x^2 + 5x)$. For $x = -\frac{1}{2}$ this gives $b = 1$. For the other two roots, $x^2 = -3x - 1$, so $2x^2 + 5x = -x - 2$ and $b = \frac{1}{2}(x+2) = \frac{1}{4}(1 \pm \sqrt{5})$. This agrees with the answer above.

Checks

- With $b = 1$, the quadratic $2x^2 + 5x + 2 = (2x+1)(x+2)$ has the root $-\frac{1}{2}$, which is a root of the cubic.
- $b = \frac{1}{4}(1+\sqrt{5}) \approx 0.8090$: the quadratic's roots are ≈ -0.3820 and -2.1180, and $-0.3820 \approx \frac{1}{2}(-3+\sqrt{5})$ is a root of the cubic.
- A computer test of 20 000 random integer choices of a, b, c, d (and of a, b, q, r) found that the condition vanishes exactly when the equations share a complex root, including the cases with $a = c$.

Common slips

- Proving only "common root $\Rightarrow$ condition" and forgetting the converse.
- Answering the $a = c$ question with "no, we divided by $a - c$". The proof breaks down, but the *result* survives, because both sides reduce to $b = d$.
- Applying part (ii) without first dividing by 2: the formula needs monic equations.

Beyond STEP

The expression in part (i) is the *resultant* of the two quadratics: a polynomial in the coefficients that vanishes exactly when the polynomials share a root. Resultants come from Sylvester's determinant, and they underlie the elimination theory used to solve systems of polynomial equations. The "divide and keep the remainder" step in part (ii) is the first step of the Euclidean algorithm for polynomials, which finds the highest common factor of two polynomials just as it does for integers. Allenby's *Rings, Fields and Groups* develops polynomial division and highest common factors in this setting.

WORKED SOLUTION

STEP III 2010, Question 5

Pure • Coordinate Geometry Construction

STEP III 2010, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q5

The key idea

Coordinates do all the work here: find R and S, write down the lines PQ and RS in intercept form, and look at where each of them crosses the line through A perpendicular to AC. They cross it at the *same* point, and that point must be T. The construction then rests on one observation: the diagonal of the given square is exactly the side we need, and the first part gives us a straight-edge-only way to draw a right angle to that diagonal.

Solution

Showing that $TA \perp AC$

Lines through $C(a,a)$ are easy to handle. The line CP joins (a,a) to $(an,0)$, so its gradient is $\dfrac{a}{a-an} = \dfrac{1}{1-n}$ (defined, since $n<1$) and its equation is

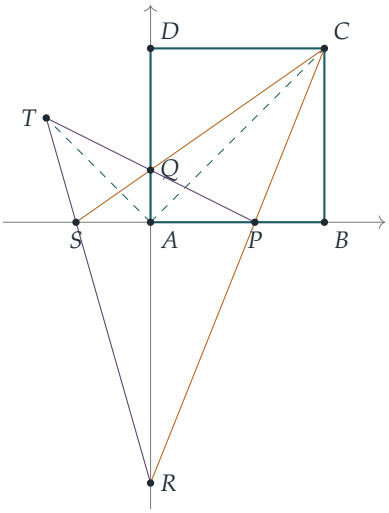
$$y = \frac{x-an}{1-n}.$$

The line DA is $x=0$, so

$$R = \left(0,\ -\frac{an}{1-n}\right).$$

Similarly CQ joins (a,a) to $(0,am)$, has gradient $1-m \neq 0$, and is $y = am + (1-m)x$. Setting $y=0$ (the line BA),

$$S = \left(-\frac{am}{1-m},\ 0\right).$$



Now use intercept form. The line PQ has intercepts an and am; the line RS has intercepts $-\frac{am}{1-m}$ and $-\frac{an}{1-n}$. So

$$\begin{aligned} PQ: &\quad \frac{x}{n}+\frac{y}{m}=a, \\ RS: &\quad -\frac{(1-m)x}{m}-\frac{(1-n)y}{n}=a. \end{aligned}$$

The line through A perpendicular to AC is $y=-x$, since AC has gradient 1. Put $y=-x$ into each equation:

$$\begin{aligned} PQ: &\quad x\Big(\frac{1}{n}-\frac{1}{m}\Big)=a, \\ RS: &\quad x\Big(-\frac{1}{m}+1+\frac{1}{n}-1\Big)=x\Big(\frac{1}{n}-\frac{1}{m}\Big)=a. \end{aligned}$$

These are the same equation. Because $m \neq n$, the bracket is non-zero, and both lines meet $y=-x$ at the single point

$$\Big(\frac{amn}{m-n},\ -\frac{amn}{m-n}\Big).$$

Ask yourself We've found a point that lies on both PQ and RS. Why must it be T?

Two distinct lines meet in at most one point. The lines PQ and RS are distinct (PQ passes through P with $x>0$ on the x-axis, while RS meets the x-axis at S with $x<0$). So any point common to both is their intersection T. Hence

$$T=\Big(\frac{amn}{m-n},\ -\frac{amn}{m-n}\Big),$$

which lies on $y=-x$. The vector $\overrightarrow{AT}$ is a multiple of $(1,-1)$ and $\overrightarrow{AC}=(a,a)$, and $(1,-1)\cdot(a,a)=0$. So TA is perpendicular to AC, as required.

Notice that the working never used $m < n$, only $m \neq n$. It also never used the actual positions of P and Q, so *any* two points P on side AB and Q on side AD, provided that $AP \neq AQ$, give a T on the perpendicular to AC at A.

The construction

A square of area $2a^2$ has side $\sqrt{2}\,a$, which is the length of the diagonal AC. So we build a square on AC as one side. In coordinates, this is the square with vertices

$$A(0,0), \quad E(a,-a), \quad F(2a,0), \quad C(a,a).$$

Its sides $\overrightarrow{AE} = (a,-a)$, $\overrightarrow{EF} = (a,a)$, $\overrightarrow{FC} = (-a,a)$ and $\overrightarrow{CA} = (-a,-a)$ all have length $\sqrt{2}\,a$, and consecutive sides are perpendicular, so its area is $2a^2$.

Ask yourself With an unmarked ruler we can only join points and extend lines. How can we possibly get a right angle?

The first part is exactly such a recipe, since every step in it joins two known points or intersects two lines. With the straight-edge:

1. Draw the diagonal AC. Mark any point P on AB and any point Q on AD, choosing them visibly unequal distances from A (no measurement is needed; we only need $AP \neq AQ$).
2. Extend CP to meet line DA at R, and extend CQ to meet line BA at S. Extend PQ and RS to meet at T.
3. Draw line TA. By the result above it is perpendicular to AC. Extend CB beyond B to meet line TA at E. (Line CB is $x = a$ and line TA is $y = -x$, so $E = (a,-a)$.)
4. Repeat steps 1–3 at the opposite corner C: mark points on CD and CB at visibly unequal distances from C and run the same construction. The half-turn about the centre of the square swaps A and C and maps the square to itself, so this gives the line through C perpendicular to AC, namely $x + y = 2a$.
5. Extend AB beyond B to meet this line at $F = (2a,0)$.

Then $AEFC$ is a square of side $AC = \sqrt{2}\,a$, so its area is $2a^2$. (Its diagonals AF and CE lie along lines AB and CB and cross at B, the centre of the new square.)

Checks

- Take $a = 1$, $m = 0.3$, $n = 0.6$. Then $R = (0,-1.5)$, $S = (-\frac{3}{7},0)$ and $T = (-0.6,0.6)$. The point T lies on the line through R and S: $-\frac{0.7}{0.3}(-0.6) - \frac{0.4}{0.6}(0.6) = 1.4 - 0.4 = 1$. It also lies on PQ: $-1 + 2 = 1$. And T is on $y = -x$.
- By computer we computed R, S and T exactly, using fractions, for 2000 random choices of a, m and n, in both orders $m < n$ and $m > n$. In every case $\overrightarrow{AT} \cdot \overrightarrow{AC} = 0$,

and the constructed $AEFC$ had four equal sides of squared length $2a^2$ meeting at right angles.

Common slips

- Solving PQ and RS simultaneously by brute force. It works, but it's messy. Testing a candidate line ($y = -x$) against both equations is far quicker.
- Forgetting to explain why the common point really is T. You need the lines to be distinct.
- In the construction, "drawing a perpendicular" or "marking off a length" without saying how. Every step must be joining points or intersecting lines.

Beyond STEP

Constructions using only a straight-edge belong to *projective geometry*, where joining and intersecting are the only operations, and ratios like the ones hidden in R, S and T are studied through cross-ratios and harmonic ranges. A classical theorem of Poncelet and Steiner says that a straight-edge alone can do everything a ruler and compass can, once a single circle and its centre are drawn; here the square plays a similar role. Coxeter's *Projective Geometry* is a beautiful introduction to the subject.

WORKED SOLUTION

STEP III 2010, Question 6

Pure • Rotations in Three Dimensions

STEP III 2010, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q6

The key idea

A rotation about an axis leaves every point on the axis where it is, and turns everything in the plane perpendicular to the axis exactly as a rotation of the plane would. So each step reduces to a two-dimensional rotation: find the plane being turned, write down an orthonormal pair of vectors in it, and rotate them. Any vector along the axis is simply left alone, and that one observation does a lot of the work for Q.

Solution

Write $\mathbf{i}, \mathbf{j}, \mathbf{k}$ for the unit vectors along the axes, so P_0, Q_0, R_0 have position vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$.

Part (i)

Rotating about the z-axis turns the x-y plane through ϕ, with $\mathbf{i}$ moving towards $\mathbf{j}$. Hence

$$P_1 = (\cos\phi,\ \sin\phi,\ 0).$$

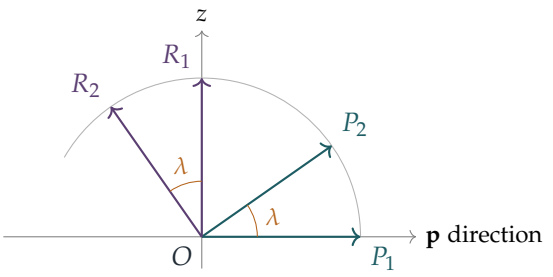
For later, the same rotation sends $\mathbf{j}$ to $Q_1 = (-\sin\phi, \cos\phi, 0)$, since Q stays a right angle ahead of P in that plane, and leaves R_0 fixed because it lies on the axis: $R_1 = (0, 0, 1)$.

Part (ii)

Let $\mathbf{p} = \overrightarrow{OP_1} = (\cos\phi, \sin\phi, 0)$. The line in the x-y plane perpendicular to OP_1 has direction

$$\mathbf{u} = (-\sin\phi,\ \cos\phi,\ 0),$$

since $\mathbf{u} \cdot \mathbf{p} = 0$ and $\mathbf{u} \cdot \mathbf{k} = 0$. The plane perpendicular to this axis is the plane spanned by the orthonormal pair $\mathbf{p}$ and $\mathbf{k}$, and the second rotation turns this plane through λ with $\mathbf{p}$ moving towards $\mathbf{k}$:



In that plane the rotation sends

$$\mathbf{p} \mapsto \cos\lambda\,\mathbf{p} + \sin\lambda\,\mathbf{k}, \qquad \mathbf{k} \mapsto -\sin\lambda\,\mathbf{p} + \cos\lambda\,\mathbf{k},$$

the second because $\mathbf{k}$ stays a right angle ahead of $\mathbf{p}$. Therefore

$$P_2 = (\cos\lambda\cos\phi,\ \cos\lambda\sin\phi,\ \sin\lambda), \qquad R_2 = (-\sin\lambda\cos\phi,\ -\sin\lambda\sin\phi,\ \cos\lambda).$$

Ask yourself Where does Q go in the second rotation?

After the first rotation Q is at $Q_1 = (-\sin\phi, \cos\phi, 0)$, which is exactly $\mathbf{u}$: it lies *on* the axis of the second rotation, so it does not move. Hence

$$Q_2 = (-\sin\phi,\ \cos\phi,\ 0).$$

As a check, P_2, Q_2, R_2 are mutually perpendicular unit vectors (for instance $\overrightarrow{OP_2} \cdot \overrightarrow{OR_2} = -\sin\lambda\cos\lambda\cos^2\phi - \sin\lambda\cos\lambda\sin^2\phi + \sin\lambda\cos\lambda = 0$), as they must be after rotations.

Part (iii)

Throughout this part we assume $P_2 \neq \pm P_0$, so that O, P_0, P_2 really do determine a plane.

Ask yourself Why is the angle of rotation the same as the angle P_2OP_0?

Since P stays in the plane OP_0P_2, the axis of the rotation must be the line through O perpendicular to that plane (a rotation moves a point in the plane perpendicular to the axis through that point). So P travels along the great circle in this plane, and every vector in the plane is turned through the angle of the rotation. That angle is therefore the angle between $\overrightarrow{OP_2}$ and $\overrightarrow{OP_0}$, and since both are unit vectors,

$$\cos\theta = \overrightarrow{OP_2} \cdot \overrightarrow{OP_0} = (\cos\lambda\cos\phi)(1) + (\cos\lambda\sin\phi)(0) + (\sin\lambda)(0) = \cos\phi\cos\lambda,$$

as required. (Turning the long way round, through $2\pi - \theta$, gives the same cosine.)

The axis is perpendicular to both $\overrightarrow{OP_2}$ and $\overrightarrow{OP_0}$, so it is along their cross product:

$$\overrightarrow{OP_2} \times \overrightarrow{OP_0} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\lambda\cos\phi & \cos\lambda\sin\phi & \sin\lambda \\ 1 & 0 & 0 \end{vmatrix} = (0,\ \sin\lambda,\ -\cos\lambda\sin\phi).$$

So a vector along the axis is

$$\begin{pmatrix} 0 \\ \sin\lambda \\ -\cos\lambda\sin\phi \end{pmatrix}.$$

With this orientation, a rotation through θ by the right-hand rule takes P_2 to P_0. Notice the axis has no x-component, as it must, since it is perpendicular to OP_0.

Checks

- If $\lambda = 0$ the second rotation does nothing, so the third just undoes the first: $\theta = \phi$ and the axis is along $(0, 0, -\sin\phi)$, the z-axis. The formulas agree.
- If $\phi = 0$ the only rotation was about the y-axis, and indeed the axis is along $(0, \sin\lambda, 0)$ and $\theta = \lambda$.
- Numerically, applying the rotations by Rodrigues' formula for 20 000 random (ϕ, λ) reproduces P_2, Q_2, R_2, and the rotation through θ about $(0, \sin\lambda, -\cos\lambda\sin\phi)$ returns P_2 to P_0. For $\phi = 0.7$, $\lambda = 0.4$, $\cos\theta \approx 0.7045$, so $\theta \approx 0.789$.

Common slips

- Rotating Q in the second step: it is on the axis, so it stays put.
- Getting the sign of R_2 wrong: as P tips up towards $\mathbf{k}$, R tips down and away from OP_1, so its horizontal part is $-\sin\lambda\,\mathbf{p}$.

Beyond STEP

The two angles here are longitude ϕ and latitude λ, and $\cos\theta = \cos\phi\cos\lambda$ is the right-angled case of the spherical cosine rule, used in navigation to find great-circle distances. More generally, the composition of two rotations is again a rotation about a single axis (Euler's rotation theorem), and the columns $\overrightarrow{OP_2}$, $\overrightarrow{OQ_2}$, $\overrightarrow{OR_2}$ form the orthogonal matrix of the combined rotation. Strang's *Introduction to Linear Algebra* treats rotation matrices and orthogonal matrices; Needham's *Visual Complex Analysis* shows how rotations of the sphere can be described with complex numbers.

WORKED SOLUTION

STEP III 2010, Question 7

Pure • Maclaurin Series

STEP III 2010, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q7

The key idea

Squaring away the awkward square root is what makes everything work: once we write $\sqrt{1-x^2}\,y'$ as a single tidy expression and differentiate again, the second-order equation falls out. After that, repeated differentiation produces a pattern in the coefficients, and induction proves it. Putting $x = 0$ into the general relation turns it into a recurrence for the Maclaurin coefficients, and for even m that recurrence hits a factor of zero, so the series stops.

Solution

The differential equation

Write $\theta = \arcsin x$, so $\sin\theta = x$ with $|\theta| < \frac{1}{2}\pi$, and $\cos\theta = \sqrt{1-x^2} > 0$ for $|x| < 1$. Then $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{1}{\sqrt{1-x^2}}$ and $y = \cos m\theta$, so by the chain rule

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{m\sin m\theta}{\sqrt{1-x^2}}, \qquad \text{that is,} \qquad \sqrt{1-x^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -m\sin m\theta.$$

Differentiate this with respect to x, using the product rule on the left:

$$\sqrt{1-x^2}\,\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{x}{\sqrt{1-x^2}}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{m^2\cos m\theta}{\sqrt{1-x^2}} = -\frac{m^2y}{\sqrt{1-x^2}}.$$

Multiplying through by $\sqrt{1-x^2}$ (which is non-zero) gives

$$(1-x^2)\,y'' - x\,y' + m^2y = 0,$$

as required. (Here and below $y^{(n)}$ means $\mathrm{d}^ny/\mathrm{d}x^n$.)

The third- and fourth-order equations

Differentiate the equation just proved. The product $(1-x^2)y''$ gives $(1-x^2)y''' - 2xy''$ and xy' gives $xy'' + y'$, so

$$(1-x^2)\,y''' - 3x\,y'' + (m^2-1)\,y' = 0.$$

Differentiating once more in the same way,

$$(1-x^2)\,y^{(4)} - 2x\,y''' - 3x\,y''' - 3y'' + (m^2-1)\,y'' = 0,$$

so

$$(1-x^2)\,y^{(4)} - 5x\,y''' + (m^2-4)\,y'' = 0.$$

The general relation

The coefficients of x run $1, 3, 5$ and the constants run $m^2-0^2,\ m^2-1^2,\ m^2-2^2$, so we conjecture that for every integer $n \geq 0$

$$(1-x^2)\,y^{(n+2)} - (2n+1)\,x\,y^{(n+1)} + (m^2-n^2)\,y^{(n)} = 0. \qquad (*)$$

We prove this by induction on n. The case $n=0$ is the equation proved first. Suppose $(*)$ holds for some n. Differentiating it,

$$\begin{aligned}(1-x^2)\,y^{(n+3)} - 2x\,y^{(n+2)} - (2n+1)\,x\,y^{(n+2)} \\ - (2n+1)\,y^{(n+1)} + (m^2-n^2)\,y^{(n+1)} = 0.\end{aligned}$$

Collecting terms, the coefficient of $x\,y^{(n+2)}$ is $-(2n+3) = -\big(2(n+1)+1\big)$ and the coefficient of $y^{(n+1)}$ is $m^2-(n^2+2n+1) = m^2-(n+1)^2$. So

$$(1-x^2)\,y^{(n+3)} - \big(2(n+1)+1\big)x\,y^{(n+2)} + \big(m^2-(n+1)^2\big)y^{(n+1)} = 0,$$

which is $(*)$ with $n+1$ in place of n. By induction, $(*)$ holds for all $n \geq 0$.

The Maclaurin series

At $x=0$ we have $\theta = 0$, so $y(0) = \cos 0 = 1$ and $y'(0) = -m\sin 0 = 0$. Putting $x=0$ in $(*)$ kills the middle term:

$$y^{(n+2)}(0) = (n^2-m^2)\,y^{(n)}(0).$$

Since $y'(0)=0$, every odd derivative vanishes at 0. For the even ones,

$$\begin{aligned}y''(0) &= (0-m^2)\cdot 1 = -m^2,\\ y^{(4)}(0) &= (2^2-m^2)(-m^2) = m^2(m^2-2^2),\\ y^{(6)}(0) &= (4^2-m^2)\,m^2(m^2-2^2) = -m^2(m^2-2^2)(m^2-4^2),\end{aligned}$$

and in general $y^{(2k)}(0) = (-1)^k m^2(m^2-2^2)(m^2-4^2)\cdots\big(m^2-(2k-2)^2\big)$. Hence the Maclaurin series is

$$y = 1 - \frac{m^2x^2}{2!} + \frac{m^2(m^2-2^2)\,x^4}{4!} - \frac{m^2(m^2-2^2)(m^2-4^2)\,x^6}{6!} + \cdots,$$

and the first three non-zero terms are $1 - \dfrac{m^2x^2}{2!} + \dfrac{m^2(m^2-2^2)x^4}{4!}$. (These are all non-zero provided $m \neq 0, \pm 2$; for those values the series has fewer terms, as the next part explains.)

Even m

Let m be an even integer, and write $|m| = 2p$. The factor $m^2 - (2p)^2$ is zero, and it appears in $y^{(2k)}(0)$ as soon as $2k - 2 \geq 2p$. So every derivative of order greater than $|m|$ is zero at $x = 0$, and the Maclaurin series stops at the $x^{|m|}$ term.

Ask yourself A series that stops at $x = 0$ only tells us about derivatives at one point. How do we know y really *equals* this polynomial for every $|x| < 1$?

We use $(*)$ again, now on the whole interval. Take $n = |m|$, so that $m^2 - n^2 = 0$, and put $v = y^{(n+1)}$. Then $(*)$ reads

$$(1 - x^2)\, v' - (2n+1)\, x\, v = 0 \qquad (|x| < 1).$$

This first-order equation has integrating factor $(1-x^2)^{(2n-1)/2}$; indeed

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big[(1-x^2)^{(2n+1)/2}\, v\Big] = (1-x^2)^{(2n-1)/2}\Big[(1-x^2)\, v' - (2n+1)\, x\, v\Big] = 0,$$

so $(1-x^2)^{(2n+1)/2}\, v$ is constant on $(-1, 1)$. Its value at $x = 0$ is $v(0) = y^{(n+1)}(0) = 0$, because $n+1$ is odd. Hence $v \equiv 0$, that is, $y^{(|m|+1)}(x) = 0$ for all $|x| < 1$, and so y is a polynomial of degree at most $|m|$. A polynomial equals its own Maclaurin series, so for $|x| < 1$

$$\cos(m \arcsin x) = 1 - \frac{m^2x^2}{2!} + \frac{m^2(m^2-2^2)\, x^4}{4!} - \cdots,$$

the right-hand side being the terminating series above.

Now let $|\theta| < \frac{1}{2}\pi$ and put $x = \sin\theta$. Then $|x| < 1$ and $\arcsin(\sin\theta) = \theta$ (this is exactly where the restriction on θ is used), so

$$\cos m\theta = 1 - \frac{m^2 \sin^2\theta}{2!} + \frac{m^2(m^2-2^2)\sin^4\theta}{4!} - \cdots,$$

a polynomial in $\sin\theta$ beginning as required.

Finally, the coefficient of $\sin^{|m|}\theta$ is $y^{(|m|)}(0)/|m|!$, which is $\pm\, m^2(m^2-2^2)\cdots\big(m^2 - (|m|-2)^2\big)/|m|!$. None of these factors is zero, since $(2j)^2 < m^2$ for $2j \leq |m| - 2$. So for $m \neq 0$ the polynomial has degree exactly $|m|$ (that is, degree m when m is a positive even integer); for $m = 0$ it is the constant 1.

Checks

- $m = 2$: the series gives $1 - 2\sin^2\theta$, which is the familiar $\cos 2\theta$.

- $m = 4$: the series gives $1 - 8\sin^2\theta + \frac{16\cdot12}{24}\sin^4\theta = 1 - 8\sin^2\theta + 8\sin^4\theta$. Directly, $\cos 4\theta = 1 - 2\sin^2 2\theta = 1 - 8\sin^2\theta\cos^2\theta = 1 - 8\sin^2\theta + 8\sin^4\theta$. They agree.
- We checked $(*)$ numerically for $n = 0, \ldots, 8$ at 200 random pairs (m, x), using Taylor expansions of $\cos(m \arcsin x)$ about each point (largest relative residual about 10^{-13}), and checked $\cos m\theta$ against the polynomial for every even m up to 12. For example, at $m = 6$, $\theta = 1$ both sides equal 0.96017.

Common slips

- Differentiating $(1 - x^2)y^{(n+2)}$ and forgetting the $-2x\,y^{(n+2)}$ term from the product rule: this is what turns $2n + 1$ into $2n + 3$.
- Assuming that a Maclaurin series which stops must equal the function. Here it does, but it needs the argument with $v = y^{(|m|+1)}$ (or another justification).
- Giving the degree as $m/2$: the polynomial contains only even powers of $\sin\theta$, but the highest power is $\sin^m\theta$.

Beyond STEP

Substituting $x = \cos\phi$ instead turns the same kind of equation into Chebyshev's equation $(1 - x^2)y'' - xy' + n^2y = 0$, whose polynomial solutions $T_n(\cos\phi) = \cos n\phi$ are the Chebyshev polynomials, central in numerical approximation. Our trick of putting $x = 0$ into the differentiated equation to get a recurrence for coefficients is the start of the *power series method* for linear ODEs, where a series terminating for special parameter values is how Legendre, Hermite and Chebyshev polynomials all arise. Boyce and DiPrima, *Elementary Differential Equations*, has a clear chapter on series solutions.

WORKED SOLUTION

STEP III 2010, Question 8

Pure • Quotient Rule Integration

STEP III 2010, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q8

The key idea

The expression $\mathrm{QR}' - \mathrm{Q}'\mathrm{R}$ is the numerator you get when you differentiate the quotient R/Q. So any integrand of the form P/Q^2 can be integrated at sight, provided we can find an R that produces the given P. In both parts the denominator tells us Q, and we find R by comparing coefficients. In part (ii) the left-hand side of the differential equation turns out to be exactly $\mathrm{Q}y' - \mathrm{Q}'y$, which is the same structure in disguise.

Solution

The general integral

By the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{R}(x)}{\mathrm{Q}(x)}\right) = \frac{\mathrm{Q}(x)\mathrm{R}'(x) - \mathrm{Q}'(x)\mathrm{R}(x)}{\big(\mathrm{Q}(x)\big)^2} = \frac{\mathrm{P}(x)}{\big(\mathrm{Q}(x)\big)^2},$$

so

$$\int \frac{\mathrm{P}(x)}{\big(\mathrm{Q}(x)\big)^2}\,\mathrm{d}x = \frac{\mathrm{R}(x)}{\mathrm{Q}(x)} + C.$$

Part (i)

Here $\mathrm{Q}(x) = 1 + 2x + 3x^2$, so $\mathrm{Q}'(x) = 2 + 6x$. With $\mathrm{R}(x) = a + bx + cx^2$ and $\mathrm{R}'(x) = b + 2cx$,

$$\begin{aligned}\mathrm{QR}' - \mathrm{Q}'\mathrm{R} &= (1 + 2x + 3x^2)(b + 2cx) - (2 + 6x)(a + bx + cx^2)\\ &= (b - 2a) + (2c - 6a)x + (2c - 3b)x^2.\end{aligned}$$

(The x^3 terms, $6cx^3$ from each product, cancel.) We need this to equal $5x^2 - 4x - 3$, so

$$b - 2a = -3, \qquad 2c - 6a = -4, \qquad 2c - 3b = 5.$$

The first two give $b = 2a - 3$ and $c = 3a - 2$. Substituting into the third,

$$2(3a-2) - 3(2a-3) = 6a - 4 - 6a + 9 = 5,$$

which holds *whatever* a is. So the third equation adds no new information, and every value of a works:

$$\mathrm{R}(x) = a + (2a-3)x + (3a-2)x^2.$$

Taking $a = 1$ gives the tidiest choice, $\mathrm{R}(x) = 1 - x + x^2$, and so

$$\int \frac{5x^2 - 4x - 3}{(1 + 2x + 3x^2)^2}\,\mathrm{d}x = \frac{1 - x + x^2}{1 + 2x + 3x^2} + C.$$

The denominator never vanishes (its discriminant is $4 - 12 < 0$), so this holds for all real x.

Non-uniqueness. We have just seen that there is a one-parameter family of valid R, so the choice is not unique: for instance $a = 0$ gives $\mathrm{R} = -3x - 2x^2$ and $a = 1$ gives $\mathrm{R} = 1 - x + x^2$, both of which work. Comparing the choice for a general a with the one for $a = 0$,

$$\big(a + (2a-3)x + (3a-2)x^2\big) - \big(-3x - 2x^2\big) = a + 2ax + 3ax^2 = a\,\mathrm{Q}(x).$$

? **Ask yourself** Why should adding a multiple of Q to R make no difference?

If we replace R by $\mathrm{R} + k\mathrm{Q}$ for a constant k, then

$$\mathrm{Q}(\mathrm{R} + k\mathrm{Q})' - \mathrm{Q}'(\mathrm{R} + k\mathrm{Q}) = \mathrm{QR}' - \mathrm{Q}'\mathrm{R} + k(\mathrm{QQ}' - \mathrm{Q}'\mathrm{Q}) = \mathrm{QR}' - \mathrm{Q}'\mathrm{R},$$

so P is unchanged. Correspondingly, the answer R/Q changes to $\mathrm{R}/\mathrm{Q} + k$: the different choices of R give antiderivatives that differ only by a constant, which is absorbed into the arbitrary constant of integration. For example,

$$\frac{1 - x + x^2}{1 + 2x + 3x^2} = \frac{-3x - 2x^2}{1 + 2x + 3x^2} + 1.$$

Part (ii)

Let $\mathrm{Q}(x) = 1 + \cos x + 2\sin x$. Then $\mathrm{Q}'(x) = -\sin x + 2\cos x$, so the coefficient of y in the equation, $\sin x - 2\cos x$, is exactly $-\mathrm{Q}'(x)$. The equation is therefore

$$\mathrm{Q}\frac{\mathrm{d}y}{\mathrm{d}x} - \mathrm{Q}'y = 5 - 3\cos x + 4\sin x.$$

On any interval where $\mathrm{Q} \neq 0$ we divide by Q^2 and recognise the left-hand side as a derivative of a quotient:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\mathrm{Q}}\right) = \frac{5 - 3\cos x + 4\sin x}{\mathrm{Q}^2}.$$

(In other words, $1/\mathrm{Q}^2$ is an integrating factor.) To integrate the right-hand side we look for an R with $\mathrm{QR}' - \mathrm{Q}'\mathrm{R} = 5 - 3\cos x + 4\sin x$. Mirroring the form of Q, try $\mathrm{R} = a + b\cos x + c\sin x$, so $\mathrm{R}' = -b\sin x + c\cos x$. Expanding,

$$\mathrm{QR}' = -b\sin x + c\cos x + (2c - b)\sin x\cos x + c\cos^2 x - 2b\sin^2 x,$$
$$\mathrm{Q}'\mathrm{R} = -a\sin x + 2a\cos x + (2c - b)\sin x\cos x + 2b\cos^2 x - c\sin^2 x.$$

Subtracting, the $\sin x\cos x$ terms cancel, and the squared terms combine as $(c - 2b)(\cos^2 x + \sin^2 x) = c - 2b$. So

$$\mathrm{QR}' - \mathrm{Q}'\mathrm{R} = (c - 2b) + (c - 2a)\cos x + (a - b)\sin x.$$

Matching with $5 - 3\cos x + 4\sin x$:

$$c - 2b = 5, \qquad c - 2a = -3, \qquad a - b = 4.$$

Subtracting the second equation from the first gives $2a - 2b = 8$, the third equation again, so as in part (i) there is a free parameter, for the same reason. Taking $b = 0$ gives $c = 5$, $a = 4$, so $\mathrm{R} = 4 + 5\sin x$. Hence

$$\frac{y}{\mathrm{Q}} = \frac{4 + 5\sin x}{1 + \cos x + 2\sin x} + C,$$

and the general solution is

$$y = 4 + 5\sin x + C(1 + \cos x + 2\sin x),$$

where C is an arbitrary constant. Although we divided by Q to find it, this function satisfies the equation for every x, as direct substitution confirms. Other choices of b just shift C: for example $b = -2$ gives $\mathrm{R} = 2 - 2\cos x + \sin x = (4 + 5\sin x) - 2\mathrm{Q}$.

Checks

- In part (i), $\int_0^1$ of the integrand should be $\frac{1}{6} - 1 = -\frac{5}{6}$ from our antiderivative; Simpson's rule gives $-0.83333\ldots$, and agreement holds over random intervals for every choice of a.
- In part (ii), substitute $y = 4 + 5\sin x$: the left-hand side is $(1 + \cos x + 2\sin x)5\cos x - (2\cos x - \sin x)(4 + 5\sin x) = 5 - 3\cos x + 4\sin x$. And $y = \mathrm{Q}$ gives $\mathrm{QQ}' - \mathrm{Q}'\mathrm{Q} = 0$, the complementary function.
- Solving the equation numerically (Runge–Kutta) from $y(0.5) = 1.7$ to $x = 1.5$ gives 3.910638, matching the formula with the corresponding C.

Common slips

- Expecting three equations in three unknowns to have a unique solution. Here they are dependent, and that is the whole point of the non-uniqueness question.

- Getting the sign of Q' wrong in part (ii): $\mathrm{Q}' = 2\cos x - \sin x$, so the y coefficient is $-\mathrm{Q}'$, not $+\mathrm{Q}'$.
- Forgetting that the constant of integration multiplies Q in the final answer: the general solution is $y = \mathrm{R} + C\mathrm{Q}$, not $y = \mathrm{R} + C$.

Beyond STEP

The equation in part (ii) is a first-order linear differential equation, and the general solution has the shape "particular solution plus multiple of the solution of the homogeneous equation", a pattern that runs through all linear problems. The expression $\mathrm{QR}' - \mathrm{Q}'\mathrm{R}$ is the Wronskian of Q and R, which plays a central role in second-order linear equations, notably in reduction of order and variation of parameters. Boyce and DiPrima, *Elementary Differential Equations*, covers both integrating factors and Wronskians.

WORKED SOLUTION

STEP III 2010, Question 9

Mechanics • Motion on a Circle

STEP III 2010, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

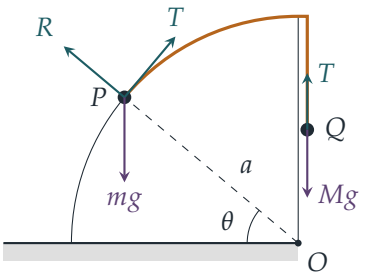
step.maths.org/questions/10-s3-q9

The key idea

Because the string is inextensible and the block is smooth, P and Q always move with the same speed, and the whole system conserves energy. That gives the speed of P as a function of θ with no differential equation to solve. Then P is moving on a circle, so resolving towards O gives the normal reaction. Staying in contact means that reaction stays non-negative, and a concavity argument shows that it's enough to check the top of the block.

Solution

The model. P and Q are particles, the string is light and inextensible, and the block is smooth, so the tension T is the same all along the string. Put P at angle θ on the arc. The string runs along the arc from P to the top of the block and then straight down beside the vertical face to Q.



Speed. When P has moved from $\theta = 0$ to θ it has travelled an arc of length $a\theta$, so Q has descended a distance $a\theta$. Both move with speed $v = a\dot{\theta}$. Meanwhile P has risen through a height $a\sin\theta$. The block is smooth and the string does no net work on the

system (it is inextensible), so energy is conserved. Starting from rest,

$$\tfrac{1}{2}(m+M)v^2 = Mga\theta - mga\sin\theta, \qquad \text{so} \qquad v^2 = \frac{2ga\,(M\theta - m\sin\theta)}{m+M}.$$

This is positive for $0 < \theta \leq \frac{1}{2}\pi$, since $M\theta > m\theta \geq m\sin\theta$, so the system really does move off from rest.

Normal reaction. P moves on a circle of radius a about O, so its acceleration towards O is v^2/a. The forces on P are as follows:

- the reaction R along OP, pointing away from O;
- the tension T along the tangent, which has no radial component;
- the weight mg, whose component towards O is $mg\sin\theta$, since OP makes angle θ with the horizontal.

Resolving towards O,

$$mg\sin\theta - R = \frac{mv^2}{a} = \frac{2mg\,(M\theta - m\sin\theta)}{m+M}.$$

Hence

$$\begin{aligned} R &= \frac{mg\big[(m+M)\sin\theta - 2M\theta + 2m\sin\theta\big]}{m+M} \\ &= \boxed{\frac{mg\big[(3m+M)\sin\theta - 2M\theta\big]}{m+M}}. \end{aligned}$$

Ask yourself R must be non-negative for every θ in $[0, \frac{1}{2}\pi]$. Do we have to search for the minimum of R?

The condition. P stays in contact exactly when $R \geq 0$ throughout, that is, when

$$f(\theta) = (3m+M)\sin\theta - 2M\theta \geq 0 \quad \text{for } 0 \leq \theta \leq \tfrac{1}{2}\pi.$$

We don't need the minimum. Since $f''(\theta) = -(3m+M)\sin\theta \leq 0$ on this interval, f is concave, so its graph lies on or above the chord joining its end points. We have $f(0) = 0$, so:

- *Necessity.* If P is still in contact at $\theta = \frac{1}{2}\pi$, then $f(\frac{1}{2}\pi) = 3m + M - \pi M \geq 0$.
- *Sufficiency.* If $f(\frac{1}{2}\pi) \geq 0$, then the chord from $(0,0)$ to $(\frac{1}{2}\pi, f(\frac{1}{2}\pi))$ lies on or above the axis. By concavity f lies on or above the chord, so $f \geq 0$ on the whole interval.

So the condition is $3m \geq (\pi - 1)M$, that is,

$$\frac{m}{M} \geq \frac{\pi - 1}{3},$$

as required. Note that $(\pi - 1)/3 \approx 0.714 < 1$, so this is consistent with $m < M$.

Checks

- At $\theta = 0$ we get $R = 0$. That's right: at the start P is at the foot of the block, the radius OP is horizontal, and P's weight is carried by the table.
- Resolving tangentially for P and vertically for Q gives $T = \dfrac{Mmg(1+\cos\theta)}{m+M} > 0$, so the string really is taut throughout.
- Numerically, with $m = 0.8$, $M = 1$ and $a = 1$ (ratio $0.8 > 0.714$), stepping the equation of motion and computing R from Newton's second law matches the formula to 10^{-13}, and R stays non-negative. With $m/M = \frac{2}{3}$, R is negative near the top.

Common slips

- Forgetting that Q also has kinetic energy, so writing $\frac{1}{2}mv^2$ instead of $\frac{1}{2}(m+M)v^2$.
- Putting the tension into the radial equation. On a smooth surface the string runs along the tangent at P.
- Checking $R \geq 0$ only at $\theta = \frac{1}{2}\pi$ without justifying why that is enough. The concavity of f is what makes it enough.

Beyond STEP

Here energy conservation stood in for solving the equation of motion $(m+M)a\ddot{\theta} = g(M - m\cos\theta)$. This is the idea of a first integral, which becomes central in Lagrangian mechanics: the string is a constraint, and one coordinate θ describes the whole system. The bead-leaving-a-sphere family of problems, of which this is one, is treated carefully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2010, Question 10

Mechanics • Small Oscillations

STEP III 2010, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q10

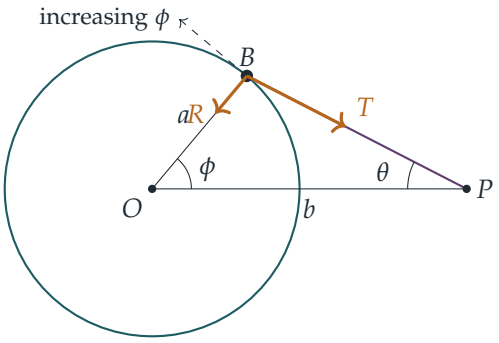
The key idea

The bead is confined to a circle, so only the component of force along the tangent to the ring changes its motion; the normal reaction (and gravity, which the ring's vertical reaction balances) never enters. Everything else is the geometry of the triangle OBP, and the sine rule turns that geometry into the string length and, for small angles, into the linear relation between θ and ϕ that the second part asks for.

Solution

The equation of motion

Model the bead as a particle on a smooth ring and the string as light and elastic. Take P on the positive x-axis and measure ϕ anticlockwise, so that B is at angle ϕ from OP, as in the figure. In triangle OBP the angle at O is ϕ, the angle at P is θ, and so the angle at B is $\pi - \theta - \phi$.



Let $l = BP$. By the sine rule in triangle OBP,

$$\frac{l}{\sin\phi} = \frac{a}{\sin\theta} = \frac{b}{\sin(\theta+\phi)},$$

using $\sin(\pi-\theta-\phi)=\sin(\theta+\phi)$. In particular $l=\dfrac{a\sin\phi}{\sin\theta}$.

The string is taut throughout, because the closest B can get to P is $b-a$, and $b-a>c$. So by Hooke's law the tension is

$$T=\frac{\lambda(l-c)}{c}=\lambda\left(\frac{a\sin\phi}{c\sin\theta}-1\right).$$

The forces in the horizontal plane are the tension T along BP and the normal reaction R of the smooth ring along the radius OB. (Vertically, the weight is balanced by the vertical part of the reaction.) The bead moves on a circle of radius a, so its acceleration along the tangent, in the direction of increasing ϕ, is $a\ddot{\phi}$. Only T has a tangential component.

The string BP makes angle $\pi-\theta-\phi$ with BO, which is the inward radius. So it makes angle $\theta+\phi$ with the outward radius, and its component along the tangent has size $T\sin(\theta+\phi)$. That component points towards P, which is the direction of *decreasing* ϕ. Resolving along the tangent,

$$ma\ddot{\phi}=-\lambda\left(\frac{a\sin\phi}{c\sin\theta}-1\right)\sin(\theta+\phi),$$

as required. (For $\phi<0$ the figure is reflected in OP; then $\theta<0$ too, and the same equation holds with signed angles.)

Small oscillations

The sine rule gave $a\sin(\theta+\phi)=b\sin\theta$ exactly. When θ and ϕ are small, $\sin x\approx x$ on both sides, so

$$a(\theta+\phi)\approx b\theta.$$

Hence $\theta\approx\dfrac{a\phi}{b-a}$ and $\theta+\phi\approx\dfrac{b\theta}{a}\approx\dfrac{b\phi}{b-a}$.

> **Ask yourself** What happens to $\dfrac{a\sin\phi}{c\sin\theta}$ as both angles go to zero? It looks like $0/0$.

That quotient is just l/c, and as B approaches the equilibrium position the length l tends to $b-a$. Equivalently, $\dfrac{a\sin\phi}{\sin\theta}\approx\dfrac{a\phi}{\theta}\approx b-a$ from the relation above. So to leading order the tension is the constant equilibrium tension

$$T_0=\lambda\left(\frac{b-a}{c}-1\right)=\frac{\lambda(b-a-c)}{c},$$

which is positive since $c<b-a$. The factor $\sin(\theta+\phi)$ is already small, so any correction to T only contributes at second order.

Substituting into the equation of motion and keeping first-order terms,

$$ma\ddot{\phi}\approx-\frac{\lambda(b-a-c)}{c}\cdot\frac{b\phi}{b-a},\qquad\text{so}\qquad\ddot{\phi}=-\omega^2\phi,\quad\omega^2=\frac{\lambda b(b-a-c)}{mac(b-a)}.$$

This is simple harmonic motion, and the period of small oscillations about $\theta = \phi = 0$ is

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{mac(b-a)}{\lambda b(b-a-c)}}.$$

Checks

- Dimensions: λ is a force, so $mac(b-a)/(\lambda b(b-a-c))$ has dimensions $\mathrm{M\,L}/(\mathrm{M\,L\,T}^{-2}) = \mathrm{T}^2$, as a period should.
- As $c \to b-a$ the equilibrium tension vanishes and the period becomes infinite: a slack string gives no restoring force, which makes sense.
- Numerically, with $a = 1$, $b = 2.5$, $c = 0.7$, $\lambda = 3$, $m = 0.5$, the formula gives a period of 1.8586. Stepping the exact motion (with the force computed directly from the position of B) from a small displacement gives the same value to six significant figures. The exact equation of motion also agrees with a direct vector resolution at random positions.

Common slips

- Getting the sign wrong: the tangential pull is towards P, that is towards $\phi = 0$, so it must appear with a minus sign.
- Taking the tension to be zero at equilibrium. The string is stretched to length $b-a > c$ there, so the tension is $\lambda(b-a-c)/c$, and it is this tension that supplies the restoring force.
- Using $\theta + \phi \approx \phi$. Here θ is of the same order as ϕ, so dropping it gives the wrong period.

Beyond STEP

The method here is the standard way to find the frequency of small oscillations: find an equilibrium, linearise the equation of motion about it, and read off ω^2. A neat feature is that the restoring force comes entirely from the *equilibrium* tension acting through the change of direction of the string, which is the same mechanism as a stretched guitar string. In a first-year course this becomes Lagrangian mechanics and normal modes; Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see small-oscillation analysis developed carefully.

WORKED SOLUTION

STEP III 2010, Question 11

Mechanics • Friction and Relative Motion

STEP III 2010, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

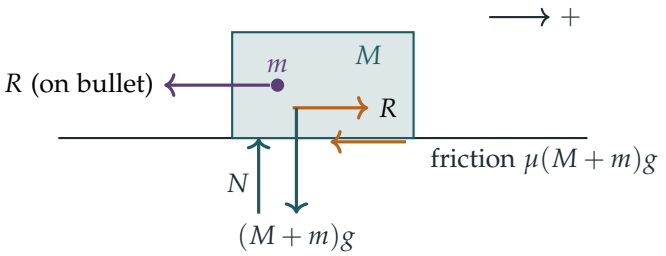
step.maths.org/questions/10-s3-q11

The key idea

Momentum is *not* conserved here, because friction from the ground acts on the block throughout, so the familiar "bullet embeds in block" shortcut is off the table. Instead, treat the bullet and the block as two separate particles, each with its own constant acceleration, linked by the equal and opposite resistance force R. Constant accelerations mean the *suvat* equations do all the work, and the relative motion is just the difference of the two.

Solution

We model the bullet and the block as particles moving along a horizontal line, with the bullet's direction of travel as positive. While the bullet is in the block, the block pushes back on the bullet with force R and, by Newton's third law, the bullet pushes the block forward with force R.



Normal reaction. Vertically nothing accelerates. The bullet is supported by the block, so the ground carries the weight of both: $N = (M+m)g$.

Friction. The forward force R on the block exceeds the greatest friction available, $\mu N = (M+m)\mu g$, since we are told $R > (M+m)\mu g$. So the block does slide, and once it moves friction takes its limiting value $\mu(M+m)g$, acting backwards.

Part (i)

The only horizontal force on the bullet is R backwards, so the bullet decelerates at R/m. The block has net forward force $R-(M+m)\mu g$, so it accelerates forwards at

$$A=\frac{R-(M+m)\mu g}{M}>0.$$

Relative to the block, the bullet's acceleration is $-\frac{R}{m}-A$, so the magnitude of its deceleration relative to the block is

$$a=\frac{R}{m}+\frac{R-(M+m)\mu g}{M},$$

as required.

Part (ii)

Relative to the block, the bullet starts with speed u and decelerates uniformly at a, so it stops relative to the block after time

$$t_1=\frac{u}{a}.$$

At that moment the block's speed is the common speed v. The block started from rest with constant acceleration A, so

$$v=At_1=\frac{Au}{a},\qquad\text{that is,}\qquad av=Au=\frac{Ru-(M+m)\mu gu}{M},$$

which is the required result.

Ask yourself Why not just use conservation of momentum to find v?

Momentum of the bullet–block system is conserved only if no external horizontal force acts. Here the ground's friction acts on the block for the whole time t_1, and removes momentum $(M+m)\mu g\,t_1$. (In a real bullet the time is tiny, so $v\approx mu/(M+m)$, but the question wants the exact value.)

Part (iii)

During time t_1 the block accelerates uniformly from rest to speed v, so its distance is the average speed times the time:

$$s_1=\frac{0+v}{2}\,t_1=\frac{uv}{2a}.$$

Part (iv)

Once the bullet has stopped moving through the block, the two move together as a single body of mass $M+m$. Its weight is still $(M+m)g$, so friction is $\mu(M+m)g$

backwards and the deceleration is μg. Starting at speed v, it slides a further

$$s_2 = \frac{v^2}{2\mu g}$$

before stopping (and then stays at rest, since no horizontal force remains to overcome static friction). The total distance is

$$s_1 + s_2 = \frac{uv}{2a} + \frac{v^2}{2\mu g} = \frac{uv}{2a} + \frac{v}{2\mu g} \cdot \frac{Au}{a} = \frac{uv}{2a\mu g}\left(\mu g + A\right),$$

where we used $v = Au/a$ from part (ii) for one factor of v.

Ask yourself How do we get rid of a and A and reach a clean answer?

Put both over the denominator mM and look for a common factor:

$$\mu g + A = \frac{M\mu g + R - (M+m)\mu g}{M} = \frac{R - m\mu g}{M},$$
$$a = \frac{MR + mR - m(M+m)\mu g}{mM} = \frac{(M+m)(R - m\mu g)}{mM}.$$

So $\dfrac{\mu g + A}{a} = \dfrac{m}{M+m}$, the factor $R - m\mu g$ cancelling (it is positive, since $R > (M+m)\mu g > m\mu g$). Hence the total distance moved by the block is

$$\frac{uv}{2\mu g} \cdot \frac{m}{M+m} = \frac{muv}{2(M+m)\mu g},$$

as required.

The case $R < (M+m)\mu g$

Now the forward push R on the block is less than the limiting friction $(M+m)\mu g$, so static friction can hold the block in place: the block never moves. The bullet simply decelerates at R/m inside a stationary block and comes to rest after penetrating a distance $\dfrac{mu^2}{2R}$ (assuming the block is thick enough), and the block stays put throughout.

Checks

- Take $m = 0.02\,\text{kg}$, $M = 2\,\text{kg}$, $u = 300\,\text{m s}^{-1}$, $\mu = 0.5$, $g = 9.8\,\text{m s}^{-2}$ and $R = 1000\,\text{N}$. Then $a \approx 50\,495\,\text{m s}^{-2}$, $v \approx 2.941\,\text{m s}^{-1}$, $s_1 \approx 0.0087\,\text{m}$, $s_2 \approx 0.8827\,\text{m}$, and $s_1 + s_2 \approx 0.8914\,\text{m}$, agreeing with $muv/\left(2(M+m)\mu g\right)$. A step-by-step numerical simulation of the two particles gives the same v and total distance.
- As $R \to \infty$ (an instant collision), $v = Au/a \to mu/(M+m)$, the momentum-conservation answer, as it should.
- Dimensions: $muv/\left((M+m)\mu g\right)$ is $(\text{speed})^2/(\text{acceleration})$, a length.

Common slips

- Using $N = Mg$ for the block, forgetting that the embedded bullet's weight is also carried by the ground.
- Conserving momentum through the impact, which ignores the friction impulse.
- Taking R as the net force on the block, rather than R minus friction.

Beyond STEP

Impulse gives a quick consistency check on the whole motion: the ground's friction impulse over the entire time the block moves must equal the bullet's initial momentum mu, since the system ends at rest. Separating a system into parts, tracking internal forces by Newton's third law and external ones carefully, is the core skill of first-year mechanics, and it leads on to the centre-of-mass frame and inelastic collisions. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas with many problems of exactly this flavour.

WORKED SOLUTION

STEP III 2010, Question 12

Probability and Statistics • Geometric Distribution Expectation

STEP III 2010, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q12

The key idea

The series S is an arithmetico-geometric series, and the trick of subtracting rS turns it into an ordinary geometric series. Once we have it, every expectation in the question is a sum of exactly this shape: the shot number grows linearly while the probability of getting that far shrinks geometrically. So the whole question comes down to writing each expectation as a sum and reading off the right values of r and d.

Solution

The series

Since $|r| < 1$, the series converges absolutely (its terms are bounded by a constant times $n|r|^n$), so we may subtract term by term. Lining up equal powers of r,

$$\begin{aligned} S &= 1 + (1+d)r + (1+2d)r^2 + (1+3d)r^3 + \cdots, \\ rS &= \quad\; r + (1+d)r^2 + (1+2d)r^3 + \cdots, \end{aligned}$$

and each coefficient of r^n with $n \geq 1$ differs by exactly d. Hence

$$(1-r)S = 1 + d\,(r + r^2 + r^3 + \cdots) = 1 + \frac{rd}{1-r},$$

summing the geometric series, which is valid as $|r| < 1$. Dividing by $1 - r \neq 0$,

$$S = \frac{1}{1-r} + \frac{rd}{(1-r)^2},$$

as required.

Arthur alone

We assume $0 < a \leq 1$ (if $a = 0$ Arthur never hits). Let N be the number of shots Arthur needs. Then $N = n$ exactly when his first $n-1$ shots miss and the nth hits, so by

independence

$$\mathrm{P}(N=n) = a'^{n-1}a, \qquad n = 1,2,3,\ldots$$

Therefore

$$\mathrm{E}(N) = \sum_{n=1}^{\infty} n\,a\,a'^{n-1} = a\big(1 + 2a' + 3a'^2 + \cdots\big).$$

The bracket is S with $r = a'$ and $d = 1$ (the coefficient of a'^n is $1+n$), and $0 \le a' < 1$, so

$$\mathrm{E}(N) = a\left(\frac{1}{1-a'} + \frac{a'}{(1-a')^2}\right) = a\left(\frac{1}{a} + \frac{a'}{a^2}\right) = \frac{a+a'}{a} = \frac{1}{a},$$

using $1 - a' = a$ and then $a + a' = 1$. (This formula also gives 1 when $a = 1$.)

The contest: who wins

From now on take $0 < a < 1$ and $0 < b < 1$, so that $0 < a'b' < 1$. Shots $1,3,5,\ldots$ are Arthur's and shots $2,4,6,\ldots$ are Boadicea's. Call one Arthur shot followed by one Boadicea shot a *round*; a round passes with nobody hitting with probability $a'b'$.

Arthur wins on shot $2k+1$ ($k = 0,1,2,\ldots$) exactly when the first k rounds pass with no hit and then Arthur hits. By independence this has probability $(a'b')^k a$, and these events are disjoint, so

$$\alpha = \sum_{k=0}^{\infty} (a'b')^k a = \frac{a}{1-a'b'}.$$

Similarly Boadicea wins on shot $2k+2$ when k rounds pass, Arthur misses and Boadicea hits, with probability $(a'b')^k a'b$. So

$$\beta = \sum_{k=0}^{\infty} (a'b')^k a'b = \frac{a'b}{1-a'b'} = \frac{(1-a)b}{1-(1-a)(1-b)}.$$

As a check, $1 - a'b' = a + b - ab = a + a'b$, so $\alpha + \beta = 1$: the contest ends with probability 1.

The contest: expected length

Let T be the number of shots in the contest. From the previous part, $T = 2k+1$ with probability $(a'b')^k a$ and $T = 2k+2$ with probability $(a'b')^k a'b$. Writing $q = a'b'$,

$$\mathrm{E}(T) = a\sum_{k=0}^{\infty}(2k+1)q^k + a'b\sum_{k=0}^{\infty}(2k+2)q^k.$$

Ask yourself How do these sums fit the series S?

The first sum is $1 + 3q + 5q^2 + \cdots$, which is S with $r = q$ and $d = 2$. The second is

twice $1 + 2q + 3q^2 + \cdots$, which is S with $r = q$ and $d = 1$. So

$$\sum_{k=0}^{\infty}(2k+1)q^k = \frac{1}{1-q} + \frac{2q}{(1-q)^2} = \frac{1+q}{(1-q)^2},$$
$$\sum_{k=0}^{\infty}(2k+2)q^k = 2\left(\frac{1}{1-q} + \frac{q}{(1-q)^2}\right) = \frac{2}{(1-q)^2}.$$

Hence

$$\mathrm{E}(T) = \frac{a(1+q) + 2a'b}{(1-q)^2}.$$

Now we compare with the target. Using $\alpha = a/(1-q)$ and $\beta = a'b/(1-q)$,

$$\frac{\alpha}{a} + \frac{\beta}{b} = \frac{1}{1-q} + \frac{a'}{1-q} = \frac{(1+a')(1-q)}{(1-q)^2}.$$

So it remains to show that $a(1+q) + 2a'b = (1+a')(1-q)$. Using $1 - q = a + a'b$,

$$(1+a')(1-q) = (1+a')(a+a'b) = a + aa' + a'b + a'^2b,$$
$$a(1+q) + 2a'b = a + aa'(1-b) + 2a'b = a + aa' - aa'b + 2a'b.$$

The difference (first minus second) is

$$a'^2b + a'b - 2a'b + aa'b = a'b\,(a' + a - 1) = 0,$$

since $a + a' = 1$. Therefore

$$\mathrm{E}(T) = \frac{\alpha}{a} + \frac{\beta}{b},$$

as required.

Checks

- If $a = 1$, Arthur hits at once: $\alpha = 1$, $\beta = 0$ and $\mathrm{E}(T) = 1$, matching $\alpha/a + \beta/b$.
- If $a = b = p$, then $\alpha/p + \beta/p = (\alpha + \beta)/p = 1/p$. That is right: the contest is then just a sequence of independent shots, each hitting with probability p, and by the first part the first hit takes $1/p$ shots on average.
- With $a = 0.3$, $b = 0.45$ we get $\alpha = 0.3/0.615 \approx 0.4878$ and $\mathrm{E}(T) = 1.7/0.615 \approx 2.764$; a simulation of 200 000 contests gave 0.486 and 2.769.

Common slips

- Forgetting that Boadicea can only win if Arthur has just missed, and writing $\beta = b/(1 - a'b')$; the missing factor a' shows up when $\alpha + \beta \neq 1$.
- Counting rounds instead of shots: a contest Boadicea wins in round $k + 1$ has $2k + 2$ shots, not $k + 1$.

Beyond STEP

Results like this are usually proved more slickly by conditioning on the first round ("first-step analysis"): if nobody hits in round one, the contest starts afresh after two shots, so $\mathrm{E}(T) = a \cdot 1 + a'b \cdot 2 + a'b'\big(2 + \mathrm{E}(T)\big)$, which gives $\mathrm{E}(T) = (1 + a')/(1 - a'b')$ in one line. First-step analysis is the basic tool for Markov chains, studied in the first or second year of a mathematics degree. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and this style of argument.

WORKED SOLUTION

STEP III 2010, Question 13

Probability and Statistics • Correlation of Linear Combinations

STEP III 2010, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q13

This question has a full guided treatment in *Beyond STEP*, Section 13.7, *Manufacturing Correlation*: https://beyondstep.keithmcnulty.org/ch13/correlation.html

The key idea

Everything here is a linear combination of independent variables with mean 0 and variance 1, so every expectation, variance and covariance can be read off from the coefficients: if $U = \sum u_k Z_k$ and $V = \sum v_k Z_k$ then $\mathrm{Cov}(U, V) = \sum u_k v_k$. The question is really building, one variable at a time, three standardised variables with any prescribed correlations, which is the Gram–Schmidt process in disguise. Each new condition pins down one new coefficient.

Solution

Throughout we use bilinearity of covariance, $\mathrm{Var}(Z_k) = 1$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$ (which holds because the Z_k are independent). In particular, for constants u_k, v_k,

$$\mathrm{Cov}\Big(\sum_k u_k Z_k,\ \sum_k v_k Z_k\Big) = \sum_{j,k} u_j v_k \,\mathrm{Cov}(Z_j, Z_k) = \sum_k u_k v_k.$$

Correlation of Z_1 and Z_2

Independence gives $\mathrm{Cov}(Z_1, Z_2) = \mathrm{E}(Z_1 Z_2) - \mathrm{E}(Z_1)\mathrm{E}(Z_2) = 0$, so $\mathrm{Corr}(Z_1, Z_2) = 0$.

The variable Y_2

Write $s = \sqrt{1 - \rho_{12}^2}$, which is real and positive since $-1 < \rho_{12} < 1$. Then $Y_2 = \rho_{12} Z_1 + s Z_2$, and by linearity of expectation

$$\mathrm{E}(Y_2) = \rho_{12}\mathrm{E}(Z_1) + s\,\mathrm{E}(Z_2) = 0.$$

By the covariance rule above,

$$\mathrm{Var}(Y_2) = \rho_{12}^2 + s^2 = \rho_{12}^2 + (1-\rho_{12}^2) = 1, \qquad \mathrm{Cov}(Y_1, Y_2) = \mathrm{Cov}(Z_1, \rho_{12}Z_1 + sZ_2) = \rho_{12}.$$

Since $\mathrm{Var}(Y_1) = \mathrm{Var}(Y_2) = 1$, the denominator in the definition is 1, and

$$\mathrm{Corr}(Y_1, Y_2) = \rho_{12}.$$

The variable Y_3

The condition $\mathrm{E}(Y_3) = 0$ holds automatically for every a, b, c, so it tells us nothing. The other three conditions give one equation each. As all of Y_1, Y_2, Y_3 have variance 1, each correlation equals the corresponding covariance.

Correlation with Y_1. $\mathrm{Cov}(Z_1,\ aZ_1 + bZ_2 + cZ_3) = a$, so

$$a = \rho_{13}.$$

Correlation with Y_2. $\mathrm{Cov}(\rho_{12}Z_1 + sZ_2,\ aZ_1 + bZ_2 + cZ_3) = \rho_{12}a + sb$, so $\rho_{12}\rho_{13} + sb = \rho_{23}$ and, since $s \neq 0$,

$$b = \frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{1-\rho_{12}^2}}.$$

Unit variance. $a^2 + b^2 + c^2 = 1$, and since $c \geq 0$ we take the non-negative root:

$$\begin{aligned} c^2 &= 1 - \rho_{13}^2 - \frac{(\rho_{23} - \rho_{12}\rho_{13})^2}{1-\rho_{12}^2} \\ &= \frac{(1-\rho_{12}^2)(1-\rho_{13}^2) - (\rho_{23} - \rho_{12}\rho_{13})^2}{1-\rho_{12}^2} \\ &= \frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1-\rho_{12}^2}, \end{aligned}$$

where in the last step the $\rho_{12}^2\rho_{13}^2$ terms cancel. Hence

$$c = \sqrt{\frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1-\rho_{12}^2}}.$$

Ask yourself What if the expression under the square root is negative?

Then no real c exists, and there are no random variables at all with these three correlations: the numbers $\rho_{12}, \rho_{13}, \rho_{23}$ are not mutually consistent. For example, $\rho_{12} = \rho_{13} = 0.9$ and $\rho_{23} = -0.9$ is impossible, because if Y_2 and Y_3 are both close to Y_1 they can't be strongly opposed to each other. The question tacitly assumes the given correlations are achievable, which is exactly the condition that this numerator is non-negative.

Final part: the X_i

Take, for $i = 1, 2, 3$,

$$X_i = \mu_i + \sigma_i Y_i,$$

with $\sigma_i > 0$ (we may take σ_i positive, since only σ_i^2 is specified). Then $\mathrm{E}(X_i) = \mu_i + \sigma_i \cdot 0 = \mu_i$ and $\mathrm{Var}(X_i) = \sigma_i^2 \mathrm{Var}(Y_i) = \sigma_i^2$. Adding constants doesn't change covariances, so

$$\mathrm{Corr}(X_i, X_j) = \frac{\sigma_i \sigma_j \,\mathrm{Cov}(Y_i, Y_j)}{\sqrt{\sigma_i^2 \sigma_j^2}} = \frac{\sigma_i \sigma_j}{|\sigma_i \sigma_j|}\,\rho_{ij} = \rho_{ij},$$

using $\sigma_i \sigma_j > 0$. So these X_i have all the required properties.

Checks

- If $\rho_{13} = \rho_{23} = 0$ we get $a = b = 0$, $c = 1$, so $Y_3 = Z_3$, independent of the others, as it should be. If $\rho_{12} = 0$ then $Y_2 = Z_2$ and $a = \rho_{13}$, $b = \rho_{23}$, $c = \sqrt{1 - \rho_{13}^2 - \rho_{23}^2}$, the obvious choice.
- Numerically, with $\rho_{12} = 0.5$, $\rho_{13} = -0.3$, $\rho_{23} = 0.2$ we get $a = -0.3$, $b \approx 0.4041$, $c \approx 0.8641$, and $a^2 + b^2 + c^2 = 1$. A simulation of 2×10^5 triples (using uniform, not normal, Z_k) gave sample correlations 0.500, -0.302, 0.200 for the X_i, with the right means and standard deviations.

Common slips

- Forgetting that $\mathrm{Var}(Y_2) = 1$ means the correlations are just covariances, and then dividing by the wrong thing.
- Writing $X_i = \mu_i + \sigma_i^2 Y_i$: the variance scales by the *square* of the multiplier.
- Taking $\sigma_i < 0$ for one variable, which flips the sign of its correlations.

Beyond STEP

The three conditions say that the matrix A of coefficients (rows $(1, 0, 0)$, $(\rho_{12}, s, 0)$, (a, b, c)) satisfies $AA^{\mathsf{T}} = R$, where R is the correlation matrix. This is the *Cholesky decomposition* of R, and it exists exactly when R is positive semi-definite; the numerator of c^2 is $\det R$. It is how statisticians simulate correlated variables in practice, and with normal Z_k it produces the multivariate normal distribution. Blitzstein and Hwang, *Introduction to Probability*, covers covariance, correlation and the multivariate normal; Strang, *Introduction to Linear Algebra*, explains positive definite matrices and the Cholesky factorisation.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

This book takes the most interesting STEP questions, from the specimen papers that preceded the first examination in 1987 to the papers of 2025, and uses each one as a doorway. Every question is solved carefully, with the questions a good student asks along the way, and then the book steps through the door: where the idea comes from, what it becomes at university, and what to read next.

It is written for talented aspiring mathematicians preparing for STEP who want more than practice papers: to start thinking like a mathematics undergraduate, and to find out what is waiting for them.

- 13 chapters across pure mathematics, mechanics, probability and statistics
- 136 past STEP questions, each with a guided solution
- The university mathematics behind every question
- Follow-up exercises with answers, and further reading

2010 Worked Solutions. The whole book, every chapter and all the worked solutions are free to read, download and share at **beyondstep.keithmcnulty.org**

Text dedicated to the public domain (CC0 1.0). STEP questions © Cambridge University Press & Assessment.

